

claude-windows / d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd.jsonl`
- SHA-256: `e2b2a6f7975630f617a6bc96b2578881bfde72890ba0a80b517d8f41b931ed5a`
- Source modified: `2026-07-05T05:40:00+00:00`
- Imported at: `2026-07-05T16:48:19+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd`

Transcript

1. user (2026-07-04T08:40:29.993Z)

Hello. Can you pick up from where the last session ended as per "MEMORY.md ? new top "? Start here" entry pointing at the handoff. session-handoff.md ? the full pickup doc. current-state.md ? a HANDOFF checkpoint." And address the review comments for all the open PRs for <https://github.com/Khelsutra/badminton-highlight-indexer/pulls> ? Also make sure to keep the north star goal of running the entire service end 2 end as the final goal and ensuring as many rough edges are captured and fixed or filed as github issues.

2. assistant (2026-07-04T08:40:34.993Z)

I'll start by reading the handoff memory files to pick up context.

3. user (2026-07-04T08:40:36.125Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

4. user (2026-07-04T08:40:37.103Z)

<tool_use_error>Directory does not exist: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.</tool_use_error>

5. assistant (2026-07-04T08:40:43.221Z)

The project memory lives under the khelsutra-guru path. Let me check there.

6. user (2026-07-04T08:40:43.959Z)

c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru\memory\working-agreement-attribution.md
c:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru\memory\candidate-segments-finetuning.md
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c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\session-handoff.md

```

7. user (2026-07-04T08:40:47.644Z)

```

1      # Memory Index
2
3      ## ? Start here
4      - [?? A1 serving: resolve PR reviews ? merge ? e2e](session-handoff.md) ? **FRESHEST,
2026-07-04. START HERE.** A1 alpha serving is DEPLOYED + LIVE behind Cloudflare Access
(`api.khelsutra.guru`). 5 bug-fix PRs opened; **462/#464 merged*; **465/#467/#468/#469 OPEN,
mergeable, CI-green, all review comments addressed, awaiting owner LGTM.** ? NEXT = watch the 4
PRs, merge each on LGTM (squash; rebase #467?#469 / #468?#469 conflicts), then run the full e2e
deploy test (verify #463 >32MiB signed-URL download + #461 upload staging), restore edge-auth,
flip A1 DONE. Full protocol + deployed state + config + gotchas in [[session-handoff]].
5      - [Working root = khelsutra-guru (multi-repo)](working-root-khelsutra-guru.md) ? **owner
directive 2026-06-29:** OPEN `Projects\khelsutra-guru` as the UI project (5 git repos), switch
sub-repo per task. Memory store MIGRATED to the khelsutra-guru key (Option B); badminton-
highlight-indexer is no longer the startup anchor. Detail + migration-verification note
[[working-root-khelsutra-guru]].
6      - [? rally finding-B promotion (P2b) + launch signoff](next-session-rally-finding-b-
promotion.md) ? **freshest, 2026-06-29.** Fixes A/B/C MERGED (`0c9c693`); B behind
`inactivity_temporal_density` (default-OFF no-op). ? NEXT = run golden A/B at SERVED, promote-
or-reject via `promote_if_served_safe`, then full suite + GO/NO-GO launch report (recommend,
don't decide). Runbook `docs/RALLY_DETECTION_FIXES_PLAN.md` S8 Q2; detail [[next-session-rally-
finding-b-promotion]].
7      - [? multi-shuttle / prominent-court detection](next-session-multishuttle-rally.md) ?
design journey (2026-06-19/20). Multi-track + court-aware selector measured vs GOLDEN: does NOT
beat single-track baseline ? NO PR; it's a targeted lever, not an F1 win. ? NEXT = the verdict-
flipping experiment: a separable-adjacent-court golden clip (`Boxhill_Doubles`, owner GPU) w/
court-membership-only selection. Detail [[next-session-multishuttle-rally]]. *(Design/scratch.)*
8      - [? batteries-included downloadable app](next-session-packaging-app.md) ? packaging

```

track. ? P3+P2b+P2c DONE 2026-06-22 (engine #310/#312, khelsutra #88) ? v1 LIGHT FEATURE-COMplete; only OWNER steps remain (clean-box acceptance + ffmpeg UX). North-star (native-WASB/training/annotation) future + gated. Doc `docs/PACKAGING\_PLAN.md` §7; detail [[packaging-app-plan]].

9 - [? khelsutra UX/l10n/runnable-tool](next-session-ux-localization.md) ? refreshed 2026-06-22. Tool BUILT + localized (6 langs); ALPHA LIVE + real-use-proven at `https://api.khelsutra.guru` (single-origin). ? NEXT = 13-issue triage backlog: async `api/compile` (#329, kills the 524), M7 Cloud-Run L4 GPU, NSFW litmus (#332) + UX/obs polish; launch gate needs native-speaker review. Full ramp-up in [[current-state]] SESSION WRAP. *(Engine/serving track = own [session-handoff](session-handoff.md)).*

10 - [? golden regression fixtures (optional)](next-session-golden-fixtures.md) ? track COMPLETE 2026-06-22 (8 PRs); clean clone runs rally-seg regression (synthetic + 6 real fixtures `fx\_r0X`), cross-OS + leak-guarded. ? NEXT (optional): (1) M4 quality-floor on real fixtures; (2) deferred video-name rename (before open-sourcing); (3) 01 protobuf / 02 key-gated. Detail [[next-session-golden-fixtures]].

11 - [? Studio media lifecycle (ingest?store?annotate?train)](project-studio-media-lifecycle.md) ? **STORAGE HALF COMPLETE 2026-07-01.** Full "add a video ? store in chosen medium (S3/GCS/Drive/local) ? free-space checked ? appears in my videos" is END-TO-END: P2 engine free-space (#407), P3 capabilities-storage (#408), P4 `POST /api/assets` (#409), P5 SPA medium picker (khelsutra #137) ? all MERGED. Annotate-in-UI (P6-P9) already shipped. 5 screencasts (04 = store-in-my-medium). **? ONLY owner-gated remains: P10/P11 (corpus promotion + train/reeval, DEFERRED D13) + `L` beta follow-ups ? do NOT build without owner lifting the gate.** Doc `khelsutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`; detail [[project-studio-media-lifecycle]].

12 - [Current state](current-state.md) ? **live micro-checkpoint** (updated every checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.** Now LIVE-only (handoff + today's checkpoints); kept read-first-short.

13 - [Current-state archive](current-state-archive.md) ? chronological checkpoint history (2026-06-06 ? 06-18, 96 blocks) split out of current-state 2026-06-19. Deep history only; NOT read each session.

14 - [Session handoff](session-handoff.md) ? current state · next steps · open decisions · ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift, session wrap) ? see working-agreement-checkpoints staleness tripwire.

15 - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read `docs/CODE\_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.

16 - [File-placement conventions](file-placement-conventions.md) ? **where new files go** (scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); **read before committing a new file.** Authoritative table = `docs/CODE\_MAP.md` §0 (PR #138); scratch/ gitignored (#136).

17

18 - [Working agreement: attribution](working-agreement-attribution.md) ? never assert fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The silver-bullet rule.

19 - [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.

20 - [Feedback: tail long-job logs](feedback-tail-long-job-logs.md) ? proactively tail progress/summaries from long-running background jobs while they run, not just the final result.

21 - [Working agreement: merging](working-agreement-merging.md) ? owner authorized Claude to MERGE its own good-engineering-fix PRs on verified-green CI (2026-06-12); product/strategy/licensing stay owner-gated.

22 - [Working agreement: tracked project docs](working-agreement-tracked-project-docs.md) ? substantial design/project work gets a **tracked project doc** (template + §7 progress tracker + definition-of-done); freeze ? track across sessions ? mark done ? archive. Owner-loved framing (2026-06-16); codified in `CLAUDE.md` + `docs/PROJECT\_DOC\_TEMPLATE.md` (PR #186).

23 - [Doc-status register + archival due-diligence](doc-status-register.md) ? owner directive (2026-06-21): `docs/DOC\_STATUS.md` is the **living doc-lifecycle register** ; **archive foundational docs only AFTER honestly updating them** (verify?flip status+completion note?update refs?keep lineage?move). Worked example = DECOUPLED #270. Codified in `CLAUDE.md` archive policy.

24 - [Lesson: verify engine-surface before scoping](lesson-verify-engine-surface-before-scoping.md) ? **anti-spiral (2026-06-21):** before scoping a doc's `[gap]`/`net-new`/"X-only" engine claim as work, RE-VERIFY against engine code at a cited `file:line` (engine drifts between sessions). `[verified]` without a `file:line` = a hypothesis. Born from B-P1; docs fixed ? khelsutra #55.

25 - [Launch Report convention](launch-report-convention.md) ? owner directive (2026-06-17): EVERY default-config/quality-bit flip ships with a documented **Launch Report**

(total-quality-impact table on both rulers + over-seg guard + honest caveats), recorded for posterity BEFORE the flip merges. Born from the Gate-A #146 flip that regressed (#151). Pairs with [[eval-dual-set-regression]].

26 - [Feedback: launch quality bar](feedback-launch-quality-bar.md) ? approaching launch (2026-06-20): tightly ****isolated one-item PRs + rising coverage****, and ****serious observability**** (full log analysis across every NEW code path; correlation/job IDs; loud failures, no silent drops). Make Observability a first-class doc section + DoD item.

27 - [Review: trust CI, don't re-run it](review-trust-ci-over-rerun.md) ? owner feedback (2026-06-25): ****don't re-run the exact CI checks locally during review**** (CI already ran them green on that commit ? wasted time); rely on green CI + a documented PR ****## Verification**** section, and spend local effort only on what CI can't cover (merge-safety, env-noise vs real defect, findings, machine-side paths). Refines the global pre-LGTM gate. Proposed the Verification convention on #378 (+offered a PR template).

28 - [Review: sign-off policy](review-signoff-policy.md) ? owner's settled rule (2026-06-25): ****sign off (written LGTM) on pure-refactoring / zero-behavior PRs once CI is green**** (confirm semantics-free via AST-identity, not re-running CI); ****HOLD logic/behavior changes for the owner**** unless authorized per-PR. Red CI blocks sign-off regardless. gh=avidullu ? written LGTM, never a formal Approve.

29 - [Gotcha: shared-checkout branch collisions](gotcha-shared-checkout-branch-collisions.md) ? spawned tasks + sibling sessions SHARE this checkout; ****branch from `origin/master` explicitly, re-verify HEAD before commit, stage explicit paths, serialize repo writes.**** The #116/#117 collision (owner upset). Global form in ~/.claude/CLAUDE.md.

30 - [Lesson: branch rename closes its open PR](lesson-branch-rename-closes-open-pr.md) ? renaming the head branch of an OPEN PR ****closes**** it (does NOT retarget); cost PR #393 ? roll-forward #394 (2026-06-29). Don't rename a branch with an open PR; rename before opening, or plan to re-create.

31 - [Gotcha: long GPU/WSL runs](gotcha-long-runs-gpu-wsl.md) ? running the WASB pipeline E2E on a big 4K video: Modern Standby kills it (keep-awake), cv2 extraction is the bottleneck (ffmpeg pre-extract), proxy-index/4K-stitch recipe + how to observe each phase.

32 - [GCP Ops Agent: always enable](gcp-ops-agent-enable.md) ? owner directive (missed TWICE): every rally GPU VM needs the Ops Agent for Observability/GPU graphs ? ****3 pieces**** (logging+monitoring APIs ON + SA metric/log-writer roles + agent install via startup-script). BAKE into `deploy/gcp` provisioning so it's automatic.

33 - [GCP VM: always DELETE, never just stop](gcp-vm-always-delete.md) ? owner cost rule (2026-06-20): a STOPPED VM still bills its disk (~\$34/mo for 200GB pd-ssd); ****DELETE**** ? \$0 (boot disk autoDelete=true). `teardown.sh` now defaults to `delete`; audit `gcloud compute disks list` for orphans. DO droplet `claude-do-rally-test` (~\$24/mo) is a shared testbed (owner-gated).

34 - [Compute = GCP only; DO dropped](compute-stack-gcp-only.md) ? ****standing decision (ADR-013, 2026-06-20):**** GCP is the SOLE compute stack (training + serving); ****DigitalOcean dropped for compute**** (GPU not enabled/credit-funded; Paperspace subscription-gated). Provision via `deploy/gcp/` only; don't relitigate DO/Paperspace without a new ADR; \$200 DO credits ? non-compute only.

35 - [Storage bucket config requirements](storage-bucket-config-requirements.md) ? owner directives (2026-07-01): bucket names via ONE config (no hardcoded literals) . split by 3 data families (user videos vs golden/dev vs scratch ? a compliance boundary, D1) . per-deployment/per-region buckets (e.g. user videos per continent). Audit: runtime single-source ?; deploy/gcp scripts had duplicated corpus/nightly literals under TWO env-var names ? ****FIXED (engine #410): `deploy/gcp/buckets.env` sourced by all 7 scripts, corpus unified under `RALLY\_CORPUS\_BUCKET` (legacy names = back-compat aliases), `.gitignore` `!` negation so the no-secret env file is tracked.**** Per-region = env override, no code change. Detail [[storage-bucket-config-requirements]].

36 - [GCS = corpus source of truth](gcs-data-source-of-truth.md) ? ****owner goal (2026-06-22):**** run ALL train+infer from cloud, ALL data in `gs://khehsutra-rally-corpus`. ****Pull from the bucket, not C:/F:/Drive.**** Map = `docs/DATA\_IN\_GCS.md`; cloud engine built (`backend.worker infer|train` by URI). Gap = ~25 MB corpus metadata local-only + trivially migratable. Pairs [[compute-stack-gcp-only]].

37 - [Cache-hygiene policy](cache-hygiene-policy.md) ? retention table (purge frames/staged vs keep CSVs/DB/reels) + the shipped tooling: self-cleaning runners (keep_frames default false), tools/cleanup_caches.py (dry-run-first), disk guard, SessionStart nudge hook. The `for e in \*` WSL footgun.

38 - [Tool: trim_video.py](tool-trim-video.md) ? reusable `scratch/trim\_video.py` for keyframe-aware LOSSLESS video trims (drop first/last N sec), `--exact` re-encode with NVENC?libx264 fallback. Reach for it instead of hand-rolling ffmpeg trims.

39 - [Decision-rigor norm](decision-rigor-norm.md) ? ****NUMBERS INFORM, FOUNDATIONS DECIDE****

(owner directive 2026-06-14, set by the Gate-A flip). Every signal/architecture change gets ablation-layer rigor (per-signal marginal ?F1 + honest CI/sign-test) AND a foundational lens (extensible/orthogonal/reversible), never numbers alone. Ties [[weak-signals-retained]] + [[paper-coauthor-aim]].

40

41 ## Project: rally segmentation pipeline

42 - [Candidate segments & fine-tuning](candidate-segments-finetuning.md) ? why raw detector candidates are persisted (detector-agnostic table); golden-eval-set goal

43 - [TrackNetV4/WASB via WSL2](tracknet-wasb-wsl2-decision.md) ? self-hosted shuttle trajectory models (not LLM APIs); WASB/TrackNetV3 = MIT, commercial-clear

44 - [Collector?indexer bridge](collector-indexer-bridge.md) ? 4-phase plan to train/eval a generic temporal rally segmenter; Phases 1?2 done, 3 in progress, 4 next

45 - [Golden-set vendoring](golden-set-vendoring.md) ? generate golden rally labels via 3rd-party vendors (Bangalore-local) + extensible regression flow; Roora selected for a 3-sport pilot, guideline + collector ingestion built, GS-1/2/4 landed, GS-3 + gold key pending

46 - [Roora guidelines Drive doc](roora-guidelines-drive-doc.md) ? the CANONICAL vendor-facing Brief+Guide Google Doc (supersedes v3/v5/v6/v7 Drive copies; v7 anomaly RESOLVED 2026-06-12); TODO markers still to fill before sending

47 - [Audio (E1) findings](audio-e1-findings.md) ? quantitative gains/losses of the classical-onset audio cue (recall 100% but discrimination ~2.2x, band-pass null) + reusable lessons for adding the NEXT signal. Audio NOT adopted for fusion; detector kept (PR #35).

48 - [Weak signals retained](weak-signals-retained.md) ? owner directive: non-significant cues (person/audio/court-bounds/future **shuttle-drop**) stay **default-OFF + retained** as scorer feature columns (no special-casing) for a growing (Roora) corpus ? never deleted on a small-n null. Measured: person ?F1 ?0.021, +audio +0.079 (both CIs span 0 at n=6).

49 - [Eval boundary tolerance](eval-boundary-tolerance.md) ? owner directive (2026-06-16): rally scoring must tolerate the ~1?1.5s **service-window** start-fuzziness ($\pm 2s$, not strict tIoU 0.5 which only allows $\pm 1.9s$ on a 5.7s rally). KEY measured finding (mp2+GX010128, n=2): local's |?start|?Gemini but |?end| 2?3x worse ? **the gap is rally-END detection**, not generic boundary precision ? next lever (R2 metric ? R4 end-rule). Reusable `scratch/at\_par.py`.

50 - [Golden corpus grew 9?15](golden-corpus-grew-9-to-15.md) ? **2026-06-22:** #316 GCS migration restored the original-6 single-court labels ? local golden eval set now **15 clips** (was 9). Trips 9-clip-pinned litmus (re-grounded PR #322); check corpus size before suspecting a code regression. Pairs [[eval-dual-set-regression]] [[gcs-data-source-of-truth]].

51 - [Eval dual-set regression](eval-dual-set-regression.md) ? owner directive (2026-06-16): keep a **RAW eval set** + a **GROWING GOLDEN-finalized set** (promote raw?golden when at-par) + a separate **golden TRAINING set** ; every quality launch must NOT regress on EITHER eval set. The ablation layer A/Bs these; wire dual-set aggregate into `run\_regression.py` when R2 lands.

52 - [Decision-snapshot regression](decision-snapshot-regression.md) ? **tracked project, decisions LOCKED, #383 merged (2026-06-25).** Deterministic **quality-neutral proof**: freeze+compare DECISION output (segments+stitch plan) off a cached trajectory in CI ? lets a genuine refactor be auto-called quality-neutral (NOT a reel-MD5). ? next = \$7 P1. Doc `docs/DECISION\_SNAPSHOT\_REGRESSION.md`.

53

54 ## Packaging / distribution

55 - [? Next session: packaging app](next-session-packaging-app.md) ? **the icebreaker/handoff to resume this track** (read first; also under ? Start here).

56 - [Batteries-included app plan](packaging-app-plan.md) ? downloadable Python app (local web service for inference/training/annotation). **11 PRs merged 2026-06-21/22** ? LIGHT-tier bundle-ready (`pip install`?`indexer --app`?bundled SPA, torch-free). Doc `docs/PACKAGING\_PLAN.md` \$7. **01+02 LOCKED** (cross-platform, LIGHT-first WASB-complete north star); 03+ gated. Pairs [[ui-driven-cloud-serving-gap]]. Don't relitigate locked calls.

57

58 ## Monetization (hosted API)

59 - [Monetization plan](monetization-plan.md) ? hosted video?highlights API on a GPU VM; licensing & per-video cost are first-class; user is Bangalore-based

60 - [Monetization research status](monetization-research-status.md) ? audit done: Ultralytics AGPL swapped out (PR #8); MonoTrack/YOLO-NAS weights avoided; codec=watch; full report in docs/

61 - [Machine-pooling idea](machine-pooling-idea.md) ? owner "food for thought" (2026-06-20): pool GPU machines (queue-on-one-host vs Cloud-Run scale-to-zero) instead of spin-up/kill per task ? born from a single-GPU-slot collision between two sessions (quota=1). Infra-track design item.

62 - [UI-driven cloud-serving gap](ui-driven-cloud-serving-gap.md) ? **owner directive

(2026-06-21): cloud highlight flow must be UI-driven + ALPHA-SHAREABLE, not SSH/CLI; surface compute options (local vs cloud) in UI at start.** ? The compute-picker + real Cloud Run L4 GPU e2e validation LANDED 2026-06-22 ? see [[cloudrun-gpu-serving-gotchas]].

63 - [Cloud Run GPU serving gotchas](cloudrun-gpu-serving-gotchas.md) ? **reusable knobs to run the WASB worker on Cloud Run L4 GPU** (region?Mumbai · `--no-gpu-zonal-redundancy` · GCS volume-mount weights · PowerShell-not-Bash `/mnt` flags · `skip\_ai\_handoff=true` no-key · digest-pin? jobs update` · `[storage-gcs]` in image · control plane needs google-cloud-run+ADC). Verified 2026-06-22 (`artifacts/cloudrun_e2e_validation/`).

64 - [khelsutra site droplet parity](khelsutra-site-droplet-parity.md) ? **RETIRED 2026-06-25.** DO droplet `claude-do-rally-test` (dual-purpose site mirror + alpha engine) kept drifting from CF ? owner retired it: data archived to GCS+local, droplet deleted (\$0), monitor removed (khelsutra #117). Follow-ups: `api.` CNAME/CF-Access cleanup. Pairs [[compute-stack-gcp-only]] [[ui-driven-cloud-serving-gap]].

65

66 ## Forward-looking

67 - [Sibling internship repo](sibling-internship-repo.md) ? khelacharya shot-intelligence layer builds ON TOP of our rally segmentation; kept off master at branch level (NOT a separate GitHub repo ? corrected 2026-06-06)

68 - [Paper co-author aim](paper-coauthor-aim.md) ? **standing side-aim**: co-author a paper on "local, cheap & efficient rally detection via multiple clever ideas". Keep `docs/QUALITY\_ITERATIONS.md` as the ablation-ready empirical backbone; honest measured results ONLY.

69

70 ## Project: muneem-ai (SEPARATE root from khelsutra-guru)

71 - [muneem-ai workspace](muneem-ai-workspace.md) ? `C:\Users\avidu\Projects\muneem-ai` = 3 repos (**muneem** Python local-first finance bookkeeper + **node-text-detection** Node/Tesseract OCR + **Email-Mining** prior art). Per-session focus switch (2026-07-01); persistent root stays khelsutra-guru. muneem CI = ruff(len 100)/mypy/pytest-cov80, vendored `ruff-bin/ruff.exe`. Detail [[muneem-ai-workspace]].

72 - [muneem grocery-replenishment SENSOR](muneem-grocery-replenishment.md) ? **GR.0?GR.5 BUILT + MERGED to main 2026-07-01 (9 PRs #100?108).** Loop = `grocery ingest`/`ingest-receipt --store` ? `pantry due`/`pantry compare`. A depletion **sensor** (owner: "as a sensor, not a delight"): per-item history (`purchase\_items`, migration v8) ? slow-staple run-out + confidence + cheapest-store; 2nd source via OCR/text receipt. Built on 206 real BigBasket line-items from **avidullu.svp@** Gmail. GR.5 = deterministic structural canonicalization; **the LEARNED Tier-1 is the ONLY unbuilt piece ? parked under \$8.5 (owner-gated).** Detail [[muneem-grocery-replenishment]].

73

8. user (2026-07-04T08:40:48.589Z)

```

1      ---
2      name: session-handoff
3      description: "Living handoff: current state, next steps, ramp-up kit, and key
decisions."
4      metadata:
5        node_type: memory
6        type: project
7      ---
8
9      # Session handoff ? khelsutra-guru
10
11      > Last refreshed: 2026-07-04. **A1 alpha serving is DEPLOYED + LIVE behind Cloudflare
Access; 5 fix PRs
12      > are in review.** This handoff hands off TWO tasks to the next session: (A) resolve the
PR reviews +
13      > merge on LGTM, then (B) run the full end-to-end deploy test. See [[current-state]].
14
15      ## ? YOU ARE HERE ? pick up (A) PR-review resolution ? (B) full e2e test
16
17      The A1 stack is live (see "Deployed state" below). Five bug-fix PRs were opened; **462
+ #464 are
18      MERGED**; **465, #467, #468, #469 are OPEN, all MERGEABLE + CI-green, every review
comment addressed,
19      awaiting the owner's LGTM.** Do NOT merge until each has an LGTM. After all four merge ?

```

```

run the e2e.
20
21  **FIRST STEPS for the next session:** `git -C C:\Users\avidu\Projects\khelsutra-
guru\badminton-highlight-indexer
22  pull --ff-only`; `gh auth status` (should be `avidullu`); read this whole file; then
start watching the
23  4 PRs (protocol below). The review watcher from the prior session was a background
PowerShell loop in a
24  session-specific scratchpad ? it does NOT carry over; just poll the PRs yourself (loop
or periodic check).
25
26  ---
27
28  ## (A) PR-REVIEW RESOLUTION ? the 4 open PRs
29
30  Repo `Khelsutra/badminton-highlight-indexer`. Reviewer = **avidullu** (the owner).
"LGTM" arrives as a
31  PR comment or a formal approval. Merge convention = **squash** (`gh pr merge <n>
--squash --delete-branch`).
32
33  | PR | branch | fixes | head | review state |
34  |---|---|---|---|---|
35  | **#465** | `fix/allowed-video-root-uri` | hosted-mode input jail vs gs:// | cd4bc73 |
change-request ADDRESSED (input_root allowlist); awaiting LGTM |
36  | **#467** | `fix/upload-gcs-staging` | closes #461 (upload?gs:// staging) | d3a9629 |
comment ADDRESSED (thread storage config); awaiting LGTM |
37  | **#468** | `fix/drain-concurrency` | closes #466 (parallel videos) | 8534077 | re-
designed; **NEW 2nd blocker OPEN (claim_due_job false-empty) ? FIX THIS FIRST** |
38  | **#469** | `fix/highlights-signed-url` | closes #463 (32 MiB download ? signed URL) |
d3b0e64 | comment ADDRESSED (thread config + extract helper); awaiting LGTM |
39
40  **What was already discussed + resolved (so you're not surprised by the threads):**
41  - **#465** ? reviewer flagged the offline suite was red
(`tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`
42  asserts hosted mode rejects gs://). Fixed in **cd4bc73**: `validate_input_path` now
takes `allowed_uri_root`
43  (wired from `storage.input_root`); a bucket URI is allowed ONLY under input_root,
arbitrary buckets
44  rejected (tenant isolation). The contract test passes unchanged. Replied on the PR.
45  - **#467** ? reviewer: `storage.put(local_src, staged_uri)` dropped the storage config.
Fixed: pass
46  `config=storage_cfg.model_dump()`. Test asserts the config reaches `put()`. Replied.
47  - **#468** ? reviewer withdrew LGTM (round 1): over-broad concurrency. **Re-designed +
pushed (8534077):**
48  `orchestration._LOCAL_COMPUTE_LANE` (BoundedSemaphore(1)) around the in-process
segmenter + compile
49  stitch (remote polls don't hold it ? overlap; local/compile serialize); broadened
50  `_cloud_dispatch_possible()` so `compute='cloud'` overrides scale too. I also OFFERED
(in a reply) to
51  make the lane *park* via `JobWaiting`/WAIT_AND_RESUME instead of *block* (starvation-
free) ? do that
52  only if the reviewer asks.
53  - **?? OPEN ? a 2nd blocker the reviewer found (round 2), NOT yet fixed. Address this
FIRST on #468:**
54  `claim_due_job()` (`backend/infrastructure/database_jobs.py:357-380`) returns `None`
when a worker
55  LOSES the CAS race (`rowcount != 1`); `_drain_due_jobs()`
(`backend/api/job_worker.py:273-276`)
56  treats `None` as "queue empty" and EXITS. So with `max_workers>1`, workers racing
the same first
57  queued row ? losers return `None` ? their drains stop even though other jobs are
queued (reviewer
58  reproduced: 8 queued + 8 claimers ? only 2/8 claimed ? dispatch capacity wasted).
59  **FIX:** make `claim_due_job()` RETRY on a lost race ? re-SELECT + re-attempt until
it either claims

```

```

60         a row OR a fresh SELECT finds no runnable rows (or return a distinct lost-race
sentinel so
61         `_drain_due_jobs` LOOPS instead of exiting). **ADD a regression test:** N queued
jobs + N concurrent
62         claimers/drains prove N DISTINCT jobs are claimed with no false-empty. Then FULL
suite + lint + push
63         + reply on #468.
64         - **#469** ? reviewer: mirror `put` + `exists` + `signed_url` dropped the storage
config. Fixed: added
65         `orchestration._storage_settings(cfg)` threaded into all three; extracted
`sIGNED_reel_url` to keep
66         `backend/main.py` under its line budget (P23/P24). Tests assert config reaches
exists/signed_url. Replied.
67
68         **Watch/merge protocol (repeat until all 4 merged):**
69         1. Poll `gh pr view <n> --json state,reviewDecision,comments,reviews` for each open PR
(every few min).
70         2. On **new change-request/comment** ? address it (fix + FULL local suite + push +
reply). ? Run the
71         **whole** `pytest tests/` suite before pushing, not just the changed file ? that's
how #465's CI-red
72         slipped through the first time (`python -m pytest tests/ -q -p no:cacheprovider`, ~2
min; + `ruff check`
73         + `mypy backend/...`).
74         3. On **LGTM/approval** for a PR ? verify `gh pr checks <n>` all pass +
`mergeable=MERGEABLE`, then
75         `gh pr merge <n> --squash --delete-branch`.
76         4. **Conflicts after a merge (expected):** #467 & #469 both touch
`backend/orchestration.py`; #468 & #469
77         both touch `backend/main.py`. When one merges, re-check the others' `mergeable`; if
`CONFLICTING`,
78         in a **worktree off origin/master** merge master into the branch, resolve, re-run the
FULL suite +
79         lint, push. (Git-safety: `git worktree add <path> <branch>`; stage explicit paths;
re-verify branch
80         before push; never `git add -A`. Shared checkout ? sibling sessions may move master.)
81
82         ---
83
84         ## (B) FULL E2E DEPLOY TEST ? run AFTER all four PRs are merged
85
86         Goal: prove the merged fixes work on a real deployed build ? the >32 MiB signed-URL
download (#463) and
87         the browser-upload?gs:// staging (#461) ? then restore the live edge-auth state and flip
A1 ? DONE.
88
89         1. **Rebuild the worker image from merged master** (has #461/#463/#465/#468 backend code
+ #462's
90         `[storage-gcs,cloud-run]`: `gcloud builds submit
--config=deploy/cloudrun/cloudbuild.yaml` (heavy
91         CUDA build, ~15 min; region asia-southeast1). ? The base Dockerfile installs
`[storage-gcs,cloud-run]`
92         but NOT the `auth` extra (PyJWT[crypto]) that M9 edge-auth needs ? the *deployed*
image gets PyJWT via
93         a derived overlay. **Recommended follow-up:** add `auth` to the base Dockerfile so a
rebuild is
94         M9-ready (a small 6th PR). For the e2e itself, edge-auth is OFF, so PyJWT isn't
required.
95         2. **Build the derived control-plane image** = base + PyJWT: context `Dockerfile` =
`FROM
96         asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest`
+
97         `RUN python3 -m pip install --no-cache-dir "google-cloud-run>=0.10,<1"
"PyJWT[crypto]>=2.8"; tag
98         `.../rally/rally-control-plane:latest`. `gcloud builds submit <ctx> --tag <img>

```

```

--region asia-southeast1`.
99      (No more paths.py/backend overlays needed once the fixes are IN the base image ? the
prior session
100      overlaid #465's OLD blanket paths.py into the live image; the rebuild replaces it
with the corrected code.)
101      3. **Deploy CP with edge-auth OFF** so you can drive it with an identity token (no CF
login needed for
102      the test). Regenerate the service YAML (config below) with `EDGE_AUTH` unset and
103      `gcloud run services replace <yaml> --region asia-southeast1`. Run gcloud from
**PowerShell**.
104      4. **Run the CUJ + verify** (private service ? auth every curl with
105      `-H "Authorization: Bearer $(gcloud auth print-identity-token)"; URL below):
106      - `POST /api/process {"video_path":"gs://khelsutra-rally-
serving/videos/MahadevpuraSingles.MP4","segmenter":"wasb_hybrid","compute":"cloud"}`
107      ? poll `/api/jobs/{id}` to `done` (L4 run ~13 min) ? note segment ids.
108      - `POST /api/compile
{"video_id":<id>,"segment_ids":[1..N],"output_filename":"e2e.mp4"}` ? poll to done.
109      - **#463 check:** `GET /api/highlights/{video_id}/e2e.mp4` should now
**307-redirect** to a
110      `storage.googleapis.com` signed URL, and `curl -L` should download the FULL >32 MiB
reel (no edge-500).
111      Also confirm the reel is mirrored at `gs://khelsutra-rally-
serving/highlights/<video_id>/e2e.mp4`.
112      - **#461 check:** upload a small mp4 via `/api/upload/init`?`/part/0`?`/complete`,
then `POST /api/process`
113      with the returned LOCAL path + `compute:cloud` ? it should STAGE to gs:// and
dispatch (not hard-fail
114      with "cloud-serving needs a cloud input"). (Signing was already verified: `gcloud
storage sign-url`
115      impersonating the compute SA produced a working URL ? the SA has
self-`serviceAccountTokenCreator`.)
116      5. **Restore edge-auth ON** (regenerate the YAML with `EDGE_AUTH=1` + redeploy) so
api.khelsutra.guru's
117      live posture is back. Verify `/api/process` sans CF JWT ? 401 and
`https://api.khelsutra.guru/api/health`
118      ? 302 to the CF login.
119      6. **Write** `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-
serving_a1-e2e-<date>.md` (a NEW
120      file ? don't clobber the existing #457 or #464 runlogs), update [[current-state]] +
this handoff, and
121      **flip A1 ? DONE** in `docs/ALPHA_LAUNCH_READINESS.md` $7 (all DoD met: endpoint +
health + edge-auth +
122      upload?process?jobs?compile?download).
123
124      **Config for the CP service (base64 into a startup wrapper via
`RALLY_APP_HOME=/tmp/apphome/config.json`):**
125      ```json
126      {"deployment":{"profile":"cloud-
serving","compute_target":"cloud_run"},"indexing":{"skip_ai_handoff":true},
127      "storage":{"backend":"gcs","output_root":"gs://khelsutra-rally-
serving","input_backend":"gcs",
128      "input_root":"gs://khelsutra-rally-serving/videos"},
129      "security":{"edge_auth_enabled":<EDGE_AUTH>,"cf_team_domain":"https://khelsutra.cloudflare
access.com",
130      "cf_access_aud":"c980da67ed888fff935fcd913ae1eb4da63f9d65805d6e1c5ca76f92f517082b",
131      "allowed_video_root":"/tmp/apphome/output"}}
132      ```
133      CP deploy (per `deploy/cloudrun/README.md` §3a): `--command /bin/bash --args` a `-c`
wrapper that
134      `printf %s '<b64>' | base64 -d > /tmp/apphome/config.json && export
RALLY_APP_HOME=/tmp/apphome && exec
135      python3 -m backend.main --server --host 0.0.0.0 --port 8080`; flags `--no-cpu-throttling
--min-instances 1
136      --cpu 2 --memory 4Gi --port 8080 --timeout 300`; env `DEPLOYMENT_PROFILE=cloud-serving,
137      CLOUD_RUN_REGION=asia-southeast1,GOOGLE_CLOUD_PROJECT=gen-lang-

```

```

client-0306026826,RALLY_ALLOWED_HOSTS=``.
138   ? `CLOUD_RUN_JOB` is a RESERVED env name ? DON'T set it (code default `rally-worker` is
correct).
139   Deploy via a `gcloud run services replace <service.yaml>` MANIFEST (shell metachars in
the wrapper break
140   through the cmd-shim otherwise). The prior session's generator `gen_service_yaml.py` +
the derived-image
141   `Dockerfile` lived in the prior scratchpad ? rebuild them from the config + notes above.
142
143   ## Deployed state (live now ? owner said KEEP running)
144   - Project `gen-lang-client-0306026826`, region `asia-southeast1`. Compute SA
145   `492532266377-compute@developer.gserviceaccount.com` (has `roles/editor` +
self-`serviceAccountTokenCreator`).
146   - **L4 Job** `rally-worker` (nvidia-l4, 8vCPU/32Gi, FUSE `khelsutra-rally-serving`?/gcs,
`WASB_WEIGHTS=/gcs/models/?`).
147   - **CP Service** `rally-control-plane` rev `00004`, edge-auth ON**, PRIVATE (IAM +
allUsers-invoker so CF can
148   reach it; app still requires the CF JWT). URL `https://rally-control-plane-2bosgeukpa-
as.a.run.app`.
149   Deployed image = `?/rally/rally-control-plane:latest` = rally-worker + google-cloud-
run + PyJWT + the
150   **OLD blanket #465 paths.py overlay** (the corrected input_root-scoped code is in PR
#465 ? the e2e
151   rebuild replaces it; harmless for the closed 4-email alpha meanwhile).
152   - **M9 LIVE:** `api.khelsutra.guru` ? CNAME `ghs.googlehosted.com` (PROXIED) via Cloud
Run domain-mapping,
153   fronted by the "KhelSutra Studio (alpha)" CF Access app (team
`https://khelsutra.cloudflareaccess.com`,
154   aud `c980da67?082b`, 4-email allowlist:
avi.dullu@sheendullu13@suneetadullu@avidullu.svp@).
155   **Add a tester** = add their email to that app's "alpha allowlist" policy (CF
dashboard or API; token at
156   `G:\My Drive\cloudflare.token.txt`, has DNS + Access-Apps edit but NOT Rulesets/Config
nor
157   `/access/organizations` ? team domain was recovered via the public certs probe). CF
acct
158   `213c12cf60de9041dd8a3521aa206472`, zone `65ef7770a6f25697843b53a093504184`. ? Google-
managed cert
159   auto-renew (~90d) can stall behind CF proxy ? switch to a CF Origin CA cert if it
does.
160
161   ## Gotchas (learned the hard way)
162   - Run **gcloud from PowerShell**, not git-bash (MSYS mangles `/gcs`, `/tmp`); quote
comma-dict flags
163   (`--add-volume "name=?,type=?"`) or the cmd-shim splits them.
164   - The CP uses per-instance **ephemeral SQLite + /tmp** ? DB/reel don't survive a
rededploy or a second
165   instance. That's WHY #463 mirrors reels to gs:// + #461 restages uploads. maxScale is
3.
166   - **Pre-merge:** run the WHOLE suite locally, not per-file (the #465 CI-red lesson).
167
168   ## Key decisions (don't relitigate)
169   - WASB-only alpha (`skip_ai_handoff=true`); config.json wins over the cloud-serving
preset (loader.py:194).
170   - Reuse the worker image as the CP (+ google-cloud-run + PyJWT); IAM-private + CF Access
is the tester auth.
171   - #468 concurrency: serialize local/compile via `_LOCAL_COMPUTE_LANE`, parallelize only
remote polls; the
172   *park*-instead-of-*block* refinement is offered to the reviewer (do it only if they
ask).
173   - A1 stays NOT-DONE until the 4 PRs merge + the e2e passes; then flip it.
174
175   ## Ramp-up kit
176   - The 4 PRs (links via `gh pr list`), their review threads, and CI. Issues:
#461/#463/#466 (fixed by the PRs).

```

```

177 - Runlog already in repo: `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-
serving_a1-cuj_2026-07-04.md`
178 (PR #464, merged) ? the full A1 deploy story.
179 - Code: `deploy/cloudrun/README.md` §2/§3/§3a + `Dockerfile` · `backend/main.py`
(routes; `api/highlights`)
180 · `backend/orchestration.py`
(`maybe_dispatch_remote`, `_execute_compile`, `signed_reel_url`, `_LOCAL_COMPUTE_LANE`)
181 · `backend/api/job_worker.py`
(`configure_drain`, `_resolve_drain_workers`, `_cloud_dispatch_possible`)
182 · `backend/storage/` (`put`/`exists`/`signed_url`; gcs.py keyless signBlob) ·
`backend/config/models.py`
183 (`deployment.max_concurrent_remote_jobs`) · `backend/utils/paths.py`
(`validate_input_path` allowed_uri_root).
184 - Repo checkout: `C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer`
(master @ 4847232 at handoff).
185

```

9. user (2026-07-04T08:40:48.773Z)

```

1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for khelsutra-guru."
4 metadata:
5   node_type: memory
6   type: project
7   originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state - khelsutra-guru
11
12 **Updated:** 2026-07-03
13
14 ## Live Checkpoint
15
16 **2026-07-04 (HANDOFF) ? A1 serving LIVE behind CF Access; 5 fix PRs in review; NEXT
SESSION picks up
17 review-resolution ? merge ? e2e.** [[session-handoff]] fully specifies the pickup.
Merged: **462**
18 (cloud-run extra), **464** (A1 runlog). OPEN + MERGEABLE + CI-green + all review
comments ADDRESSED,
19 awaiting owner LGTM: **465** (input-jail input_root allowlist), **467** (upload?gs://
staging, closes
20 #461), **468** (drain concurrency ? RE-DESIGNED after LGTM-withdrawal:
`_LOCAL_COMPUTE_LANE` serializes
21 local/compile, only remote polls parallelize; closes #466; ? offered a *park*-vs-*block*
follow-up to the
22 reviewer), **469** (32MiB download ? gs:// signed URL, closes #463; keyless signBlob
VERIFIED). Do NOT
23 merge without LGTM; run the WHOLE suite before any fix-push (the #465 CI-red lesson).
After all 4 merge ?
24 rebuild worker image from master + derived PyJWT layer ? deploy (edge-auth OFF) ? CUJ
verifying the >32MiB
25 signed-URL download + the upload staging ? restore edge-auth ON ? flip A1 DONE.
Deployed: rev 00004,
26 api.khelsutra.guru LIVE (CF Access, 4-email allowlist). See [[session-handoff]].
27
28 ---
29
30
31 **2026-07-04 (earlier) ? A1 CORE SERVING DEPLOYED + gs:// CUJ PROVEN end-to-end.**
Deployed the L4 job
32 `rally-worker` + control-plane service `rally-control-plane` (PRIVATE/IAM, `--no-cpu-
throttling` +
33 min-instances 1) on Cloud Run asia-southeast1. Verified boot (no crash ? config.json's
34 `skip_ai_handoff=true` beats the cloud-serving preset per loader.py:194;

```

```

storage.backend=gcs avoids
35     STORAGE_INCOHERENT), health, capabilities. **gs:// CUJ green:** `/api/process` (gs://
video, wasb_hybrid,
36     compute=cloud) ? L4 WASB fully-local ? **48 rallies** ? `/api/compile` ? **valid 166 MiB
1080p30 reel,
37     244s** (ffprobe-confirmed; delivered to owner). 3 hosted-serving bugs surfaced+handled:
**PR #462**
38     (image lacked `google-cloud-run` dispatch client ? worked around with a derived CP
image, durable fix in
39     PR), **461** (browser-upload not staged to gs://), **463** (`/api/highlights` 500s >32
MiB Cloud Run
40     cap). Runlog **PR #464**. **M9 edge-auth ENABLED + verified** (rev 00004: image +
PyJWT/google-cloud-run + the
41     PR #465 URI-jail fix overlaid; `/api/process` sans CF JWT ? 401, `/api/health` ? 200).
Found+fixed **PR
42     #465** (allowed_video_root jail realpath-rejected gs://, which edge-auth would've
broken). **A1 NOT flipped
43     DONE** ? remaining: CF **DNS routing** for api.khelsutra.guru (owner dashboard: allUsers
+ domain-map/CF
44     origin-rule + re-point the dead-tunnel CNAME) + #461/#463. See [[session-handoff]].
45     Windows: run gcloud from PowerShell (MSYS mangles `/gcs`); quote comma-dict flags.
46
47     ---
48
49
50     **2026-07-04 (later) ? A1 deploy IN PROGRESS, context handoff. See [[session-handoff]]
for the exact next commands.** De-risked + half-executed: GCP resources recreated (serving
bucket + weights + test video + gemini secret + `rally` repo), worker image building, compile-
reliability fix MERGED (#460), config designed (WASB-only, gcs output_root, base64 in the
handoff). Clean state: 0 jobs / 0 services deployed, ~$0 idle. Next session: create L4 job ?
deploy CP service (`--no-cpu-throttling --min-instances 1`) ? health ? CUJ (bucket path first) ?
CF auth ? DEPLOY_REPORT ? flip A1 DONE. This session also merged #456/#457/#459/#460, reviewed
#451/#452/#458/#62, filed the SPA upload-guard bug.
51
52     ---
53
54
55     **2026-07-04 ? A1 infra proven + GPU optimization DONE + all cloud torn down.** First
real Cloud Run
56     L4 serving run of the owned WASB detector (fully-local, zero Gemini) worked end-to-end
via the M6
57     dispatch shim. **Key finding: the L4 is CPU-FEED-bound, not GPU-bound (GPU <1% on every
config).**
58     Measured on MahadevpuraSingles (13418 frames, L4/8vCPU): baseline 727.9s ?
**batch64+prefetch4 =
59     506.1s (?30.5%/1.44x, parity BYTE-IDENTICAL)**; torch cu118 arch bump = 0.8% noise
(discarded, +traj
60     numeric drift); batch128 = OOM (24GB ceiling). **Shipped: PR #457** (batch/prefetch
knobs + tests +
61     `DEPLOY_REPORT_cloud-serving_2026-07-04.md`) MERGED to master `c5fc27c` on 11/11 green
CI; earlier
62     **456** (loud AI-key startup check + unbuffered worker logs) also merged. Remaining ~5x
needs a
63     PARALLEL-FEED change (scoped in report §3: chunk video, 1 DataLoader worker/chunk; then
more-vCPU
64     helps). **ALL GCP resources deleted** (4 jobs, rally image repo, khelsutra-rally-serving
bucket,
65     gemini secret) ? $0-residual audit clean; kept owner's rally-corpus/-nightly. Detail
66     [[wasb-l4-perf-cpu-feed-bound]]. **A1 full serving deploy (control plane + CF-auth flip)
NOT done ?
67     deferred when owner pivoted to optimize+teardown; CF token at G:\My
Drive\cloudflare.token.txt is
68     VERIFIED-active for when it resumes.**
69
70     ---

```

71
72
73 **2026-07-03 (later) ? PR #452 (A2 golden-manifest portability) REVIEWED, awaiting author fixes.** Codex-authored `backend/eval/golden\_manifest.py` resolver + 8 consumers strict-wired + tests/docs. Gates reproduced green on head `5a91701` (16057/6 skip, cov 85.44%, ruff+mypy clean; the 5 served-Gate-A owner-box tests now RUN ? the A2 promise works). 4 findings posted (review 4627357796): **fusion_audio.py:38 left on raw json.load** (missed consumer, breaks A7 audio augment on stale paths ? top finding, in review body since the file isn't in the diff); test guard `\_golden\_specs` swallows ManifestPathError ? silent skip-all with generic reason; basename-shadow hardening (corpus verified collision-free today); DEFAULT_MANIFEST duplicated vs rally_seg_eval. Design note: \$9 Q4 producer-side fix stays open. Local checkout left ON the PR branch pending fix verification. **A1 pre-flight (owner lifted spend gate, \$300):** gcloud authed (project `gen-lang-client-0306026826`, 0 services/jobs deployed, `rally` registry + corpus buckets exist), Gemini key at `G:\My Drive\gemini.api\_key.txt` VERIFIED working (never print it; ? Secret Manager), CF token at `G:\My Drive\cloudflare.token.txt` **VERIFIED active** (zone khelsutra.guru readable, DNS read OK ? stale `api.`/`app.` CNAMEs exist from June deploy; Access API OK ? existing Access app "KhelSutra Studio (alpha)" on app.khelsutra.guru to mirror for api.; write scopes prove at first edit). **A1 pre-flight ALL GREEN ? A1 execution STARTED 2026-07-03.** SPA ingest screen exists (khelsutra `App.ingest` + /api/process/jobs client); khelsutra #329 sync-compile 524 is in A1 scope.

74
75 **2026-07-03 ? PR #451 (alpha readiness tracker) reviewed ? fixed ? verified ? OWNER-MERGED as `5eb8755` (squash); local PR branch deleted, all 5 repos synced.** ? Next grab = **A2 corpus portability** (ungated; manifest confirmed stale: points at dead `C:\Users\avidu\Projects\Annotation Setup\...`, data now under `khelsutra-guru/` + WSL + F:) and the **A6 ablation circular-import fix**. A3's records half is doable post-A2 (newer-9 source MP4s located on F: ? GX/BXH/mahadevpura folders); its vault/upload half stays owner-gated. **A7 is ALSO grabbable locally post-A2** (verified 2026-07-03: RTX 2070 Super 8GB + torch 2.6.0+cu124 CUDA=True in the repo env; all 9 trajectories in `khelsutra-guru/Annotation Setup/Trajectories/`; `fusion\_golden` person pass is windows-only single-pass + `fusion\_audio --augment` is CPU; newer-9 schema = windowing-only candidates, needs regeneration not upconvert; 2 sources GX010137/GX010141 still to pin on F:, Boxhill_Doubles?BXH-file mapping to confirm). Codex-authored `docs/ALPHA\_LAUNCH\_READINESS.md` (cross-repo alpha tracker; \$7 = execution source of truth) + index edits. Claude review posted 8 verified findings inline (table-breaking raw pipe in A12, ambiguous cross-repo "D13", missing CDS-M11 cross-link, spurious A4?A6 dep, analysis-derived D3/D4 presented as owner-locked, non-reproducible \$3 gen0 command, `GOLDEN\_VIDEOS` path inconsistency, stale-at-merge A0 status). Author fixed **all 8** in amended `41ce99c` (also added observability/job-failure triage to A13). Verified line-by-line on the new head + full local gates: suite **1595? / 11 skip**, cov **85.20%** (floor 70), `ruff` + `mypy backend training` clean, `git diff --check` clean, branch = exactly 1 commit over `origin/master e4914d1`. Product/strategy doc ? **merge stays owner-gated** (per working-agreement-merging). After merge, the real alpha gates: A1 (serving endpoint, owner spend/CF), A2 (corpus portability ? next concrete PR per \$8), A10/A11 (ship-gate target + WASB-C3, owner/legal). Known real bug tracked as A6: `backend.eval.ablation`?`ablation\_pdf` circular import (fix = defer `export\_pdf` import into `main()`).

76
77 ---

78
79 The 2026-07-02 health audit remediation is **COMPLETE + ARCHIVED**. After the sibling session drove P0?P28 to done, a follow-up session (2026-07-02 PM) **independently re-verified every P0?P2 fix in `origin/master` code at file:line** (all CONFIRMED, CI green on all 5 heads), then closed the last confirmed-but-untracked residuals and archived the umbrella doc. **No open remediation PRs; all 5 repos have 0 non-default branches, 0 orphaned worktrees, 0 dirty files** (full workflow/worktree/branch cleanup done).

80
81 Archived tracker (was live, now archived):

82
83 `badminton-highlight-indexer\docs\archives\past\_projects\AUDIT\_REPORT\_khelsutra-guru\_2026-07-02-COMPLETED-2026-07-02.md`

84
85 Final repo heads:

86
87 | Repo | Head |
88 |---|---|
89 | `badminton-highlight-indexer` | `ac6184b` |

```

90 | `khelsutra` | `1639804` |
91 | `rally-annotator` | `ef9e088` |
92 | `sports-data-collector` | `bfc014f` |
93 | `sports-obsreport` | `a0c7252` |
94
95 ## Follow-up session residuals (2026-07-02 PM) ? all merged on green
96
97 - `sports-data-collector#61` ? **R1-41**: crawler skips the whole match when any per-set
CSV fails (was silently ingesting partial rally data) + regression test. Suite 294?, cov 81.33%.
98 - `#441` ? **R1-24/R1-30**: `git rm --cached` the 7 tracked `scratch/*.py` (they defied
the `scratch/` ignore-contract; secret-leak vector). Files kept on disk, now ignored.
99 - `#442` ? **R1-11/R1-26**: pin `ruff==0.15.16` / `mypy==2.1.0` in ci + nightly
(verified green at those versions ? zero churn).
100 - `#446` ? archived the umbrella audit doc + repointed all inbound refs
(README/DOC_STATUS/NEXT_STEPS/test). ? gotcha: first commit only carried the `git mv` because
`git add` aborted on the already-moved old path ? a second commit landed the content edits;
link-check green after.
101 - Filed remaining owner-gated/future items as GitHub issues: **443** (CORS default
`['']`, R1-12) . **444** (corpus Phase-2 upload decision) . **445** (GCS eval/litmus consume
`gs://` by URI). Already-open: #301 (retention automation), #327 (path-jail gs://), #332 (NSFW),
#384 (CI single-source).
102
103 ## Latest Completed Items
104
105 - `#438`: Roora pilot non-video GCS archive complete; raw videos remain excluded.
106 - `sports-data-collector#56`: Roora pilot golden-label fixtures committed with tests.
107 - `sports-data-collector#60`: six owned golden source MP4 MD5/path manifest committed;
uploads remain owner-gated.
108 - `khelsutra#150`: retired-droplet docs refreshed from current master; stale `#141`
closed superseded.
109 - `#439` and `#440`: tracker/docs final reconciliation complete and merged.
110 - Closed superseded stale PRs: `sports-data-collector#54`, `sports-data-collector#55`,
`khelsutra#141`.
111 - Root non-git audit snapshot replaced with a pointer to the live engine tracker.
112
113 ## Verification
114
115 - Engine `#439` and `#440` CI green, including full offline suite with coverage floor,
ruff, mypy, wheel smoke, markdown links, leak/license checks, and cross-OS golden fixtures.
116 - Engine docs local checks: `python -m tools.check_md_links`; `git diff --check`.
117 - Sports-data-collector `#56`: merge-result local full suite `293 passed, 2 skipped`;
coverage `82.02%` against floor `80`; CI green.
118 - Sports-data-collector `#60`: local full suite `290 passed, 2 skipped`; coverage
`82.02%` against floor `80`; CI green.
119 - Khelsutra `#150`: local `site` tests `95 passed`; web typecheck; web coverage gate;
web e2e/contract smoke `5 passed`; CI green.
120 - Prior UI rows include screencast/Playwright evidence under
`C:\Users\avidu\Projects\khelsutra-guru\khelsutra-screencast\_raw`.
121
122 ## Leftovers
123
124 | Area | Leftover | Status |
125 |---|---|---|
126 | Corpus videos | Decide/upload large source videos: full 4K originals vs proxies, and
wider archive scope. | Owner-gated ? **issue #444**. |
127 | GCS eval/litmus | Make eval/litmus consume `gs://` data directly. | Future project ?
**issue #445**. |
128 | CORS default | `allow_origins=['']` lets any site script the localhost appliance. |
Security ? **issue #443**. |
129 | Hosted deployment | Enable upload retention in live infra and clean up
Cloudflare/alpha. | Owner infra/timing ? **301** + CF issues. |
130 | Product backlog | Non-audit feature/quality issues. | Normal backlog. |
131
132 ## Resume Rule
133

```

134 Before any future work: `git pull --ff-only` in the target repo, branch from
`origin/<base>`, re-check branch/HEAD before staging, and stage explicit paths only.
135

10. user (2026-07-04T08:41:06.112Z)

Already up to date.
4847232 docs(runlog): A1 alpha serving deploy ? hosted gs:// CUJ report (#464)
5832acb deploy: add cloud-run extra so the reused worker image can dispatch as control plane
(#462)
a5d6246 serving: re-queue interrupted compile jobs on restart (#329 reliability) (#460)
master...origin/master

11. user (2026-07-04T08:41:07.716Z)

? Logged in to github.com account avidullu (keyring)
- Active account: true

12. assistant (2026-07-04T08:41:13.922Z)

Repo is current at `4847232` (matches the handoff) and gh is authed as `avidullu`. Let me set up
task tracking and fetch the live state of all open PRs and their review threads.

13. user (2026-07-04T08:41:18.027Z)

headRefName	mergeable number		
reviewDecision title			
-----		-----	-----

fix/highlights-signed-url	MERGEABLE	469	fix(cloud-serving): serve large reels
via a gs:// signed UR?			
fix/drain-concurrency	MERGEABLE	468	fix(cloud-serving): parallelize the
drain under cloud_run (?)			
fix/upload-gcs-staging	MERGEABLE	467	fix(cloud-serving): stage browser-
uploaded local files to g?			
fix/allowed-video-root-uri	MERGEABLE	465	fix(hosted): allowed_video_root jail
must not reject storag?			

14. user (2026-07-04T08:41:26.456Z)

Task #1 created successfully: Fix #468 second blocker: claim_due_job false-empty on lost CAS
race

15. user (2026-07-04T08:41:28.738Z)

Task #2 created successfully: Sweep all 4 open PRs (#465/#467/#468/#469) for new review comments
and address them

16. user (2026-07-04T08:41:36.330Z)

Task #3 created successfully: Run full e2e deploy test after all 4 PRs merge

17. user (2026-07-04T08:41:39.357Z)

Task #4 created successfully: Capture rough edges: fix small ones, file GitHub issues for the
rest

18. user (2026-07-04T08:41:45.696Z)

Updated task #2 status

19. user (2026-07-04T08:41:53.350Z)

```
{"headRef0id":"cd4bc7382a631d1e2a41594e78f33883556844a0","mergeable":"MERGEABLE","number":465,"reviewDecision":null,"state":"OPEN","threads":[]}  
{ "headRef0id":"d3a9629c9e1894304a4f97ccd631dd5221fc54fc","mergeable":"MERGEABLE","number":467,"reviewDecision":null,"state":"OPEN","threads":[]}  
{ "headRef0id":"8534077b79794ecef7c4f6559a708c5bfe382498","mergeable":"MERGEABLE","number":468,"reviewDecision":null,"state":"OPEN","threads":[]}  
{ "headRef0id":"d3b0e64b467ccc243545ddccee9e73e316a5e732","mergeable":"MERGEABLE","number":469,"reviewDecision":null,"state":"OPEN","threads":[]}
```

20. user (2026-07-04T08:42:06.606Z)

===== PR #465 =====

```
{"comments":[{"author":"avidullu","body":"Blocking: the full offline suite is currently red on this PR, and the failure is tied to the behavior change here rather than CI  
infra.\n\n`tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri` still asserts that `gs://`, `s3://`, and `gdrive://` inputs return 400 when `allowed_video_root` is set. This PR changes `validate_input_path()` so storage URIs skip the local filesystem jail and now return 202, so the suite fails with:\n\n`AssertionError: gs://b/x.mp4 -\u003e 202; assert 202 == 400`\n\nPlease reconcile that existing hosted-mode contract test with the new security boundary be", "createdAt":"2026-07-04T07:50:34Z"}],{"author":"avidullu","body":"Thanks ? agreed on the security boundary, and I took the `only trusted/configured buckets` route you outlined. Fixed in cd4bc73:\n\n- `validate_input_path` now gains `allowed_uri_root` (wired from `storage.input_root` at `/api/process`). In hosted mode a bucket URI is allowed only when it's under that configured input_root; an arbitrary bucket ? or any URI when no `input_root` is set ? is rejected (with a separator-boundary check so `gs://b/videos-evil` isn't treated as under `gs://b/videos`). Local paths keep the realpath jail.\n- So `test_process_hosted_mode_rejects_nonlocal_uri` passes  
**u", "createdAt":"2026-07-04T07:59:13Z"}], "reviews":[]}
```

===== PR #467 =====

```
{"comments":[{"author":"avidullu","body":"Potential blocker for non-default storage configs: the new staging upload drops the active storage settings.\n\nIn `backend/orchestration.py`, the local-upload path builds `staged_uri` from `storage.input_root` but calls:\n\n`storage.put(local_src, staged_uri)`\n\nwithout passing the current storage config. Other storage operations in the repo pass the model dump into `storage.fetch/put/list_uris` so backend-specific settings survive. Without that here, staging ignores configured settings such as `storage.s3.endpoint_url` / region / addressing style for R2, DO Spaces, or MinIO, and also any GCS proj", "createdAt":"2026-07-04T07:50:45Z"}],{"author":"avidullu","body":"Good catch ? fixed. The staging call now threads the active storage settings:\n\n`storage.put(local_src, staged_uri, config=storage_cfg.model_dump() if storage_cfg is not None else None)`\n\nso `get_storage` builds the configured backend (S3 `endpoint_url`/region/addressing for R2·DO·MinIO; GCS project/client) rather than a fresh default. Extended the staging test to assert the config dict (`input_root`/`input_backend`) actually reaches `put()`. Full offline suite green (1628 passed).", "createdAt":"2026-07-04T08:03:09Z"}],{"author":"avidullu","body":"I found one more config-shape blocker in the staging gate.\n\nThe new local-upload staging path decides whether staging is possible with:\n\n`python\ninput_root = storage_cfg.input_root\ninput_backend = storage_cfg.input_backend or \"local\"\n\nif not input_root or input_backend == \"local\":\n    return _failed(...)\n\nThat rejects a valid config where `storage.input_root` is itself an explicit cloud URI and `input_backend` is left at its default. The storage layer already treats an explicit root like `gs://in-bucket/videos` as backend-bearing (`resolve_input_uri` joins under `input_root` before consul", "createdAt":"2026-07-04T08:37:47Z"}], "reviews":[]}
```

===== PR #468 =====

```
{"comments":[{"author":"avidullu","body":"Withdrawing my earlier LGTM pending a logic clarification/fix.\n\nOn re-review, this changes the global job-drain executor size based only on `deployment.compute_target == \"cloud_run\"`, but the jobs flowing through that executor are not all remote Cloud Run detect jobs:\n\n- On a `cloud_run` server, compile jobs (`kind='compile'`) and explicit `compute='local'` process jobs will now also run concurrently, even though the stated rationale for raising the worker count is that remote Cloud Run dispatch/poll is I/O-bound and does not consume the local GPU/Gemini path. That weakens the existing seria", "createdAt":"2026-07-04T08:08:11Z"}],{"author":"avidullu","body":"Thanks ? you're right, the global-executor-size approach was too broad. Reworked to gate on the effective per-job compute path, not the server default:\n\n1. Local work no longer parallelizes. Added `orchestration._LOCAL_COMPUTE_LANE` (a `BoundedSemaphore(1)`) held around the two CPU/GPU-heavy in-process paths ? the in-process segmenter (`process_video`) and the ffmpeg compile stitch. A remote Cloud Run dispatch is a bucket-poll and never holds the lane, so N remote jobs
```

```

overlap while compile + `compute='local'` jobs stay 1-at-a-time (single-box GPU + rate-fragile
Gemini invariant pr", "createdAt": "2026-07-04T08:26:20Z"}, {"author": "avidullu", "body": "I found
one more blocker before I can re-LGTM.\n\n`claim_due_job()` is CAS-safe for avoiding double-
claims, but under parallel drain workers the losers return `None`
(`backend/infrastructure/database_jobs.py:357-380`). `_drain_due_jobs()` then treats `None` as
`"queue empty"` and exits (`backend/api/job_worker.py:273-276`). With `max_workers \u003e 1`,
multiple workers can select the same first queued row; one updates it, and the losers stop even
though later jobs are still queued.\n\nI reproduced locally on this head with 8 queued jobs and
8 concurrent claimers:\n\n```\n[None, None, None, None, 'j0',
N", "createdAt": "2026-07-04T08:32:24Z"}, {"author": "avidullu", "body": "Good find, and thanks for
the concrete repro ? that's a real false-empty: a CAS-loser returning `None` is
indistinguishable from an empty queue, so parallel drains bail early. This work is being handed
to a continuing session; the exact next action captured for it is: make `claim_due_job()` retry
on a lost race (re-SELECT + re-attempt until it claims a row or a fresh SELECT is genuinely
empty ? or return a distinct lost-race sentinel so `_drain_due_jobs()` loops), plus a regression
test with N queued jobs + N concurrent claimers proving N distinct claims and no false-empty.
Will fix + re-reqe", "createdAt": "2026-07-04T08:34:58Z"}], "reviews": []}
===== PR #469 =====
{"comments": [{"author": "avidullu", "body": "Potential blocker for configured object stores: the
new mirrored-reel path does not pass the active storage config to the storage layer.\n\nTwo new
call sites use only the URI:\n\n- `backend/orchestration.py`: `storage.put(out_path,
reel_uri)`\n- `backend/main.py`: `storage.exists(reel_uri)` and `storage.signed_url(reel_uri,
method='GET')`\n\nThat bypasses backend-specific settings from `cfg.storage`, unlike the
existing fetch/put/list paths that pass a storage settings dict. For S3-compatible stores this
drops `endpoint_url`, region, and addressing style, so R2/D0 Spaces/MinIO signed URLs can be
gen", "createdAt": "2026-07-04T07:50:56Z"}, {"author": "avidullu", "body": "Fixed. Added
`orchestration._storage_settings(cfg)` (`model_dump()`) and threaded it into all three calls ?
the mirror `storage.put(...)` and `storage.exists(...)` + `storage.signed_url(...)` ? so the
configured backend (S3 `endpoint_url`/region; GCS project/client/`signed_url_ttl`) is used
instead of a fresh default. Also extracted the exists+sign logic into
`orchestration.signed_reel_url(...)` so `backend/main.py` stays under its line budget (P23/P24)
rather than growing the god-module. Tests now assert the config dict reaches
`exists()`/`signed_url()`. Full offline suite green (1629
passed", "createdAt": "2026-07-04T08:12:52Z"}, {"author": "avidullu", "body": "One more issue: the new
`storage.signed_url()` helper currently ignores the backend's configured
`signed_url_ttl`. \n\nEven though the active storage config is now threaded into
`signed_reel_url()`, the helper signature defaults `ttl=3600` and always passes that value into
the backend:\n\n```\npython\ndef signed_url(..., ttl: int = 3600, ...):\n    return
get_storage(...).signed_url(ref, ttl=ttl, method=method)\n```\n\nBoth `GCSStorage` and
`S3Storage` only fall back to their configured `self._ttl` when `ttl` is falsey (`ttl` or
`self._ttl`), so the configured TTL never gets a chance to apply for
`/api/high", "createdAt": "2026-07-04T08:37:47Z"}], "reviews": []}

```

21. user (2026-07-04T08:42:26.976Z)

I found one more config-shape blocker in the staging gate.

The new local-upload staging path decides whether staging is possible with:

```

```python
input_root = storage_cfg.input_root
input_backend = storage_cfg.input_backend or "local"
if not input_root or input_backend == "local":
 return _failed(...)
```

```

That rejects a valid config where `storage.input\_root` is itself an explicit cloud URI and `input\_backend` is left at its default. The storage layer already treats an explicit root like `gs://in-bucket/videos` as backend-bearing (`resolve\_input\_uri` joins under `input\_root` before consulting `input\_backend`), and the model docs call out `input\_root="gs://bucket/videos" + "x.mp4" => "gs://bucket/videos/x.mp4"`.

I reproduced on this head with:

```

```json

```

```
{
 "deployment": {"compute_target": "cloud_run"},
 "storage": {
 "backend": "gcs",
 "input_root": "gs://in-bucket/videos",
 "output_root": "gs://out-bucket/results"
 }
}
...
```

A local upload still fails pre-dispatch with "needs a cloud input ... or a cloud storage.input\_root" because `input_backend` remains the default `local`, even though `input_root` is already a `gs://` destination.

Suggested fix: classify the configured `input_root` itself (e.g. `storage.parse_uri(input_root).backend` in `{"gcs", "s3", "gdrive-api"}` / at least not `local`) instead of relying on `input_backend`, and add a regression test where `input_root="gs://..."` is set without `input_backend="gcs"`.

===== 469 =====

One more issue: the new `storage.signed_url()` helper currently ignores the backend's configured `signed_url_ttl`.

Even though the active storage config is now threaded into `signed_reel_url()`, the helper signature defaults `ttl=3600` and always passes that value into the backend:

```
```python
def signed_url(..., ttl: int = 3600, ...):
    return get_storage(...).signed_url(ref, ttl=ttl, method=method)
```
```

Both `GCSStorage` and `S3Storage` only fall back to their configured `self._ttl` when `ttl` is falsey (`ttl` or `self._ttl`), so the configured TTL never gets a chance to apply for `/api/highlights`.

I reproduced locally on this head with an injected GCS client and `config={"gcs": {"signed_url_ttl": 123}}`; calling `storage.signed_url("gs://bucket/highlights/vid/reel.mp4", config=..., method="GET")` passed an expiration of `3600` seconds, not `123`.

Suggested fix: make the helper's `ttl` optional/`None` and only pass an explicit `ttl` when the caller provides one, or pass `ttl=0/None` through so the backend uses its configured `signed_url_ttl`. Please add a small regression test that `/api/highlights` / `signed_reel_url()` honors a non-default `storage.gcs.signed_url_ttl` (and ideally S3 too, since it has the same pattern).

===== 468 (blocker comment) =====

I found one more blocker before I can re-LGTM.

`claim_due_job()` is CAS-safe for avoiding double-claims, but under parallel drain workers the losers return `None` (`backend/infrastructure/database_jobs.py:357-380`). `_drain_due_jobs()` then treats `None` as "queue empty" and exits (`backend/api/job_worker.py:273-276`). With `max_workers > 1`, multiple workers can select the same first queued row; one updates it, and the losers stop even though later jobs are still queued.

I reproduced locally on this head with 8 queued jobs and 8 concurrent claimers:

```
```text
[None, None, None, None, 'j0', None, None, 'j1'] -> only 2/8 claims
```
```

For long Cloud Run bucket-poll jobs this can leave dispatch capacity unused and queued jobs waiting until another submit/wake or a running worker finally frees up, which undermines the concurrency improvement this PR is trying to provide.

Suggested fix: make `claim\_due\_job()` retry on `rowcount != 1` until it either successfully claims a row or a fresh SELECT finds no runnable rows, or return a distinct lost-race sentinel so `\_drain\_due\_jobs()` loops instead of treating it as empty. Please add a regression test with concurrent claimers/drains over N queued jobs proving there is no false-empty result / that N distinct jobs can be claimed.

## 22. user (2026-07-04T08:43:07.468Z)

```
__pycache__
.claude
.github
.mypy_cache
.pytest_cache
.ruff_cache
.vscode
artifacts
backend
deploy
docs
eval_baselines
output
scratch
scripts
tests
tools
training
--- venv? ---
False
--- which python for repo ---
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer
4847232 [master]
C:/Users/avidu/AppData/Local/Temp/bhi-review-b42cb9b03635404488ddb2da340a7e2a/pr465
cd4bc73 (detached HEAD)
C:/Users/avidu/AppData/Local/Temp/bhi-review-b42cb9b03635404488ddb2da340a7e2a/pr467
d3a9629 (detached HEAD)
C:/Users/avidu/AppData/Local/Temp/bhi-review-b42cb9b03635404488ddb2da340a7e2a/pr469
d3b0e64 (detached HEAD)
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/cfg-wt 20b2d5a [claude/wasb-
config-validation]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/compile-wt b09a1d4 [claude/compile-
reliability]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-pre-wt 355dad6 (detached HEAD)
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-wt 5de1ef2 (detached HEAD)
C:/Users/avidu/Projects/khelsutra-guru/wt-465fix
cd4bc73 [fix/allowed-video-root-uri]
C:/Users/avidu/Projects/khelsutra-guru/wt-drain-concurrency
8534077 [fix/drain-concurrency]
C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download
d3b0e64 [fix/highlights-signed-url]
C:/Users/avidu/Projects/khelsutra-guru/wt-upload-staging
d3a9629 [fix/upload-gcs-staging]
```

## 23. assistant (2026-07-04T08:43:17.102Z)

Prior session's worktrees for all four PR branches already exist at the right heads ? I'll reuse them. Let me verify the Python environment and that the worktrees are clean.

## 24. user (2026-07-04T08:43:20.741Z)

Python 3.13.13

```

pytest 9.0.3
--- wt-drain-concurrency ---
--- wt-upload-staging ---
--- wt-signed-download ---
--- wt-465fix ---

```

## 25. user (2026-07-04T08:43:34.287Z)

```

=== claim_due_job (database_jobs.py) ===
 (f"+{max(0, int(delay_sec))} seconds", progress, job_id),
 f"Failed to set job {job_id} waiting",
)

def seconds_until_next_due(self) -> Optional[float]:
 """Seconds until the soonest parked ('waiting') job becomes due, so the worker can
 schedule a wake-up (EXECUTION_POLICY.md P3 self-scheduling). Returns 0.0 if one is
 already due, or None if nothing is waiting. A NULL ``next_poll_at`` ? due now (0.0)."""
 try:
 with self._connect() as conn:
 row = (
 conn.cursor()
 .execute(
 "SELECT MIN(CASE WHEN next_poll_at IS NULL THEN 0 ELSE "
 " (julianday(next_poll_at) - julianday('now')) * 86400.0 END) AS secs "
 "FROM jobs WHERE status='waiting'"
)
 .fetchone()
)
 if row is None or row["secs"] is None:
 return None
 return max(0.0, float(row["secs"]))
 except Exception as e:
 logger.error(f"Failed to compute next-due delay: {e}")
 return None

def claim_due_job(self) -> Optional[Dict[str, Any]]:
 """Atomically claim ONE runnable job ? 'queued', or 'waiting' whose ``next_poll_at`` is
 due ?
 flipping it to 'running'. Compare-and-set on (id, status) so concurrent workers never
 double-claim (the DB row-claim is the only coordination needed; no master). Earliest-due
 first. Returns the claimed job dict, or None if nothing is runnable."""
 try:
 with self._connect() as conn:
 cur = conn.cursor()
 row = cur.execute(
 "SELECT id, status FROM jobs WHERE status='queued' "
 "OR (status='waiting' AND (next_poll_at IS NULL OR next_poll_at <=
CURRENT_TIMESTAMP)) "
 "ORDER BY COALESCE(next_poll_at, created_at) ASC LIMIT 1"
).fetchone()
 if not row:
 return None
 cur.execute(
 "UPDATE jobs SET status='running', started_at=CURRENT_TIMESTAMP "
 "WHERE id=? AND status=?",
 (row["id"], row["status"]),
)
 conn.commit()
 if cur.rowcount != 1:
 return None # lost the race to a concurrent worker
 return self.get_job(row["id"])
 except Exception as e:
 logger.error(f"Failed to claim a due job: {e}")
 return None

```

```

def fail_stale_jobs(self, reason: str = "interrupted by server restart") -> int:
 """Crash recovery: any job still queued/running from a previous process is dead
 (the executor is in-process). Mark failed so clients see a terminal state, not a
 hang. Returns the number of jobs swept.

 NOTE: ``recover_stale_jobs`` is the preferred startup call ? it RE-QUEUES stale
 ``compile`` jobs (idempotent CPU stitch, safe to retry) instead of failing them. This
 blanket-fail method stays for callers/tests that want the old behaviour."""
 try:
 with self._connect() as conn:
 cur = conn.cursor()
 cur.execute(
 "UPDATE jobs SET status='failed', error=?, finished_at=CURRENT_TIMESTAMP "
 "WHERE status IN ('queued', 'running')",
 (reason,),
)

```

## 26. user (2026-07-04T08:43:35.446Z)

```

=== _drain_due_jobs (job_worker.py) ===
 """M11 data-flywheel (COMPUTE_DECOUPLED_SERVING M11, D12). DEFAULT-OFF + INERT: a no-op
 unless
 ``config.flywheel.enabled`` AND attested consent ? and consent is a **faked hard-deny**
 today
 (``flywheel.consent_attested`` returns False for every job until counsel + the consent
 schema
 land), so this captures **nothing** in production. When activated it captures the consented
 job's
 artifacts into the per-video workspace (? later labeling ? the golden training set).

 Best-effort ? never fails a completed job because the flywheel hiccuped. ``job`` is the
 claimed
 row (carries ``owner_id``); ``result`` is the pipeline output."""

 flywheel_cfg = getattr(config, "flywheel", None)
 if not flywheel_cfg or not getattr(flywheel_cfg, "enabled", False):
 return
 try:
 from backend import flywheel

 flywheel.run_for_job(job, result, config)
 except Exception as e: # never wedge a completed job on the flywheel
 logger.warning("Flywheel for job %s failed (non-fatal): %s", job.get("id"), e)

def _drain_due_jobs(config: Any, db: Any) -> int:
 """One worker tick: claim and run every currently-runnable job (queued, or waiting whose
 `next_poll_at` is due), earliest-due first, until the table has nothing runnable.

 Each parked job (`JobWaiting` ? status 'waiting') drops out of this drain and is picked
 up by a FUTURE drain once due; we schedule that follow-up drain via `_schedule_drain` so
 a single-worker box keeps making progress with no central timer (the DB is the clock).
 Returns the number of jobs that reached a terminal/parked state this tick.
 """
 import backend.main as _main

 ran = 0
 while True:
 job = db.claim_due_job()
 if job is None:
 break
 ran += 1
 _main._run_claimed_job(job, config, db)
 # If anything is parked-but-not-yet-due, arrange a wake-up so it resumes without a new
 # submit. On the single-box executor this is a delayed re-drain; multi-worker just relies

```

```

on the next submit/drain. Best-effort ? never raises into the worker.
_main._schedule_next_drain_if_waiting(config, db)
return ran

def _schedule_next_drain_if_waiting(config: Any, db: Any) -> None:
 """If any job is parked in 'waiting', arm a re-drain for when the soonest one is due, so a
 self-scheduled job resumes with no further client submit. The drain itself re-checks
 due-ness, so an early wake is harmless and a clamped wake for a far-future job is fine."""
 import backend.main as _main

 try:
 delay = db.seconds_until_next_due()
 except Exception as e: # pragma: no cover - defensive; never wedge the worker
 logger.error(f"Could not compute next-due delay: {e}")
 return
 if delay is None:
 return # nothing waiting
 # Clamp so a far-future job doesn't pin a thread forever; the drain re-checks due-ness.
 _main._arm_wake_timer(max(0.0, min(delay, _main._MAX_DRAIN_WAKE_SEC)), config, db)

def _arm_wake_timer(delay_sec: float, config: Any, db: Any) -> None:
 """Arm a single daemon timer that re-submits a drain after ``delay_sec`` (no central
 scheduler ? the DB is the clock). Isolated as the sole timer-creating seam so tests can
 monkeypatch it to a no-op and assert the *scheduling decision* without real waiting."""
 import backend.main as _main

 timer = threading.Timer(
 delay_sec,

```

## 27. assistant (2026-07-04T08:43:43.707Z)

Now the #467 staging gate and #469 signed\_url code.

## 28. user (2026-07-04T08:43:48.547Z)

```

=== staging gate context (orchestration.py) ===
189: def _failed(msg: str, ran: bool) -> Tuple[Dict[str, Any], bool]:
228: return _failed(
247: return _failed(
253: # possible when a CLOUD input_root is configured; without one we genuinely can't stage ?
fail.
254: input_root = (getattr(storage_cfg, "input_root", "") or "").strip()
255: input_backend = (
256: getattr(storage_cfg, "input_backend", "local") or "local"
258: if not input_root or input_backend == "local":
259: return _failed(
260: "cloud-serving needs a cloud input (e.g. gs://?), or a cloud storage.input_root to "
266: staged_uri = f"{input_root.rstrip('/')}/{video_id}/{os.path.basename(local_src)}"
273: staged_uri,
282: staged_uri,
286: return _failed(
287: f"cloud-serving: failed to stage the uploaded file to {input_root}: {e}",
293: req.video_path = staged_uri
294: input_ref = storage.resolve_input_ref(staged_uri, storage_cfg)
296: video_id, staged_uri, "processed", [r.model_dump() for r in val_results]
300: return _failed(
332: return _failed(f"cloud dispatch error: {e}", True)
334: return _failed(f"cloud job failed (returncode {result.returncode}).", True)
355: return _failed(

```

## 29. user (2026-07-04T08:43:51.398Z)

```

=== storage __init__ signed_url helper ===
184: def put(
199: def exists(uri: str, *, config: Optional[Dict[str, Any]] = None) -> bool:
205: def signed_url(
209: ttl: int = 3600,
216: return get_storage(ref.backend, config=config).signed_url(ref, ttl=ttl, method=method)
=== orchestration signed_reel_url ===
695: def _storage_settings(cfg) -> Optional[Dict[str, Any]]:
697: GCS project/client/signed_url_ttl), or None. Threaded into storage calls so a non-default
backend
703: def signed_reel_url(cfg, video_id: str, name: str) -> Optional[str]:
711: settings = _storage_settings(cfg)
713: return storage.signed_url(reel_uri, config=settings, method="GET")
788: storage.put(out_path, reel_uri, config=_storage_settings(cfg))

```

### 30. user (2026-07-04T08:44:00.816Z)

```

 input_ref = storage.resolve_input_ref(req.video_path, storage_cfg)
except ValueError as e:
 return _failed(
 f"cloud-serving: unresolvable input '{req.video_path}': {e}", False
)
if input_ref.backend == "local":
 # A browser upload lands on THIS box's local disk; the Cloud Run job (a different
machine)
 # can't read it. Stage it UP to the input bucket first, then dispatch from there (#461).
Only
 # possible when a CLOUD input_root is configured; without one we genuinely can't stage ?
fail.
 input_root = (getattr(storage_cfg, "input_root", "") or "").strip()
 input_backend = (
 getattr(storage_cfg, "input_backend", "local") or "local"
).strip()
 if not input_root or input_backend == "local":
 return _failed(
 "cloud-serving needs a cloud input (e.g. gs://?), or a cloud storage.input_root
to "
 "stage local uploads into; a path local to this machine can't be read by the
Cloud "
 "Run job.",
 False,
)
 local_src = input_ref.key # resolved local path of the upload
 staged_uri = f"{input_root.rstrip('/')}/{video_id}/{os.path.basename(local_src)}"
 try:
 notify("staging upload to bucket")
 # Thread the active storage settings so backend-specific config survives (S3
endpoint_url/
 # region/addressing for R2·D0·MinIO; GCS project/client) ? not a fresh default
client.
 storage.put(
 local_src,
 staged_uri,
 config=(
 storage_cfg.model_dump() if storage_cfg is not None else None
),
)
 except Exception as e: # noqa: BLE001 ? a staging failure is a clean job failure, not a
500
 logger.error(
 "[API] cloud-serving: failed to stage %s -> %s: %s",
 local_src,
 staged_uri,
 e,
 exc_info=True,

```

```

)
 return _failed(
 f"cloud-serving: failed to stage the uploaded file to {input_root}: {e}",
 False,
)
 # Repoint BOTH the dispatch and the DB video record at the staged bucket object: the L4
job # DETECTs from it, and a later /api/compile re-fetches the same source (the local upload
is # gone once this instance recycles). add_video is an UPSERT ? it only updates filepath.
 req.video_path = staged_uri
 input_ref = storage.resolve_input_ref(staged_uri, storage_cfg)
 db.add_video(
 video_id, staged_uri, "processed", [r.model_dump() for r in val_results]
)
 out_root = (getattr(storage_cfg, "output_root", "") or "").strip()
 if not out_root:
 return _failed(

```

### 31. user (2026-07-04T08:44:01.978Z)

```

def _backend_class(name: str):
 """Lazy-resolve a backend class so cloud SDKs aren't imported until needed."""
 if name == "local":
 from backend.storage.local import LocalStorage

 return LocalStorage
 if name == "s3":
 from backend.storage.s3 import S3Storage

 return S3Storage
 if name == "gcs":
 from backend.storage.gcs import GCSStorage

 return GCSStorage
 if name == "gdrive":
 from backend.storage.gdrive import GDriveStorage

 return GDriveStorage
 if name == "gdrive-api":
 from backend.storage.gdrive_api import GDriveApiStorage

 return GDriveApiStorage
 raise ValueError(
 f"unknown storage backend: {name!r} (expected local|s3|gcs|gdrive|gdrive-api)"
)

def get_storage(
 backend: Optional[str] = None,
 *,
 config: Optional[Dict[str, Any]] = None,
 **overrides,
) -> StorageBackend:
 """Construct a backend. ``backend`` (or ``config['backend']``, default 'local') selects
 the implementation; its per-backend settings come from ``config[backend]`` merged with
 ``overrides``. Settings carry NO secrets ? only endpoints/regions/paths/source-selectors."""
 cfg = config or {}
 name = backend or cfg.get("backend", "local")
 # Per-backend settings live under the backend name; tolerate the hyphen?underscore form so a
 # hyphenated backend ("gdrive-api") can read a pydantic field that can't contain a hyphen
 # ("gdrive_api"). Hyphen-free names are unaffected (the first lookup wins).
 settings: Dict[str, Any] = dict(
 cfg.get(name) or cfg.get(name.replace("-", "_")) or {}
)

```

```

settings.update(overrides)
return _backend_class(name)(**settings)

def fetch(
 uri: str,
 dst: "str | os.PathLike[str] | None" = None,
 *,
 config: Optional[Dict[str, Any]] = None,
 overwrite: bool = False,
) -> str:
 """Resolve ``uri`` to a local file path. For ``local`` this is an identity passthrough
 (no copy); for s3/gcs it downloads to ``dst``."""
 ref = parse_uri(uri)
 return get_storage(ref.backend, config=config).fetch_to_local(
 ref, dst, overwrite=overwrite
)

def put(
 local_path: "str | os.PathLike[str]",
 uri: str,
 *,
 config: Optional[Dict[str, Any]] = None,
 content_type: Optional[str] = None,
) -> str:
 """Upload/copy ``local_path`` to ``uri``; return ``uri``."""
 ref = parse_uri(uri)
 get_storage(ref.backend, config=config).put(
 local_path, ref, content_type=content_type
)
 return uri

def exists(uri: str, *, config: Optional[Dict[str, Any]] = None) -> bool:
 """True if the object at ``uri`` exists (a bucket object or a local file)."""
 ref = parse_uri(uri)
 return bool(get_storage(ref.backend, config=config).exists(ref))

def signed_url(
 uri: str,
 *,
 config: Optional[Dict[str, Any]] = None,
 ttl: int = 3600,
 method: str = "GET",
) -> str:
 """A time-limited signed URL for ``uri`` ? the client fetches bytes DIRECTLY from the store,
 off
 the app's response path (so a large object isn't streamed back through the server; #463).
 Backends
 without object signing (e.g. local) raise ``NotImplementedError``."""
 ref = parse_uri(uri)
 return get_storage(ref.backend, config=config).signed_url(ref, ttl=ttl, method=method)

def list_uris(
 prefix_uri: str, *, config: Optional[Dict[str, Any]] = None

```

## 32. user (2026-07-04T08:44:13.688Z)

```

=== parse_uri ===
 raw,
 input_backend=sc.get("input_backend", "local"),
 input_root=sc.get("input_root", ""),

```

```

)
return parse_uri(uri)

def input_is_local(raw: str, storage_config: Any = None) -> bool:
 """True if ``raw`` resolves (under the input config) to a LOCAL filesystem path ? i.e. a
 plain on-disk file the caller can stat/validate directly. False for a bucket / Drive URI
 that must be fetched first. Raises ``ValueError`` for an unsupported scheme (same as
 ``resolve_input_ref``)."""
 return resolve_input_ref(raw, storage_config).backend == "local"

def acquire_local_input(
 raw: str, storage_config: Any = None, *, scratch_parent: "str | None" = None
) -> str:
 """Resolve ``raw`` (a path or storage URI) to a LOCAL file path ready for processing.

 Local / bare paths are returned UNCHANGED (identity passthrough, no copy ? today's behaviour
 byte-for-byte). A bucket / Drive source (``gs://``/``s3://``/``gdrive://``, or a bare name
 under a
 non-local ``input_root``/``input_backend``) is downloaded into a fresh scratch dir and the
 local
 path returned. The caller owns successful remote scratch copies; use
 ``cleanup_acquired_local_input`` when the consumer has finished reading the returned
 path."""
 sc = _storage_dict(storage_config)
 ref = resolve_input_ref(raw, sc)
 if ref.backend == "local":
 return (
 ref.key
) # identity ? no temp, no copy, byte-identical to passing the path directly
 scratch = tempfile.mkdtemp(prefix="rally-input-", dir=scratch_parent)
 try:
 return fetch(ref.uri, os.path.join(scratch, ref.name or "in.mp4"), config=sc)
 except Exception:
 shutil.rmtree(scratch, ignore_errors=True)
 raise

def cleanup_acquired_local_input(
 raw: str, local_path: str, storage_config: Any = None
) -> None:
 """Delete a remote scratch dir produced by ``acquire_local_input``.

 Local inputs are identity paths and must never be removed. Remote acquisitions are
 downloaded
 into a fresh ``rally-input-*`` directory; deleting that directory is safe once the caller
 has
 finished hashing/validating/stitching/segmenting the returned file.
 """
 try:
 if input_is_local(raw, storage_config):
 return
 except ValueError:
 return
 if not local_path:
 return
 scratch = os.path.dirname(os.path.abspath(local_path))
 if os.path.basename(scratch).startswith("rally-input-") and os.path.isdir(scratch):
 shutil.rmtree(scratch, ignore_errors=True)

=== GCS signed_url ===
25: signed_url_ttl: int = 3600,
28: self._ttl = signed_url_ttl

```

```

76: def signed_url(self, ref, *, ttl=3600, method="GET") -> str:
78: exp = timedelta(seconds=ttl or self._ttl)
=== S3 signed_url ===
30: signed_url_ttl: int = 3600,
33: self._ttl = signed_url_ttl
90: def signed_url(self, ref, *, ttl=3600, method="GET") -> str:
95: ExpiresIn=ttl or self._ttl,

```

### 33. user (2026-07-04T08:45:32.307Z)

```

=== tests mentioning claim_due_job ===
C:\Users\avidu\Projects\khehsutra-guru\wt-drain-concurrency\tests\test_job_self_scheduling.py
C:\Users\avidu\Projects\khehsutra-guru\wt-drain-concurrency\tests\test_job_worker_loop.py
C:\Users\avidu\Projects\khehsutra-guru\wt-drain-concurrency\tests\test_server_entrypoint.py
=== tests mentioning staging / staged_uri (467 wt) ===
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\tests\test_api_cloud_dispatch.py
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\tests\test_freespace.py
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\tests\test_main_module_budget.py
=== tests mentioning signed_url (469 wt) ===
C:\Users\avidu\Projects\khehsutra-guru\wt-signed-download\tests\test_storage.py
C:\Users\avidu\Projects\khehsutra-guru\wt-signed-download\tests\test_upload_download.py
=== CI lint/type commands ===
ci.yml:16: # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
ci.yml:17: # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
ci.yml:21: name: Ruff lint
ci.yml:29: - name: Install ruff
ci.yml:30: # Pinned so an upstream ruff release can't redden every PR overnight; bump
deliberately.
ci.yml:31: run: python -m pip install ruff==0.15.16
ci.yml:32: - name: Ruff check
ci.yml:33: run: ruff check . --output-format github
ci.yml:66: # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
ci.yml:68: # up. Typed core deps are installed so mypy sees their real signatures
ci.yml:71: name: Mypy (lenient baseline)
ci.yml:79: - name: Install mypy + typed core deps

```

### 34. user (2026-07-04T08:45:38.883Z)

```

- name: Install mypy + typed core deps
 # mypy pinned alongside ruff so a type-checker release can't redden CI overnight.
 run: python -m pip install mypy==2.1.0 pydantic fastapi uvicorn python-dotenv google-
genai numpy
- name: Mypy
 run: mypy backend training

Fast guard against the 7fbb0 class of breakage: a SyntaxError or
module-level import error shipping while the suite stays green.
Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
on machines without the GPU stack (test_import_smoke stubs what's absent).
import-smoke:
 name: Syntax sweep + import smoke (no GPU stack)
 runs-on: ubuntu-latest
 timeout-minutes: 10
 steps:
 - uses: actions/checkout@v6
 - uses: actions/setup-python@v6
 with:
 python-version: "3.13"
 - name: Install minimal deps

```

### 35. user (2026-07-04T08:46:40.511Z)

Updated task #1 status

### 36. user (2026-07-04T08:47:59.385Z)

Workflow launched in background. Task ID: wmdmpc7il

Summary: Fix the 3 open review blockers on PRs #467/#468/#469 in their existing worktrees, then adversarially verify each fix

Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\subagents\workflows\wf\_8c732492-138

Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\workflows\scripts\fix-pr-review-blockers-wf\_8c732492-138.js

(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:

"C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\workflows\scripts\fix-pr-review-blockers-wf\_8c732492-138.js"} to iterate without resending the script.)

Run ID: wf\_8c732492-138

To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\workflows\scripts\fix-pr-review-blockers-wf\_8c732492-138.js", resumeFromRunId: "wf\_8c732492-138"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

### 37. user (2026-07-04T08:48:31.933Z)

Monitor started (task b8qxd3ys, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

### 38. user (2026-07-04T08:48:34.490Z)

#466: cloud-serving: control-plane drain serializes videos (max\_workers=1) ? parallel

/api/process queue behind each other

#463: cloud-serving: /api/highlights download 500s for reels >32 MiB (Cloud Run response limit)

#461: cloud-serving: browser-uploaded local files not staged to gs:// before L4 dispatch (upload CUJ fails)

#448: Detach post-upload processing into a durable background job UX

#447: Add API support to delete a video from the Studio library

#445: Make eval/litmus consumers fetch gs:// labels/trajectories/videos by URI (not local open())

#444: Corpus Phase-2: decide + execute the large source-video upload to GCS (owner-gated)

#443: [security] Default CORS allow\_origins=['\*'] lets any site script the localhost appliance (R1-12)

#384: ci: single-source the shared offline guard lanes + coverage floor via a workflow\_call reusable

#349: WASB native: decode is CPU-bound & single-threaded in streaming mode ? L4 GPU ~80% idle (no NVDEC/parallel decode)

#332: [safety] Content-safety pre-flight litmus: flag non-sports / NSFW video BEFORE sending it to Gemini

#331: [perf] Hosted processing is CPU-bound (ffmpeg/cv2) ? measured per-phase breakdown + levers (GPU/NVENC, parallel extraction, async compile)

#328: Surface the per-video runlog / execution trace to the user (endpoint + UI), not just server logs

#327: Allow gs:// / s3:// / gdrive-api:// ingest on an exposed (jailed) box ? scheme-exempt recognized storage backends from the path-jail

#326: Hosted-app observability: request tracing + replay for debugging production CUJ failures

#301: T11 follow-up: run the upload retention sweep as an automated scheduled job (hosted alpha)

#175: Shared cloud container (OneDrive) for GPU-extractor golden data ? sync GPU box ? dev sessions

#172: Calibrate the ship-gate F1@tIoU target (currently provisional ? NOT owner-ratified)

### 39. assistant (2026-07-04T08:48:40.145Z)

Monitor armed. Now the Dockerfile `auth` extra follow-up ? let me check the base image install

line on master.

40. user (2026-07-04T08:48:44.210Z)

**Cloud Run GPU **\*\*Job\*\*** image for the rally worker (COMPUTE\_DECOUPLED\_SERVING**

#

**Runs `python -m backend.worker infer ?` INLINE on an L4 (detect + window + stitch co  
CPU/wall + egress are metered into one job ? D4). Built + pushed + verified by the own  
M7 real run (the \$100/?10k budget, D17); CI does NOT build this (a CUDA image is too  
dispatch shim is proven with a fake client (tests/test\_compute\_backend.py).**

#

**Two python stacks, deliberately separate (a framework conflict otherwise ? same split  
native GPU box, deploy/digitalocean/gpu\_setup.sh):**

- \* the MAIN app env (this repo + ffmpeg) drives the pipeline + stitching;**
- \* a conda `wasb` env (py3.8 + torch 1.11.0+cu113) runs the WASB shuttle detector as  
(the native runner shells out to \$WASB\_PYTHON), so the app's torch never co-insta**

#

**? WASB-C3: the pretrained badminton weights are academic-data-trained ? provenanc  
commercial sharing. They are NOT baked into this image; stage them at runtime (a b  
mount) and point \$WASB\_WEIGHTS at them. Resolve WASB-C3 before any public, n**

FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04

```
ENV DEBIAN_FRONTEND=noninteractive \
 PYTHONUNBUFFERED=1 \
 PIP_NO_CACHE_DIR=1 \
 CONDA_DIR=/opt/conda \
 WASB_REPO_DIR=/opt/models/WASB-SBDT
```

**ffmpeg (stitch/decode) + DejaVu fonts (the watermark filter) + git/curl for the WASB en**

```
RUN apt-get update && apt-get install -y --no-install-recommends \
 ffmpeg fonts-dejavu-core git curl ca-certificates build-essential \
 python3 python3-pip python3-venv \
&& rm -rf /var/lib/apt/lists/*
```

**--- WASB detector env (micromamba + conda-forge ? BSD-licensed, NO Anaconda To  
Enforces ADR-005 (every dependency ? code, data, weights, AND the build toolchain  
MIT/Apache/BSD): micromamba (BSD-3) + conda-forge (community) replaces Minicon  
`defaults` channels (the ?200-employee Terms-of-Service trap flagged in PACKAGING  
MAMBA\_ROOT\_PREFIX=\$CONDA\_DIR (=opt/conda) keeps the env at \$CONDA\_DIR/e  
is UNCHANGED. Same verified torch-1.11.0+cu113 WASB stack; the pip wheels are un**

```
ENV MAMBA_ROOT_PREFIX=/opt/conda
```

```
RUN curl -sSL https://micro.mamba.pm/api/micromamba/linux-64/latest \
 | tar -xj -C /usr/local bin/micromamba \
&& micromamba create -y -q -n wasb -c conda-forge python=3.8 \
&& git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO_DIR" \
&& "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
 torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url
https://download.pytorch.org/whl/cu113 \
&& ([-f "$WASB_REPO_DIR/requirements.txt"] && "$CONDA_DIR/envs/wasb/bin/python" -m pip
```

```
install -q -r "$WASB_REPO_DIR/requirements.txt" || true) \
 && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
 hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml matplotlib
"numpy<1.24"
```

### --- Main application env ---

```
WORKDIR /app
COPY pyproject.toml README.md ./
COPY requirements-trackers.txt ./
COPY backend ./backend
COPY training ./training
```

**Install with the [storage-gcs,cloud-run] extras. storage-gcs: the cloud worker PULLS the input and PUSHES the gs:// outputs/done-marker through backend.storage (google-cloud-storage lazy-imported there); without it the job dies at the first gs:// fetch with `ModuleNotFoundError: google.cloud`. cloud-run: this SAME image is reused as the deployed CONTROL PLANE for the L4 worker Job via google.cloud.run\_v2 (backend/compute/cloud\_run.py, lazy-imported there); without it the control plane fails every dispatch with "google-cloud-run is required for CloudRunClient client". Both are real Cloud Run failures the mock-client tests can't see (the SDKs are not in the D13/P12: `trackers` (ByteTrack's new home) is a git-only dep (broken PyPI wheel ? see [roboflow/trackers#474](https://github.com/roboflow/trackers/issues/474)) so it's NOT pulled by `pip install .`; install it out-of-band before the main env, or the person-cue path crashes with ModuleNotFoundError. The immutable env is single-sourced in requirements-trackers.txt.**

```
RUN python3 -m pip install --upgrade pip \
 && python3 -m pip install -r requirements-trackers.txt \
 && python3 -m pip install ".[storage-gcs,cloud-run]"
```

**The native runner finds the WASB env + repo by env (no `conda activate` needed); we use the native runtime (WASB-C3). detector\_impl=native = the Linux-native WASB runner; cuda stay disabled. PYTHONUNBUFFERED: without it the worker's stdout/stderr sit in the block buffer until the process exit, so Cloud Logging shows a single end-of-run burst instead of live progress (observed on the first real M7 runs, 2026-07-03) ? a running job looks stalled and mid-run failures log no output.**

```
ENV PYTHONUNBUFFERED=1 \
 WASB_PYTHON=/opt/conda/envs/wasb/bin/python \
 WASB_REPO_DIR=/opt/models/WASB-SBDT \
 WASB_WEIGHTS=/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar \
 RALLY_DETECTOR_IMPL=native
```

**The dispatcher (CloudRunCompute) supplies the per-execution args:**

**infer --video-uri gs://? --out-uri gs://? --done-uri gs://? --segmenter wasb\_hybrid --set**

```
ENTRYPOINT ["python3", "-m", "backend.worker"]
=== extras in pyproject ===
```

**Test + lint/type tooling ? matches the CI lanes (.github/workflows/ci.yml).**

```
dev = [
 "pytest",
 "pytest-cov",
 "httpx", # fastapi TestClient transport
 "ruff",
 "mypy",
 "PyJWT[crypto]>=2.8", # M9 edge-auth: Cf-Access JWT verify (RS256) ? tests + CI
]
```

**M9 edge-auth (Cloudflare Access). DEFAULT-OFF: only needed when security.edge\_auth\_jwt` is lazy-imported, so the base runtime works without this; install it for hosted/multi**

```
auth = ["PyJWT[crypto]>=2.8"]
```

**Per-video observability reports (SOFT dependency: the pipeline degrades to a no-op if absent). Pinned to the canonical private repo's immutable commit SHA; fetched over HTTPS. If you lack access, simply omit this group ? reporting tests skip via pytest.importorskip. For local dev against a sibling checkout:**

**pip install -e ../sports-obsreport --no-build-isolation**

```
reporting = [
 "obsreport @ git+https://github.com/Khelsutra/sports-
 obsreport.git@a0c7252c6e4159f8702360bea1cb46e5148b5c9f",
]
```

**`gpu` is DOCUMENTATION ONLY ? it is intentionally empty. torch/torchvision CUDA wheels require `--index-url`/`--extra-index-url`, which PEP 621 cannot express, so they are NOT declarable here. Install them out-of-band instead:**

**# CUDA 12.4 box:**

**pip install torch>=2.12.0 torchvision>=0.27.0 --index-url https://download.pytorch.org**

**# CPU-only box (also what CI uses):**

**pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu**

```
gpu = []
```

**Storage backends (docs/DECOUPLED\_COMPUTE/). The default `local` backend needs these. Cloud SDKs are imported lazily, so `import backend.storage` works without the install the extra for the bucket you use. (boto3's S3 client covers AWS + Cloudflare R2 DO Spaces + MinIO via endpoint\_url.) `gdrive://` needs NO extra ? it reads a locally MOUNTED Google Drive (Google Drive for Desktop) as plain files; see backend/storag**

```
storage-s3 = ["boto3>=1.34"]
storage-gcs = ["google-cloud-storage>=2.16"]
```

**41. user (2026-07-04T08:49:18.176Z)**

```
Preparing worktree (new branch 'fix/dockerfile-auth-extra')
branch 'fix/dockerfile-auth-extra' set up to track 'origin/master'.
HEAD is now at 4847232 docs(runlog): A1 alpha serving deploy ? hosted gs:// CUJ report (#464)
4847232 docs(runlog): A1 alpha serving deploy ? hosted gs:// CUJ report (#464)
fix/dockerfile-auth-extra
```

**42. user (2026-07-04T08:49:23.704Z)**

```
55 # Install with the [storage-gcs,cloud-run] extras. storage-gcs: the cloud worker PULLS
the gs://
56 # input and PUSHES the gs:// outputs/done-marker through backend.storage (google-cloud-
storage is
57 # lazy-imported there); without it the job dies at the first gs:// fetch with
`ModuleNotFoundError`:
58 # google.cloud`. cloud-run: this SAME image is reused as the deployed CONTROL PLANE,
which dispatches
59 # the L4 worker Job via google.cloud.run_v2 (backend/compute/cloud_run.py, lazy-
imported); without it
```

```

60 # the control plane fails every dispatch with "google-cloud-run is required for
CloudRunCompute's real
61 # client". Both are real Cloud Run failures the mock-client tests can't see (the SDKs
are faked in CI).
62 # D13/P12: `trackers` (ByteTrack's new home) is a git-only dep (broken PyPI wheel ?
63 # roboflow/trackers#474) so it's NOT pulled by `pip install .`; install it out-of-band
before
64 # the main env, or the person-cue path crashes with ModuleNotFoundError. The immutable
git SHA is
65 # single-sourced in requirements-trackers.txt.
66 RUN python3 -m pip install --upgrade pip \
67 && python3 -m pip install -r requirements-trackers.txt \
68 && python3 -m pip install ".[storage-gcs,cloud-run]"
69
70 # The native runner finds the WASB env + repo by env (no `conda activate` needed);
weights come in at
71 # runtime (WASB-C3). detector_impl=native = the Linux-native WASB runner; cuda stays the
default.
72 # PYTHONUNBUFFERED: without it the worker's stdout/stderr sit in the block buffer until
process
73 # exit, so Cloud Logging shows a single end-of-run burst instead of live progress
(observed on the
74 # first real M7 runs, 2026-07-03) ? a running job looks stalled and mid-run failures log
nothing.
75 ENV PYTHONUNBUFFERED=1 \
76 WASB_PYTHON=/opt/conda/envs/wasb/bin/python \
77 WASB_REPO_DIR=/opt/models/WASB-SBDT \
78 WASB_WEIGHTS=/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar \
79 RALLY_DETECTOR_IMPL=native

```

#### 43. user (2026-07-04T08:49:41.214Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-dockerfile-auth\deploy\cloudrun\Dockerfile has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 44. user (2026-07-04T08:49:50.061Z)

```

75: --cpu 2 --memory 4Gi --port 8080 --allow-unauthenticated \
83: Reserve `--allow-unauthenticated` for the pre-CF-auth smoke test only; lock ingress to the
CF proxy

```

#### 45. user (2026-07-04T08:49:56.368Z)

```

`compute_target=cloud_run`). `get_compute_backend(config)` then builds `CloudRunCompute` from
env:
```bash
export DEPLOYMENT_PROFILE=cloud-serving
export CLOUD_RUN_JOB=rally-worker CLOUD_RUN_REGION=$REGION GOOGLE_CLOUD_PROJECT=$PROJECT
A `worker infer` (or the API job path) with a gs:// video now DISPATCHES to the Job in
running in-process; the container runs the SAME worker INLINE (compute_target forc
python -m backend.worker infer --video-uri gs://<bucket>/videos/clip.mp4 --out-uri gs://<bucket>
--segmenter wasb_hybrid
...

**The AI handoff needs its key IN THE JOB env** (`GEMINI_API_KEY` ? mount a Secret Manager
secret
on the job), or run fully-local with `--set indexing.skip_ai_handoff=true` (candidate windows
emitted as rallies ? the eval-parity mode). With neither, the segmenter used to bail to a silent
0-rally "success" **after** the paid GPU pass; the cloud-serving startup checks now fail this at
boot (`AI_HANDOFF_KEY_MISSING`, rc 2) before the video is even pulled.

```

3a. Control-plane SERVICE deploy ? required flags for reliable async jobs

The API server (`/api/process`, `/api/compile` ? #329) runs its ****job worker IN-PROCESS****: the

```

request returns `202` and a background thread drains the job (GPU detection dispatches to the
Job;
the reel stitch runs on the control plane itself). On Cloud Run that background thread only
makes
progress while the instance has CPU, so a scale-to-zero **service** deploy MUST set:
```bash
gcloud run deploy rally-control-plane --image <image> --region $REGION \
 --no-cpu-throttling \ # CPU stays allocated between requests ? the drain thread
finishes
 --min-instances 1 \ # one always-warm instance drains reliably (cheap CPU; NOT the D7
GPU pool)
 --cpu 2 --memory 4Gi --port 8080 --allow-unauthenticated \
 --set-env-vars DEPLOYMENT_PROFILE=cloud-serving,CLOUD_RUN_JOB=rally-
worker,CLOUD_RUN_REGION=$REGION,GOOGLE_CLOUD_PROJECT=$PROJECT
```

```

Without `--no-cpu-throttling`, a `/api/compile` returns 202 and then ****stalls**** ? the reel never finishes and startup crash-recovery would mark it terminal. `min-instances 1` does NOT contradict D7 (that was the ~\$500/mo warm ****GPU**** pool); this is a ~1-vCPU CPU instance. Defense-in-depth is in the code: `Database.recover\_stale\_jobs()` ****re-queues**** a compile interrupted by a restart (idempotent stitch) rather than failing it, so even an unlucky scale-down mid-stitch self-heals on the next drain. Reserve `--allow-unauthenticated` for the pre-CF-auth smoke test only; lock ingress to the CF proxy (M9) before inviting testers.

4. M7 acceptance (the owner runlog)

Capture a `DEPLOY\_REPORT\_cloud-serving\_<date>.md` under `docs/COMPUTE\_DECOUPLED\_SERVING/runlogs/` (mirrors the truly-local template): the exact commands, rally counts, `gpu\_active\_seconds`/`wall\_seconds`/`bytes\_out`, a reel smoke-test (`ffmpeg -f null -` exit 0 + a 5-frame contact sheet), gotchas, and the ****\$0** teardown audit**.

Gotchas

- The dispatched container ****must run inline**** ? `CloudRunCompute` forces `compute\_target=local\_gpu` + `indexing.detector\_impl=native` in the per-execution args so the Job never re-dispatches itself.
- Completion is a ****bucket marker**** (`<out\_uri>/\_dispatch/<token>.done`), not the Cloud Run API status

46. assistant (2026-07-04T08:50:04.290Z)

The README never documented the overlay workaround, so the Dockerfile change plus a one-line note in §3a is the complete fix. Let me add that note, then commit and push.

47. user (2026-07-04T08:50:07.098Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

48. user (2026-07-04T08:50:13.362Z)

```

80      D7 (that was the ~$500/mo warm **GPU** pool); this is a ~1-vCPU CPU instance. Defense-
in-depth is in
81      the code: `Database.recover_stale_jobs()` **re-queues** a compile interrupted by a
restart (idempotent
82      stitch) rather than failing it, so even an unlucky scale-down mid-stitch self-heals on
the next drain.
83      Reserve `--allow-unauthenticated` for the pre-CF-auth smoke test only; lock ingress to
the CF proxy

```

```
84      (M9) before inviting testers.
85
86      ## 4. M7 acceptance (the owner runlog)
87      Capture a `DEPLOY_REPORT_cloud-serving_<date>.md` under
```

49. user (2026-07-04T08:50:19.973Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-dockerfile-auth\deploy\cloudrun\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

50. user (2026-07-04T08:50:30.011Z)

```
fix/dockerfile-auth-extra
4847232 docs(runlog): A1 alpha serving deploy ? hosted gs:// CUJ report (#464)
[fix/dockerfile-auth-extra 9ed5d42] deploy(cloudrun): bake the auth extra (PyJWT) into the base
image so an edge-auth control plane needs no derived overlay
 2 files changed, 9 insertions(+), 3 deletions(-)
9ed5d42 deploy(cloudrun): bake the auth extra (PyJWT) into the base image so an edge-auth
control plane needs no derived overlay
```

51. user (2026-07-04T08:50:51.213Z)

```
remote:
remote: Create a pull request for 'fix/dockerfile-auth-extra' on GitHub by visiting:
remote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/fix/dockerfile-
auth-extra
remote:
branch 'fix/dockerfile-auth-extra' set up to track 'origin/fix/dockerfile-auth-extra'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]      fix/dockerfile-auth-extra -> fix/dockerfile-auth-extra
https://github.com/Khelsutra/badminton-highlight-indexer/pull/470
```

52. user (2026-07-04T08:51:09.822Z)

```
[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
--- md links ---
[check-md-links] clean ? no broken internal links in live docs.
```

53. assistant (2026-07-04T08:51:16.013Z)

md-links is clean. The pip output was suppressed by `--quiet` ? let me confirm the resolve actually succeeded.

54. user (2026-07-04T08:51:25.863Z)

```
Requirement already satisfied: pyjwt>=2.8 in
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages
(from pyjwt[crypto]>=2.8; extra ==
"auth"->badminton-highlight-indexer==0.1.0) (2.13.0)
Requirement already satisfied: google-cloud-run<1,>=0.10 in
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages
(from badminton-highlight-indexer==0.1.0)
(0.16.0)
Requirement already satisfied: google-api-core<3.0.0,>=2.11.0 in
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
google-api-core[grpc]<3.0.0,>=2.11.0->google-cloud-run<1,>=0.10->badminton-highlight-
indexer==0.1.0) (2.31.0)
Requirement already satisfied: google-auth!=2.24.0,!2.25.0,<3.0.0,>=2.14.1 in
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
google-cloud-run<1,>=0.10->badminton-highlight-indexer==0.1.0) (2.53.0)
Requirement already satisfied: grpcio<2.0.0,>=1.33.2 in
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
```

```
google-cloud-run<1,>=0.10->badminton-highlight-indexer==0.1.0) (1.81.1)
Requirement already satisfied: proto-plus<2.0.0,>=1.22.3 in
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages (from
google-cloud-run<1,>=0.10->badminton-highlight-indexer==0.1.0) (1.28.0)
exit=
```

55. user (2026-07-04T08:52:04.069Z)

Updated task #4 status

56. user (2026-07-04T08:52:15.000Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/471>

57. user (2026-07-04T08:52:39.365Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/472>

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/473>

58. user (2026-07-04T09:02:57.696Z)

```
<retrieval_status>timeout</retrieval_status>
```

```
<task_id>wdmdpc7il</task_id>
```

```
<task_type>local_workflow</task_type>
```

```
<status>running</status>
```

59. assistant (2026-07-04T09:03:07.765Z)

PR #465 just got the owner's LGTM. Per the handoff protocol: read the full comment, verify checks + mergeability, then squash-merge.

60. user (2026-07-04T09:03:10.661Z)

avidullu @ 2026-07-04T08:55:42Z:

LGTM. I re-reviewed the updated approach and verified it locally on a current-`origin/master` merge of this PR.

Verified gates:

```
```text
```

```
python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70
```

**1634 passed, 10 skipped; coverage 85.40% (floor 70%)**

```
ruff check .
```

**All checks passed**

```
mypy backend training
```

**Success: no issues found in 147 source files**

```
```
```

Coverage comparison was like-for-like in temp worktrees: `origin/master` measured 85.39%, and the current-master merge of this PR measured 85.40%, so no regression. The prior hosted-mode contract issue is resolved by scoping storage URIs to `storage.input\_root` while preserving the local path jail.

61. user (2026-07-04T09:03:20.567Z)

Built wheel smoke (install + launch) pass 1m9s <https://github.com/Khelsutra/badminton->

```
highlight-indexer/actions/runs/28699728277/job/85115619212
Dependency license scan (no copyleft in runtime deps) pass 31s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28699728277/job/85115619221
Full test suite (offline, mocked) pass 2m28s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28699728277/job/85115619203
Golden fixtures cross-OS (macos-latest) pass 25s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28699728277/job/85115619233
Golden fixtures cross-OS (ubuntu-latest) pass 24s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28699728277/job/85115619220
Golden fixtures cross-OS (windows-latest) pass 1m6s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28699728277/job/85115619219
Leak scan (PII / attribution guard) pass 7s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28699728277/job/85115619197
Markdown link check (live docs) pass 7s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28699728277/job/85115619207
Mypy (lenient baseline) pass 33s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28699728277/job/85115619216
Ruff lint pass 9s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28699728277/job/85115619213
Syntax sweep + import smoke (no GPU stack) pass 17s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28699728277/job/85115619208
mergeable=MERGEABLE state=CLEAN
```

62. user (2026-07-04T09:03:33.416Z)

(PowerShell completed with no output)

63. user (2026-07-04T09:03:44.206Z)

```
state=MERGED mergedAt=2026-07-04T09:03:32Z
--- re-check remaining PRs mergeable after master moved ---
#467: UNKNOWN UNKNOWN
#468: UNKNOWN UNKNOWN
#469: UNKNOWN UNKNOWN
#470: UNKNOWN UNKNOWN
```

64. user (2026-07-04T09:05:56.623Z)

```
<retrieval_status>success</retrieval_status>

<task_id>wdmdpc7il</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>
{
  "summary": "Fix the 3 open review blockers on PRs #467/#468/#469 in their existing worktrees,
then adversarially verify each fix",
  "agentCount": 6,
  "logs": [],
  "result": [
    {
      "pr": 468,
      "fix": {
        "done": true,
        "files_changed": [
          "C:/Users/avidu/Projects/khelsutra-guru/wt-drain-
concurrency/backend/infrastructure/database_jobs.py",
```

```

"C:/Users/avidu/Projects/khelsutra-guru/wt-drain-
concurrency/tests/test_job_worker_loop.py"
],
"tests_added": [
    "tests/test_job_worker_loop.py :: test_concurrent_claimers_claim_all_jobs_distinctly"
],
"suite": "1634 passed, 10 skipped, 8 warnings in 112.63s (0:01:52)",
"ruff": "ruff 0.15.16: All checks passed!",
"mypy": "mypy 2.1.0: Success: no issues found in 147 source files (4 pre-existing
annotation-unchecked notes, unrelated to this change)",
"summary": "Implemented the reviewer's preferred fix: claim_due_job() in
backend/infrastructure/database_jobs.py now loops on a lost CAS race ? re-SELECT the head
runnable row; if none, return None (None now truly means nothing-runnable); attempt the CAS
UPDATE; on rowcount==1 return the claimed job, otherwise retry. The loop is deliberately
unbounded (every lost race means another worker claimed that row, so the runnable set shrinks ?
lock-free global progress; a cap would reintroduce the false-empty), and it reuses the single
_connect() connection safely because the SELECT is an autocommit read and each CAS commits
before the next iteration, so no write lock is held across retries (the 30s busy_timeout covers
momentary contention). The docstring explains the retry, the new None semantics, and cites the
#466/#468 parallel-drain design. _drain_due_jobs() in backend/api/job_worker.py needed NO change
and was not touched ? its \"if job is None: break\" is now correct because None means genuinely
empty. Added the reviewer-required regression test: 8 queued jobs, 8 claimer threads released
together by a threading.Barrier, each calling db.claim_due_job() once; asserts all 8 results are
non-None (no false-empty), the 8 claimed ids are exactly the 8 distinct queued jobs (no double-
claim), all rows end up 'running', and a subsequent claim returns None (truly empty). Verified
the test catches the pre-fix bug (5/5 failures with the fix stashed) and is stable with the fix
(15/15 passes under concurrent sibling suites); no 'database is locked' errors surfaced, so no
extra contention handling was needed beyond the existing busy_timeout.",
"pr_reply_notes": "Fixed as suggested (first option): claim_due_job() now retries the
SELECT->CAS loop on rowcount != 1 instead of returning None, so None is only returned when a
fresh SELECT finds no runnable row ? a lost race can never masquerade as queue-empty. The retry
loop is unbounded on purpose: a lost race implies another worker successfully claimed that row,
so the runnable set strictly shrinks and global progress is guaranteed (a retry cap could
reintroduce the false-empty you reproduced). The loop reuses one connection; the SELECT is an
autocommit read and each CAS commits before the next attempt, so no write lock is held across
iterations. _drain_due_jobs() needed no change ? its None-means-empty exit is now correct by
construction. Regression test added
(tests/test_job_worker_loop.py::test_concurrent_claimers_claim_all_jobs_distinctly): 8 queued
jobs, 8 claimer threads synchronized on a threading.Barrier, each claiming once; asserts all 8
return non-None and the claimed ids are exactly the 8 distinct jobs, then that a 9th claim
returns None. Evidence: with the fix stashed the new test reproduces your finding and fails 5/5
runs (losers return None); with the fix it passes 15/15 consecutive runs, no SQLite 'database is
locked' under the 8-thread race (the existing 30s busy_timeout covers contention). Full suite
1634 passed / 10 skipped; ruff 0.15.16 and mypy 2.1.0 clean. The park-vs-block
(JobWaiting/WAIT_AND_RESUME) refinement was intentionally left out per the separate
discussion.",
"problems": ""
},
"verdict": {
    "addresses_blocker": true,
    "test_fails_without_fix": true,
    "issues": [],
    "verdict": "approve",
    "notes": "Fix implements the reviewer's preferred remedy: claim_due_job()
(backend/infrastructure/database_jobs.py:373-393) retries a lost CAS by re-SELECTing the head
runnable row, so None is returned only when a fresh SELECT finds nothing runnable;
_drain_due_jobs() correctly needed no change. Unbounded retry is safe: rowcount==0 implies the
(id,status) pair changed, so the runnable set shrinks per retry (global progress), and each
iteration commits before retrying so no write lock spans retries. PROVEN the regression test
bites: at PR head 8534077 (throwaway detached worktree, only the test file copied)
test_concurrent_claimers_claim_all_jobs_distinctly failed with \"lost-race leaked None:
['j0','j1',None,'j2',None,None,None,None]\" ? the blocker's exact failure mode; throwaway
worktree removed. With the fix: new test passed 5/5 consecutive runs (no flakiness), module
16/16, full suite 1634 passed/10 skipped, ruff clean, mypy clean (2 pre-existing notes), CI
coverage command passed at 85.41% vs 70% floor. Not verifiable locally: real multi-instance

```

Cloud Run behavior, but the threaded SQLite CAS race is exactly the reviewer's repro. No modifications made to the fix agent's changes; nothing committed or pushed."

```
}
},
{
  "pr": 467,
  "fix": {
    "done": true,
    "files_changed": [
      "C:/Users/avidu/Projects/khelsutra-guru/wt-upload-staging/backend/orchestration.py",
      "C:/Users/avidu/Projects/khelsutra-guru/wt-upload-
staging/tests/test_api_cloud_dispatch.py"
    ],
    "tests_added": [
      "tests/test_api_cloud_dispatch.py ::
test_cloud_stages_when_input_root_is_cloud_and_input_backend_default"
    ],
    "suite": "1629 passed, 10 skipped, 8 warnings in 128.78s (0:02:08)",
    "ruff": "ruff 0.15.16 ? \"All checks passed!\" (python -m ruff check ., from worktree
root)",
    "mypy": "mypy 2.1.0 ? \"Success: no issues found in 147 source files\" (python -m mypy
backend training; 4 pre-existing annotation-unchecked notes only)",
    "summary": "Replaced the staging gate's predicate in backend/orchestration.py (~line
254-267, inside the input_ref.backend == 'local' branch): instead of \"not input_root or
input_backend == 'local'\", the gate now classifies the configured input_root itself via
storage.parse_uri(input_root).backend, failing only when that is 'local' (empty root or an
unsupported-scheme root are treated as local, the latter via a caught ValueError). This is the
truthful gate because staged_uri is built by joining under input_root and storage.put() routes
by parse_uri(staged_uri).backend ? so an explicit cloud root (gs://, s3://) stages fine with
input_backend at its default, exactly the reviewer's repro. The legacy shape the old gate
accepted (bare/local-looking input_root + explicit non-local input_backend) was broken
downstream anyway: put() would route to LocalStorage (a copy the Cloud Run job can't read) and
the follow-up resolve_input_ref(staged_uri) would double-join the bare staged path under
input_root, so classifying the root removes a lie rather than a capability. Added the required
regression test mimicking the existing staging test's fixture pattern (input_root='gs://in-
bucket/videos', input_backend asserted still at default 'local', local upload -> storage.put
called with the gs:// staged URI and the Cloud Run dispatch continues from it). The existing
negative test (no input_root -> 'needs a cloud input' failure, no put, no dispatch) still passes
unchanged. Changes are left uncommitted in the worktree as instructed.",
    "pr_reply_notes": "Fixed ? good catch. The gate now classifies the configured root
itself: storage.parse_uri(input_root).backend must be non-local (empty or unsupported-scheme
roots count as local), and storage.input_backend is no longer consulted. Your exact repro shape
(storage.input_root='gs://in-bucket/videos' with input_backend left at its default) now stages
the local upload to gs://in-bucket/videos/<video_id>/<name> and dispatches from the staged URI.
Why parse_uri(input_root) rather than \"input_root set AND (root cloud OR input_backend non-
local)\": the staged URI is built by joining under input_root and storage.put() routes by
parse_uri(staged_uri).backend, so staging genuinely works only when the root itself carries a
cloud scheme. The legacy shape the old gate admitted (bare local-looking input_root + explicit
input_backend='gcs') was never functional downstream ? put() would route to LocalStorage (a copy
the Cloud Run job can't read) and resolve_input_ref(staged_uri) would then double-join the bare
staged path under input_root ? so the new gate makes the check truthful instead of preserving a
broken acceptance. The _failed message is unchanged and remains accurate (\"a cloud
storage.input_root to stage local uploads into\"). Regression test added: tests/test_api_cloud_d
ispatch.py::test_cloud_stages_when_input_root_is_cloud_and_input_backend_default (asserts
input_backend is still the default 'local', storage.put is called with the gs:// staged URI, and
the dispatch --video-uri is the staged URI); the existing negative test (no input_root -> hard
fail, no put/dispatch) still passes. Verified locally on d3a9629 + this fix: full suite 1629
passed / 10 skipped in 129s; ruff 0.15.16 clean; mypy 2.1.0 clean (147 files).",
    "problems": ""
  },
  "verdict": {
    "addresses_blocker": true,
    "test_fails_without_fix": true,
    "issues": [],
    "verdict": "approve",
  }
}
```

"notes": "Verified adversarially and could not refute. (1) Gate logic proven truthful against the storage layer: staged_uri is joined under input_root and storage.put routes by parse_uri(staged_uri).backend, so classifying the root itself (backend/orchestration.py:260-267) is the correct predicate; empty root and unsupported-scheme root (ValueError) correctly map to local->fail. (2) Regression test proven to bite: at unfixed PR head d3a9629 in a throwaway detached worktree it fails with the blocker's exact failure ('failed' != 'success', pre-dispatch 'needs a cloud input'); throwaway removed. (3) In the real worktree with the fix: test file 26 passed; full suite 1629 passed / 10 skipped (matches agent's claim); ruff clean; mypy clean (4 pre-existing notes only). (4) Behavior change beyond the blocker ? rejecting local-looking input_root + explicit non-local input_backend ? is a correctness improvement (that shape staged to LocalStorage, unreadable by Cloud Run) and matches the reviewer's suggested fix. Non-blocking notes: gdrive:// (mounted) root still passes the gate though Cloud Run can't read it ? same pre-existing hole as the old input_backend='gdrive' path, and the reviewer explicitly accepted 'at least not local'; coverage-vs-base was not measured (not part of this verification's scope). Changes remain uncommitted in C:/Users/avidu/Projects/khelsutra-guru/wt-upload-staging as instructed."

```

    }
  },
  {
    "pr": 469,
    "fix": {
      "done": true,
      "files_changed": [
        "C:/Users/avidu/Projects/khelsutra-guru/wt-signed-
download/backend/storage/__init__.py",
        "C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/base.py",
        "C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/gcs.py",
        "C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/s3.py",
        "C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/tests/test_storage.py",
        "C:/Users/avidu/Projects/khelsutra-guru/wt-signed-
download/tests/test_upload_download.py"
      ],
      "tests_added": [
        "tests/test_storage.py :: test_module_signed_url_honors_configured_gcs_ttl",
        "tests/test_storage.py :: test_module_signed_url_honors_configured_s3_ttl",
        "tests/test_storage.py :: test_module_signed_url_explicit_ttl_overrides_config",
        "tests/test_upload_download.py ::
test_highlights_signed_url_honors_configured_gcs_ttl"
      ],
      "suite": "1633 passed, 10 skipped, 8 warnings in 126.22s (0:02:06)",
      "ruff": "ruff 0.15.16: \"All checks passed!\" (python -m ruff check ., from worktree
root)",
      "mypy": "mypy 2.1.0: \"Success: no issues found in 147 source files\" (python -m mypy
backend training; 4 pre-existing annotation-unchecked notes only)",
      "summary": "Changed the module-level storage.signed_url() helper
(backend/storage/__init__.py) from ttl: int = 3600 to ttl: Optional[int] = None so a caller that
doesn't pass ttl no longer shadows the operator's configured signed_url_ttl; with ttl=None the
backends' existing \"ttl or self._ttl\" fallback applies. For consistency (and so mypy stays
green when the helper passes Optional[int] through the ABC), StorageBackend.signed_url in
backend/storage/base.py and the GCSStorage/S3Storage method defaults were changed from ttl=3600
to ttl=None; docstrings on the helper and the ABC now document the None-means-configured-TTL
contract citing #469. Audited all callers: the only production caller of any backend
.signed_url() is the module helper, and the helper's only production caller is
orchestration.signed_reel_url() (backend/orchestration.py:713), which passes no ttl and
therefore now honors the configured TTL ? verified end-to-end by a new /api/highlights test that
injects a fake GCS client via cfg.storage.gcs and asserts the 307 redirect location carries
ttl=123. Added the reviewer's exact GCS repro, an S3 equivalent (ExpiresIn=456), and an
explicit-ttl-overrides-config test; the fake GCS blob's generate_signed_url now embeds the
expiration seconds in the URL so TTL plumbing is assertable. backend/main.py was not touched
(line budget unaffected).",
      "pr_reply_notes": "Fixed ? the helper was indeed forcing ttl=3600 into the backend, so
the configured signed_url_ttl never applied. storage.signed_url() now defaults ttl=None and
passes it through unchanged; the backends' existing \"ttl or self._ttl\" fallback then applies
their configured signed_url_ttl. For signature consistency (and mypy), StorageBackend.signed_url
and the GCSStorage/S3Storage overrides also default ttl=None now ? an explicit int still

```

```

overrides. Audited all callers: the module helper is the only production caller of the backend
methods, and signed_reel_url() is the helper's only production caller (it passes no ttl), so
/api/highlights now honors the configured TTL with no other behavior change; nothing relied on
the implicit 3600. Regression tests added: (1) your exact repro ? injected GCS client +
config={'gcs': {'signed_url_ttl': 123}} ? the signer receives a 123s expiration
(tests/test_storage.py::test_module_signed_url_honors_configured_gcs_ttl); (2) same for S3 ?
ExpiresIn=456 (::test_module_signed_url_honors_configured_s3_ttl); (3) end-to-end
/api/highlights with a fake GCS client injected through cfg.storage.gcs ? 307 redirect URL
carries ttl=123
(tests/test_upload_download.py::test_highlights_signed_url_honors_configured_gcs_ttl); (4)
explicit ttl=999 still overrides the configured 123
(::test_module_signed_url_explicit_ttl_overrides_config). Verified locally: full suite 1633
passed / 10 skipped; ruff 0.15.16 clean; mypy 2.1.0 clean.",
    "problems": ""
},
"verdict": {
    "addresses_blocker": true,
    "test_fails_without_fix": true,
    "issues": [],
    "verdict": "approve",
    "notes": "Fix is exactly the reviewer's suggested approach: storage.signed_url() helper
ttl default changed to None (backend/storage/__init__.py), with matching None defaults on
StorageBackend.signed_url (base.py), GCSStorage (gcs.py), and S3Storage (s3.py); the backends'
existing 'ttl or self._ttl' fallback now lets configured signed_url_ttl apply, and constructor
defaults (3600) preserve unconfigured behavior. Caller audit confirms signed_reel_url
(backend/orchestration.py:713) is the sole production caller and passes no ttl. Bite proof at
unfixed head d3b0e64 (throwaway worktree, test files only): GCS test failed with ttl=3600 vs 123
(reviewer's exact repro), S3 test failed with X-Amz-Expires=3600 vs 456, /api/highlights e2e
test failed with redirect location ?ttl=3600 ? 3 of 4 new tests fail; the 4th (explicit-ttl-
overrides-config) passes at head by design as an over-correction guard, not a regression test.
Reproduced locally in the real worktree: both test files 59/59; full suite 1633 passed / 10
skipped; ruff clean; mypy clean (4 pre-existing annotation-unchecked notes); no coverage floor
configured in the repo to check. GCS IAM-signBlob fallback path uses the same expiration, so
both signing paths honor config. Non-issue: explicit ttl=0 still falls back to configured TTL
('ttl or self._ttl') ? pre-existing, degenerate input, consistent with the reviewer's
suggestion."
}
},
"workflowProgress": [
    {
        "type": "workflow_phase",
        "index": 1,
        "title": "Fix"
    },
    {
        "type": "workflow_phase",
        "index": 2,
        "title": "Verify"
    },
    {
        "type": "workflow_agent",
        "index": 1,
        "label": "fix:#468",
        "phaseIndex": 1,
        "phaseTitle": "Fix",
        "agentId": "ae958a1af2ef5ef54",
        "model": "claude-fable-5",
        "state": "done",
        "startedAt": 1783154879438,
        "queuedAt": 1783154879396,
        "attempt": 1,
        "lastToolName": "StructuredOutput",
        "lastToolSummary": "1634 passed, 10 skipped, 8 warnings in 112.63s (0:01:52)",
        "promptPreview": "You are fixing a code-review blocker on PR #468 (branch fix/drain-

```

```

concurrency) of Khelsutra/badminton-highlight-indexer.\nYour worktree:
C:/Users/avidu/Projects/khelsutra-guru/wt-drain-concurrency (must be on fix/drain-concurrency
at 8534077)\n\nENVIRONMENT (read carefully):\n- Windows 11 box; use the PowerShell tool for
shell commands (or Bash for POSIX one-liners). Global Python 3.13.13 has the full?",
  "lastProgressAt": 1783155305778,
  "tokens": 74775,
  "toolCalls": 21,
  "durationMs": 426339,
  "resultPreview": "{\n\"done\":true,\n\"files_changed\":[\n\"C:/Users/avidu/Projects/khelsutra-
guru/wt-drain-
concurrency/backend/infrastructure/database_jobs.py\",
\"C:/Users/avidu/Projects/khelsutra-
guru/wt-drain-
concurrency/tests/test_job_worker_loop.py\"],\n\"tests_added\":[\n\"tests/test_job_worker_loop.py ::
test_concurrent_claimers_claim_all_jobs_distinctly\",
\"suite\":\n\"1634 passed, 10 skipped, 8
warnings in 112.63s (0:01:52)\",\n\"ruff\"?\"
},
{
  "type": "workflow_agent",
  "index": 2,
  "label": "fix:#467",
  "phaseIndex": 1,
  "phaseTitle": "Fix",
  "agentId": "a47789bf8f991c707",
  "model": "claude-fable-5",
  "state": "done",
  "startedAt": 1783154879441,
  "queuedAt": 1783154879396,
  "attempt": 1,
  "lastToolName": "StructuredOutput",
  "lastToolSummary": "1629 passed, 10 skipped, 8 warnings in 128.78s (0:02:08)",
  "promptPreview": "You are fixing a code-review blocker on PR #467 (branch fix/upload-gcs-
staging) of Khelsutra/badminton-highlight-indexer.\nYour worktree:
C:/Users/avidu/Projects/khelsutra-guru/wt-upload-staging (must be on fix/upload-gcs-staging at
d3a9629)\n\nENVIRONMENT (read carefully):\n- Windows 11 box; use the PowerShell tool for shell
commands (or Bash for POSIX one-liners). Global Python 3.13.13 has the full?",
  "lastProgressAt": 1783155192129,
  "tokens": 69376,
  "toolCalls": 29,
  "durationMs": 312688,
  "resultPreview": "{\n\"done\":true,\n\"files_changed\":[\n\"C:/Users/avidu/Projects/khelsutra-
guru/wt-upload-staging/backend/orchestration.py\",
\"C:/Users/avidu/Projects/khelsutra-guru/wt-
upload-
staging/tests/test_api_cloud_dispatch.py\"],\n\"tests_added\":[\n\"tests/test_api_cloud_dispatch.py
:: test_cloud_stages_when_input_root_is_cloud_and_input_backend_default\",
\"suite\":\n\"1629
passed, 10 skipped, 8 warnings in 128.78s (0:02:08)\",\n\"ru?\"
},
{
  "type": "workflow_agent",
  "index": 3,
  "label": "fix:#469",
  "phaseIndex": 1,
  "phaseTitle": "Fix",
  "agentId": "af636c5ecb7b6056d",
  "model": "claude-fable-5",
  "state": "done",
  "startedAt": 1783154879442,
  "queuedAt": 1783154879396,
  "attempt": 1,
  "lastToolName": "StructuredOutput",
  "lastToolSummary": "1633 passed, 10 skipped, 8 warnings in 126.22s (0:02:06)",
  "promptPreview": "You are fixing a code-review blocker on PR #469 (branch fix/highlights-
signed-url) of Khelsutra/badminton-highlight-indexer.\nYour worktree:
C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download (must be on fix/highlights-signed-url
at d3b0e64)\n\nENVIRONMENT (read carefully):\n- Windows 11 box; use the PowerShell tool for
shell commands (or Bash for POSIX one-liners). Global Python 3.13.13 has t?",

```

```

    "lastProgressAt": 1783155222869,
    "tokens": 78299,
    "toolCalls": 32,
    "durationMs": 343427,
    "resultPreview": "{\\"done\\":true,\\"files_changed\\":[\\\"C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/__init__.py\\\",\\\"C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/base.py\\\",\\\"C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/gcs.py\\\",\\\"C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download/backend/storage/s3.py\\\",\\\"C:/Users/avidu/Projects/khelsutr?\
  },
  {
    "type": "workflow_agent",
    "index": 4,
    "label": "verify:#467",
    "phaseIndex": 2,
    "phaseTitle": "Verify",
    "agentId": "aa58f4ccaf67eca9d",
    "model": "claude-fable-5",
    "state": "done",
    "startedAt": 1783155192154,
    "queuedAt": 1783155192133,
    "attempt": 1,
    "lastToolName": "StructuredOutput",
    "lastToolSummary": "approve",
    "promptPreview": "You are an ADVERSARIAL verifier for a fix to PR #467 (branch fix/upload-gcs-staging) of Khelsutra/badminton-highlight-indexer. A fix agent just left UNCOMMITTED changes in the worktree C:/Users/avidu/Projects/khelsutra-guru/wt-upload-staging. Your default stance: try to REFUTE that the fix is correct and complete.\\n\\nThe blocker it must resolve:\\nREVIEWER BLOCKER (verbatim, from PR #467):\\n\\\"I found on?\",
    "lastProgressAt": 1783155956530,
    "tokens": 70706,
    "toolCalls": 23,
    "durationMs": 764376,
    "resultPreview": "{\\"addresses_blocker\\":true,\\"test_fails_without_fix\\":true,\\"issues\\":[
],\\"verdict\\":\\"approve\\\",\\"notes\\":\\"Verified adversarially and could not refute. (1) Gate logic proven truthful against the storage layer: staged_uri is joined under input_root and storage.put routes by parse_uri(staged_uri).backend, so classifying the root itself (backend/orchestration.py:260-267) is the correct predicate; empty root an?\"
  },
  {
    "type": "workflow_agent",
    "index": 5,
    "label": "verify:#469",
    "phaseIndex": 2,
    "phaseTitle": "Verify",
    "agentId": "ac9028b4eb07da1e2",
    "model": "claude-fable-5",
    "state": "done",
    "startedAt": 1783155222877,
    "queuedAt": 1783155222875,
    "attempt": 1,
    "lastToolName": "StructuredOutput",
    "lastToolSummary": "approve",
    "promptPreview": "You are an ADVERSARIAL verifier for a fix to PR #469 (branch fix/highlights-signed-url) of Khelsutra/badminton-highlight-indexer. A fix agent just left UNCOMMITTED changes in the worktree C:/Users/avidu/Projects/khelsutra-guru/wt-signed-download. Your default stance: try to REFUTE that the fix is correct and complete.\\n\\nThe blocker it must resolve:\\nREVIEWER BLOCKER (verbatim, from PR #469):\\n\\\"One mo?\",
    "lastProgressAt": 1783155534447,
    "tokens": 58026,
    "toolCalls": 16,
    "durationMs": 311570,
    "resultPreview": "{\\"addresses_blocker\\":true,\\"test_fails_without_fix\\":true,\\"issues\\":[
],\\"verdict\\":\\"approve\\\",\\"notes\\":\\"Fix is exactly the reviewer's suggested approach: storage.signed_url() helper ttl default changed to None (backend/storage/__init__.py), with

```

matching None defaults on StorageBackend.signed_url (base.py), GCSStorage (gcs.py), and S3Storage (s3.py); the backends' existing 'ttl or self._ttl' fallback now?"

```
    },
    {
      "type": "workflow_agent",
      "index": 6,
      "label": "verify:#468",
      "phaseIndex": 2,
      "phaseTitle": "Verify",
      "agentId": "a6adca732bba00df3",
      "model": "claude-fable-5",
      "state": "done",
      "startedAt": 1783155305787,
      "queuedAt": 1783155305785,
      "attempt": 1,
      "lastToolName": "StructuredOutput",
      "lastToolSummary": "approve",
      "promptPreview": "You are an ADVERSARIAL verifier for a fix to PR #468 (branch fix/drain-
concurrency) of Khelsutra/badminton-highlight-indexer. A fix agent just left UNCOMMITTED changes
in the worktree C:/Users/avidu/Projects/khelsutra-guru/wt-drain-concurrency. Your default
stance: try to REFUTE that the fix is correct and complete.\n\nThe blocker it must
resolve:\nREVIEWER BLOCKER (verbatim, from PR #468):\n\"claim_du?\",
      "lastProgressAt": 1783155754897,
      "tokens": 54872,
      "toolCalls": 15,
      "durationMs": 449110,
      "resultPreview": "{\n  \"addresses_blocker\":true,\n  \"test_fails_without_fix\":true,\n  \"issues\":[
],\n  \"verdict\":\n    \"approve\",\n    \"notes\":\n      \"Fix implements the reviewer's preferred remedy:
claim_due_job() (backend/infrastructure/database_jobs.py:373-393) retries a lost CAS by re-
SELECTing the head runnable row, so None is returned only when a fresh SELECT finds nothing
runnable; _drain_due_jobs() correctly needed no change. Unbounded retry ?\"
    }
  ],
  "totalTokens": 406054,
  "totalToolCalls": 136
}
</output>
```

65. user (2026-07-04T09:06:11.575Z)

```
diff --git a/backend/infrastructure/database_jobs.py b/backend/infrastructure/database_jobs.py
index a9e18fb..8310f07 100644
--- a/backend/infrastructure/database_jobs.py
+++ b/backend/infrastructure/database_jobs.py
@@ -358,25 +358,38 @@ class DatabaseJobsMixin:
     """Atomically claim ONE runnable job ? 'queued', or 'waiting' whose ``next_poll_at`` is
    due ?
        flipping it to 'running'. Compare-and-set on (id, status) so concurrent workers never
        double-claim (the DB row-claim is the only coordination needed; no master). Earliest-
    due
    - first. Returns the claimed job dict, or None if nothing is runnable."""
    + first. Returns the claimed job dict, or None ONLY when a fresh SELECT finds nothing
    + runnable ? never for a lost race.
    +
    + A lost CAS RETRIES instead of returning: under the parallel drain (#466) several
workers
    + can SELECT the same head-of-queue row, and if the losers reported None the caller
    + (``_drain_due_jobs``) would read it as "queue empty" and stop with runnable jobs still
    + queued ? stranding dispatch capacity until some later submit/wake. The retry loop is
    + deliberately UNBOUNDED: every lost race means another worker successfully claimed that
    + row, so the runnable set shrank and global progress is guaranteed (lock-free progress);
    + a retry cap would just reintroduce the false-empty. One connection serves all attempts
    ?
    + the SELECT is an autocommit read and each CAS commits before the next iteration, so no
    + write lock is ever held across a retry (busy_timeout covers momentary contention)."""
```

```

try:
    with self._connect() as conn:
        cur = conn.cursor()
        row = cur.execute(
-         "SELECT id, status FROM jobs WHERE status='queued' "
-         "OR (status='waiting' AND (next_poll_at IS NULL OR next_poll_at <=
CURRENT_TIMESTAMP)) "
-         "ORDER BY COALESCE(next_poll_at, created_at) ASC LIMIT 1"
-         ).fetchone()
-         if not row:
-             return None
-         cur.execute(
-         "UPDATE jobs SET status='running', started_at=CURRENT_TIMESTAMP "
-         "WHERE id=? AND status=?",
-         (row["id"], row["status"]),
-         )
-         conn.commit()
-         if cur.rowcount != 1:
-             return None # lost the race to a concurrent worker
+         while True:
+             row = cur.execute(
+             "SELECT id, status FROM jobs WHERE status='queued' "
+             "OR (status='waiting' AND (next_poll_at IS NULL OR next_poll_at <=
CURRENT_TIMESTAMP)) "
+             "ORDER BY COALESCE(next_poll_at, created_at) ASC LIMIT 1"
+             ).fetchone()
+             if not row:
+                 return None # nothing runnable ? a TRUE empty, not a lost race
+             cur.execute(
+             "UPDATE jobs SET status='running', started_at=CURRENT_TIMESTAMP "
+             "WHERE id=? AND status=?",
+             (row["id"], row["status"]),
+             )
+             conn.commit()
+             if cur.rowcount == 1:
+                 break # claimed it
+             # Lost the race: the winner flipped this row; re-SELECT for the next one.
        return self.get_job(row["id"])
    except Exception as e:
        logger.error(f"Failed to claim a due job: {e}")
diff --git a/tests/test_job_worker_loop.py b/tests/test_job_worker_loop.py
index e84d2a6..fe9c6f9 100644
--- a/tests/test_job_worker_loop.py
+++ b/tests/test_job_worker_loop.py
@@ -13,13 +13,16 @@ Covered:
- a `JobWaiting` parks the job (no terminal state) and a later drain (once due) resumes it;
- not-yet-due parked jobs are left alone and a wake-timer is armed for them;
- the wake-scheduling decision (`_schedule_next_drain_if_waiting`) is asserted with the
- timer seam stubbed ? no real waiting, no leaked threads.
+ timer seam stubbed ? no real waiting, no leaked threads;
+- concurrent claimers (#466/#468 parallel drain) never double-claim AND never report a
+ false-empty: a lost CAS race retries inside `claim_due_job`, so None means truly empty.

The real-timer seam (`_arm_wake_timer`) is monkeypatched to a recorder in every test that
can park a job, so nothing actually sleeps and no daemon timer outlives the test.
"""

import sqlite3
+import threading

import pytest

@@ -222,6 +225,44 @@ def test_claim_orders_earliest_due_first(env):
    assert set(order) == {"a", "b"}

```

```

+# --- concurrent claimers: no false-empty, no double-claim (#466/#468 parallel drain) -----
+
+
+def test_concurrent_claimers_claim_all_jobs_distinctly(tmp_path):
+    """Regression for the #468 review blocker: under the parallel drain, N claimers racing
+    over N queued jobs can SELECT the same head-of-queue row; only one CAS wins. Before the
+    lost-race retry, each loser returned None ? which `_drain_due_jobs` reads as "queue
+    empty" ? so a single tick stranded runnable jobs (observed 2/8 claimed). Now every
+    claimer must walk away with a job while any is runnable: all N results are non-None and
+    the N claimed ids are distinct."""
+    db = Database(str(tmp_path / "race.db"))
+    n = 8
+    for i in range(n):
+        db.create_job(f"j{i}", f"/v{i}.mp4")
+
+    barrier = threading.Barrier(n)
+    results: list = [None] * n
+
+    def _claim(slot: int) -> None:
+        barrier.wait() # release all claimers together so they genuinely race on the head row
+        results[slot] = db.claim_due_job()
+
+    threads = [threading.Thread(target=_claim, args=(i,)) for i in range(n)]
+    for t in threads:
+        t.start()
+    for t in threads:
+        t.join()
+
+    claimed_ids = [r["id"] if r else None for r in results]
+    # No false-empty: with N runnable jobs, no claimer may come back empty-handed.
+    assert all(r is not None for r in results), f"lost-race leaked None: {claimed_ids}"
+    # No double-claim: the N winners are exactly the N distinct queued jobs.
+    assert sorted(claimed_ids) == [f"j{i}" for i in range(n)]
+    for i in range(n):
+        assert db.get_job(f"j{i}")["status"] == "running"
+    assert db.claim_due_job() is None # and None still means TRULY empty afterwards
+
+
+    # --- seconds_until_next_due (the wake-delay computation) -----

```

M backend/infrastructure/database_jobs.py
M tests/test_job_worker_loop.py

66. assistant (2026-07-04T09:06:20.819Z)

#468's diff is clean and correct. Now the other two.

67. user (2026-07-04T09:06:22.891Z)

```

diff --git a/backend/orchestration.py b/backend/orchestration.py
index 46be449..4f9d4ea 100644
--- a/backend/orchestration.py
+++ b/backend/orchestration.py
@@ -251,11 +251,20 @@ def _maybe_dispatch_remote(
    # A browser upload lands on THIS box's local disk; the Cloud Run job (a different
    machine)
    # can't read it. Stage it UP to the input bucket first, then dispatch from there
    (#461). Only
    # possible when a CLOUD input_root is configured; without one we genuinely can't stage
    ? fail.
+    # Classify the ROOT ITSELF (parse_uri), not storage.input_backend: the staged URI below
    is
+    # built by joining under input_root, and storage.put routes by
    parse_uri(staged_uri).backend ?

```

```

+         # so an explicit cloud root (gs://?, s3://?) stages fine with input_backend left at its
+         # default, while a bare/local-looking root can't stage no matter what input_backend
says
+         # (resolve_input_uri only consults input_backend when input_root is EMPTY, and an empty
root
+         # gives us nothing to build the staged URI under).
        input_root = (getattr(storage_cfg, "input_root", "") or "").strip()
-        input_backend = (
-            getattr(storage_cfg, "input_backend", "local") or "local"
-        ).strip()
-        if not input_root or input_backend == "local":
+        try:
+            root_backend = (
+                storage.parse_uri(input_root).backend if input_root else "local"
+            )
+        except ValueError: # unsupported scheme in input_root ? nowhere real to stage to
+            root_backend = "local"
+        if root_backend == "local":
            return _failed(
                "cloud-serving needs a cloud input (e.g. gs://?), or a cloud storage.input_root
to "
                "stage local uploads into; a path local to this machine can't be read by the
Cloud "
diff --git a/tests/test_api_cloud_dispatch.py b/tests/test_api_cloud_dispatch.py
index ee36ca1..7b67de0 100644
--- a/tests/test_api_cloud_dispatch.py
+++ b/tests/test_api_cloud_dispatch.py
@@ -274,6 +274,43 @@ def test_cloud_stages_local_upload_to_bucket(cloud_env, tmp_path,
monkeypatch):
    assert row is not None and row["filepath"] == dest

+def test_cloud_stages_when_input_root_is_cloud_and_input_backend_default(
+    cloud_env, tmp_path, monkeypatch
+):
+    """input_root='gs://?' with input_backend left at its DEFAULT ('local') is a valid cloud
config ?
+    the storage layer classifies an explicit root by its scheme (resolve_input_uri joins under
+    input_root before ever consulting input_backend), so the staging gate must too (PR #467
review).
+    Regression: the gate previously required input_backend != 'local' and hard-failed this
shape."""
+    _db, _bucket, _mp = cloud_env
+    m.global_config.storage.input_root = "gs://in-bucket/videos"
+    # input_backend deliberately NOT set ? stays at the model default 'local'
+    assert m.global_config.storage.input_backend == "local"
+    puts: list = []
+
+    def _fake_put(src, uri, **kw):
+        puts.append((str(src), uri))
+        return uri
+
+    monkeypatch.setattr(m.storage, "put", _fake_put)
+    fake = _FakeCloudClient()
+    _inject_fake(monkeypatch, fake)
+    local_path = str(tmp_path / "upload.mp4")
+
+    out = m._execute_processing(
+        m.ProcessRequest(video_path=local_path, segmenter="wasb_hybrid")
+    )
+
+    assert out["status"] == "success" # NOT the 'needs a cloud input' pre-dispatch failure
+    assert len(puts) == 1
+    src, dest = puts[0]
+    assert src == local_path

```

```

+ assert dest.startswith("gs://in-bucket/videos/") and dest.endswith("/upload.mp4")
+ # dispatch continued from the STAGED gs:// URI
+ assert len(fake.calls) == 1
+ argv = fake.calls[0]
+ assert argv[argv.index("--video-uri") + 1] == dest
+
+
def test_cloud_local_upload_without_input_root_still_fails(cloud_env, tmp_path, monkeypatch):
    """No cloud input_root configured ? we genuinely can't stage a local upload ? hard-fail (no
    dispatch), rather than dispatching a path the Cloud Run job can't read."""
M backend/orchestration.py
M tests/test_api_cloud_dispatch.py

```

68. user (2026-07-04T09:06:33.009Z)

(PowerShell completed with no output)

69. user (2026-07-04T09:06:38.222Z)

```

C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py:75: def
resolve_input_uri(
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-76- raw: str,
*, input_backend: str = "local", input_root: str = ""
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-77-) -> str:
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-78-
"""Resolve a user-supplied input reference into a storage URI (COMPUTE_DECOUPLED_SERVING M2).
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-79-
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-80-
Precedence (first match wins):
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-81- 1. An
explicit ``scheme://`` on ``raw`` (``gs://``, ``s3://``, ``local://``, ``gdrive://``) ? as-is.
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-82- 2. An
ABSOLUTE local path (``/data/x.mp4``, ``C:\\?``, ``C:/?``) ? as-is (always local).
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-83- 3. A
bare/relative name + a non-empty ``input_root`` ? joined under it
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-84-
(``gs://bucket/videos`` + ``x.mp4`` ? ``gs://bucket/videos/x.mp4``).
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-85- 4. A
bare/relative name + ``input_backend != "local"`` ? prefixed (``gcs`` + ``bucket/x`` ?
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-86-
``gcs://bucket/x``).
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-87- 5.
Otherwise ? ``raw`` unchanged (a relative/bare LOCAL path ? today's behaviour).
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-88-
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-89- So the
default (``input_backend="local"`` , ``input_root=""``) is an identity passthrough for every
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-90- path used
today, preserving byte-for-byte behaviour.
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-91- ""
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-92- s =
str(raw)
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-93- if
_SCHEME_RE.match(s): # 1. explicit scheme wins
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-94- return s
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-95- # 2. an
anchored LOCAL path is always local, never joined. Detect this OS-INDEPENDENTLY ?
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-96- #
os.path.isabs is platform- AND Python-version-dependent (on Windows + py3.13 a POSIX-style
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-97- #
"/data/x" is NOT absolute). Anchored == a leading "/" or "\", or a Windows drive prefix "X:".
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-98- if
s.startswith(("/", "\\")) or (len(s) >= 2 and s[1] == ":" and s[0].isalpha()):
C:\Users\avidu\Projects\kheIsutra-guru\wt-upload-staging\backend\storage\ref.py-99- return s

```

```

C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-100-   if
input_root: # 3. join a relative name under the configured root
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-101-
return input_root.rstrip("/") + "/" + s.lstrip("/")
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-102-   if (
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-103- input_backend and input_backend != "local"
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-104- ): # 4.
prefix a bucket/key with the declared backend
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-105- return f"{input_backend}://{s}"
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-staging\backend\storage\ref.py-106- return s
# 5. bare/relative local path ? unchanged

```

70. user (2026-07-04T09:06:50.052Z)

```

diff --git a/backend/storage/__init__.py b/backend/storage/__init__.py
index a5958cc..38889de 100644
--- a/backend/storage/__init__.py
+++ b/backend/storage/__init__.py
@@ -206,12 +206,16 @@ def signed_url(
    uri: str,
    *,
    config: Optional[Dict[str, Any]] = None,
-   ttl: int = 3600,
+   ttl: Optional[int] = None,
    method: str = "GET",
) -> str:
    """A time-limited signed URL for ``uri`` ? the client fetches bytes DIRECTLY from the
store, off
    the app's response path (so a large object isn't streamed back through the server; #463).
Backends
-   without object signing (e.g. local) raise ``NotImplementedError``.
+   without object signing (e.g. local) raise ``NotImplementedError``.
+
+   ``ttl=None`` (the default) defers to the backend's configured ``signed_url_ttl`` from
+   ``config`` ? a hardcoded default here would silently shadow the operator's setting (#469).
+   An explicit ``ttl`` overrides it for callers that need a specific expiry."""
    ref = parse_uri(uri)
    return get_storage(ref.backend, config=config).signed_url(ref, ttl=ttl, method=method)

diff --git a/backend/storage/base.py b/backend/storage/base.py
index 74fea15..2a82245 100644
--- a/backend/storage/base.py
+++ b/backend/storage/base.py
@@ -54,9 +54,11 @@ class StorageBackend(ABC):
    def delete(self, ref: StorageRef) -> None: ...

    def signed_url(
-       self, ref: StorageRef, *, ttl: int = 3600, method: str = "GET"
+       self, ref: StorageRef, *, ttl: "int | None" = None, method: str = "GET"
    ) -> str:
-       """A time-limited URL handing the bytes to a browser/LLM. Not every backend can
+       """A time-limited URL handing the bytes to a browser/LLM. ``ttl=None`` (the default)
+       means "use the backend's configured ``signed_url_ttl``" ? so operator config wins
+       unless a caller explicitly needs a different expiry (#469). Not every backend can
+       sign (``local`` cannot; Drive only emulates) ? callers must treat
+       ``NotImplementedError`` as expected and fall back to ``fetch_to_local``."""
        raise NotImplementedError(f"{type(self).__name__} cannot produce signed URLs")

diff --git a/backend/storage/gcs.py b/backend/storage/gcs.py
index 261d72c..a0834ce 100644
--- a/backend/storage/gcs.py
+++ b/backend/storage/gcs.py
@@ -73,7 +73,7 @@ class GCSStorage(StorageBackend):
    def delete(self, ref) -> None:

```

```

        self._blob(ref).delete()

-     def signed_url(self, ref, *, ttl=3600, method="GET") -> str:
+     def signed_url(self, ref, *, ttl=None, method="GET") -> str:
        blob = self._blob(ref)
        exp = timedelta(seconds=ttl or self._ttl)
        creds = getattr(self._client, "_credentials", None)
diff --git a/backend/storage/s3.py b/backend/storage/s3.py
index 6c09d15..a9ae7e0 100644
--- a/backend/storage/s3.py
+++ b/backend/storage/s3.py
@@ -87,7 +87,7 @@ class S3Storage(StorageBackend):
     def delete(self, ref) -> None:
        self._client.delete_object(Bucket=ref.bucket, Key=ref.key)

-     def signed_url(self, ref, *, ttl=3600, method="GET") -> str:
+     def signed_url(self, ref, *, ttl=None, method="GET") -> str:
        op = "get_object" if method.upper() == "GET" else "put_object"
        return self._client.generate_presigned_url(
            op,
diff --git a/tests/test_storage.py b/tests/test_storage.py
index 8d7b427..11f1875 100644
--- a/tests/test_storage.py
+++ b/tests/test_storage.py
@@ -250,7 +250,10 @@ class _FakeBlob:
        self.store.pop((self.bucket, self.name), None)

        def generate_signed_url(self, version, expiration, method):
-         return f"https://gcs.example/{self.bucket}/{self.name}?version={version}"
+         # Expose the expiration so TTL-plumbing tests can assert the configured
+         # signed_url_ttl actually reaches the signer (#469).
+         ttl = int(expiration.total_seconds())
+         return f"https://gcs.example/{self.bucket}/{self.name}?version={version}&ttl={ttl}"

class _FakeGCSClient:
@@ -309,6 +312,39 @@ def test_gcs_backend_roundtrip_with_fake_client(tmp_path):
    assert not be.exists(ref)

+## ----- module signed_url helper: TTL plumbing
+## (#469)
+## The helper's ttl default must be None so the backend's configured signed_url_ttl applies ?
+## a hardcoded helper default would silently shadow the operator's config for every caller
+## that doesn't pass an explicit ttl (e.g. /api/highlights via signed_reel_url).
+
+def test_module_signed_url_honors_configured_gcs_ttl():
+    # Reviewer repro on #469: config={'gcs': {'signed_url_ttl': 123}} must reach the signer.
+    url = storage.signed_url(
+        "gcs://bucket/highlights/vid/reel.mp4",
+        config={"gcs": {"client": _FakeGCSClient(), "signed_url_ttl": 123}},
+        method="GET",
+    )
+    assert "ttl=123" in url
+
+def test_module_signed_url_honors_configured_s3_ttl():
+    url = storage.signed_url(
+        "s3://bucket/highlights/vid/reel.mp4",
+        config={"s3": {"client": _FakeS3Client(), "signed_url_ttl": 456}},
+    )
+    assert "X-Amz-Expires=456" in url
+

```

```

+def test_module_signed_url_explicit_ttl_overrides_config():
+    url = storage.signed_url(
+        "gs://bucket/x",
+        config={"gcs": {"client": _FakeGCSClient(), "signed_url_ttl": 123}},
+        ttl=999,
+    )
+    assert "ttl=999" in url
+
+
+    # ----- Google Drive (local mount)
+    # The backend reads a locally MOUNTED Google Drive (Google Drive for Desktop). A tmp_path
+    # folder stands in for the mount's "My Drive" root ? no network, no Drive API, no extra deps.
diff --git a/tests/test_upload_download.py b/tests/test_upload_download.py
index b258a13..14b78d6 100644
--- a/tests/test_upload_download.py
+++ b/tests/test_upload_download.py
@@ -329,6 +329,35 @@ def test_highlights_redirects_to_signed_url_when_reel_in_bucket(env,
monkeypatch
    assert seen.get("sign_cfg") is not None and seen["sign_cfg"].get("output_root") ==
"gs://b/out"

+def test_highlights_signed_url_honors_configured_gcs_ttl(env):
+    """/api/highlights must sign with the operator's storage.gcs.signed_url_ttl, not a
+    hardcoded
+    3600 (#469 review). Exercises the REAL storage layer end-to-end (config ? get_storage ?
+    GCSStorage.signed_url) with an injected fake client ? only the GCS SDK is faked."""
+
+    class _Blob:
+        def __init__(self, bucket, key):
+            self.bucket, self.key = bucket, key
+
+        def exists(self):
+            return self.key == "out/highlights/vid469/reel.mp4"
+
+        def generate_signed_url(self, version, expiration, method):
+            ttl = int(expiration.total_seconds())
+            return f"https://gcs.example/{self.bucket}/{self.key}?ttl={ttl}"
+
+    class _Client:
+        def bucket(self, name):
+            return type("_B", (), {"blob": lambda _s, key: _Blob(name, key)})(())
+
+    client, db, _tmp, cfg = env
+    cfg.storage.output_root = "gs://b/out"
+    cfg.storage.gcs = {"client": _Client(), "signed_url_ttl": 123}
+    db.add_video("vid469", "gs://b/in/x.mp4", "processed", [])
+    r = client.get("/api/highlights/vid469/reel.mp4", follow_redirects=False)
+    assert r.status_code == 307
+    assert r.headers["location"].endswith("?ttl=123")
+
+
+def test_highlights_falls_back_to_local_when_no_bucket_reel(env, monkeypatch):
+    """No mirrored bucket object (or a signing hiccup) ? serve the local file, unchanged
+    behaviour."""
+    client, db, _tmp, cfg = env
+    M backend/storage/__init__.py
+    M backend/storage/base.py
+    M backend/storage/gcs.py
+    M backend/storage/s3.py
+    M tests/test_storage.py
+    M tests/test_upload_download.py

```

All three diffs are correct and in-style. Committing each on its branch (with branch/HEAD re-verification), then pushing.

72. user (2026-07-04T09:07:14.336Z)

fix/drain-concurrency
8534077 review: gate concurrency on effective compute path, not the server default (#466)
[fix/drain-concurrency 21ee141] fix(jobs): retry claim_due_job on a lost CAS race so None means truly empty
2 files changed, 71 insertions(+), 17 deletions(-)
21ee141 fix(jobs): retry claim_due_job on a lost CAS race so None means truly empty

73. user (2026-07-04T09:07:18.574Z)

fix/upload-gcs-staging
d3a9629 review: thread active storage config into the staging put (#461)
[fix/upload-gcs-staging c9bf37e] fix(serving): staging gate classifies input_root by scheme, not input_backend
2 files changed, 50 insertions(+), 4 deletions(-)
c9bf37e fix(serving): staging gate classifies input_root by scheme, not input_backend

74. user (2026-07-04T09:07:22.580Z)

fix/highlights-signed-url
d3b0e64 review: thread active storage config into mirror/exists/sign; extract helper (#463)
[fix/highlights-signed-url ce450e5] fix(storage): signed_url helper defers to the backend's configured signed_url_ttl
6 files changed, 78 insertions(+), 7 deletions(-)
ce450e5 fix(storage): signed_url helper defers to the backend's configured signed_url_ttl

75. user (2026-07-04T09:07:37.098Z)

To <https://github.com/Khelsutra/badminton-highlight-indexer.git>
8534077..21ee141 fix/drain-concurrency -> fix/drain-concurrency
To <https://github.com/Khelsutra/badminton-highlight-indexer.git>
d3a9629..c9bf37e fix/upload-gcs-staging -> fix/upload-gcs-staging
To <https://github.com/Khelsutra/badminton-highlight-indexer.git>
d3b0e64..ce450e5 fix/highlights-signed-url -> fix/highlights-signed-url

76. assistant (2026-07-04T09:07:43.802Z)

All three pushed. Now posting the replies to the reviewer on each PR.

77. user (2026-07-04T09:08:09.134Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/468#issuecomment-4881411828>
<https://github.com/Khelsutra/badminton-highlight-indexer/pull/467#issuecomment-4881411894>
<https://github.com/Khelsutra/badminton-highlight-indexer/pull/469#issuecomment-4881411950>

78. user (2026-07-04T09:08:18.542Z)

Updated task #1 status

79. user (2026-07-04T09:08:23.993Z)

#467: MERGEABLE UNSTABLE head=c9bf37e
#468: MERGEABLE UNSTABLE head=21ee141
#469: MERGEABLE UNSTABLE head=ce450e5
#470: MERGEABLE UNSTABLE head=9ed5d42

80. user (2026-07-04T09:08:47.031Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-

state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

81. user (2026-07-04T09:09:26.873Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

82. user (2026-07-04T09:09:39.492Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

83. user (2026-07-04T09:09:52.650Z)

```
Exit code 1
--- PR #467 ---
pass: 10
pending: 1
--- PR #468 ---
pass: 10
pending: 1
--- PR #469 ---
pass: 10
pending: 1
--- PR #470 ---
fail: 1
pass: 10
```

84. assistant (2026-07-04T09:09:57.875Z)

PR #470 has a CI failure ? let me see which check and why.

85. user (2026-07-04T09:10:01.146Z)

```
Exit code 1
Full test suite (offline, mocked)      fail      2m39s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28701060041/job/85119025538
```

86. user (2026-07-04T09:10:16.306Z)

Full test suite (offline, mocked)	Run full suite (with coverage floor)
2026-07-04T08:53:31.2768857Z	assert
"FROM nvidia/cuda" in text #	
GPU base (CUDA)	
Full test suite (offline, mocked)	Run full suite (with coverage floor)
2026-07-04T08:53:31.2769549Z	assert
"ffmpeg" in text #	
stitch/decode	
Full test suite (offline, mocked)	Run full suite (with coverage floor)
2026-07-04T08:53:31.2770439Z	assert
"fonts-dejavu" in text # the	
watermark filter font	
Full test suite (offline, mocked)	Run full suite (with coverage floor)
2026-07-04T08:53:31.2771300Z	assert
"torch==1.11.0+cu113" in text #	
the verified WASB stack	
Full test suite (offline, mocked)	Run full suite (with coverage floor)
2026-07-04T08:53:31.2772087Z	assert
"WASB-SBDT" in text # the	
shuttle detector repo	
Full test suite (offline, mocked)	Run full suite (with coverage floor)

```

2026-07-04T08:53:31.2772891Z          assert
'ENTRYPOINT ["python3", "-m", "backend.worker"]'
in text
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.2773705Z          assert
"WASB_PYTHON=/opt/conda/envs/wasb/bin/python" in text
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.2778035Z          # on the
first gs:// fetch (ModuleNotFoundError: google.cloud) or
on dispatch (google-cloud-run
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.2780673Z >          assert
".[storage-gcs,cloud-run]" in text
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.2783137Z E          assert
'.[storage-gcs,cloud-run]' in '# Cloud
Run GPU **Job** image for
the rally worker (COMPUTE_DECOUPLED_SERVING
M6/M7,
ADR-012/D16).\n#\n# Runs `python...? --out-uri
gs:/// --done-uri gs:/// --segmenter
wasb_hybrid --set ?\nENTRYPOINT
["python3", "-m", "backend.worker"]\n'
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.2785277Z
tests/test_compute_backend.py:696: AssertionError
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.8559829Z FAILED
tests/test_compute_backend.py::test_dockerfile_has_required_directives -
assert '.[storage-gcs,cloud-run]' in '#
Cloud
Run GPU **Job** image for
the rally worker (COMPUTE_DECOUPLED_SERVING
M6/M7, ADR-012/D16).\n#\n# Runs `python...?
--out-uri gs:/// --done-uri gs:/// --
segmenter wasb_hybrid --set ?\nENTRYPOINT
["python3", "-m", "backend.worker"]\n'
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:31.8562445Z 1 failed, 1622
passed, 13 skipped, 8 warnings
in 82.82s (0:01:22)
Full test suite (offline, mocked)      Run full suite (with coverage floor)
2026-07-04T08:53:33.3511333Z ##[error]Process
completed with exit code 1.

```

87. user (2026-07-04T09:10:28.508Z)

```

assert "WASB_PYTHON=/opt/conda/envs/wasb/bin/python" in text
# The cloud worker reads the gs:// input + writes the gs:// outputs/done-marker through
# backend.storage (storage-gcs); and when this image is reused as the deployed control plane
it
# dispatches the L4 Job via google.cloud.run_v2 (cloud-run). A bare `pip install .` boots
but dies
# on the first gs:// fetch (ModuleNotFoundError: google.cloud) or on dispatch (google-cloud-
run
# missing) ? real Cloud Run failures the mock-client tests can't see. Lock both extras in
structurally.
assert ".[storage-gcs,cloud-run]" in text

```

```

def test_dockerfile_uses_micromamba_not_anaconda_tos():
    """P3a: the WASB env is built with micromamba + conda-forge (BSD, no ToS), NOT Miniconda +

```

88. user (2026-07-04T09:10:34.949Z)

```

670         )
671         assert rc == 2
672         assert not (out / "_dispatch").exists()
673
674
675         # ----- Dockerfile structural guard
676         (no build)
677
678         def test_dockerfile_has_required_directives():
679             """CI can't build a CUDA image, so guard the Dockerfile's load-bearing directives
680             structurally:
681             the GPU base, ffmpeg + DejaVu fonts (the watermark), the WASB conda env, and the
682             worker entrypoint."""
683             from pathlib import Path
684
685             text = Path("deploy/cloudrun/Dockerfile").read_text(encoding="utf-8")
686             assert "FROM nvidia/cuda" in text # GPU base (CUDA)
687             assert "ffmpeg" in text # stitch/decode
688             assert "fonts-dejavu" in text # the watermark filter font
689             assert "torch==1.11.0+cu113" in text # the verified WASB stack
690             assert "WASB-SBDT" in text # the shuttle detector repo
691             assert 'ENTRYPOINT ["python3", "-m", "backend.worker"]' in text
692             assert "WASB_PYTHON=/opt/conda/envs/wasb/bin/python" in text
693             # The cloud worker reads the gs:// input + writes the gs:// outputs/done-marker
694             through
695             # backend.storage (storage-gcs); and when this image is reused as the deployed
696             control plane it
697             # dispatches the L4 Job via google.cloud.run_v2 (cloud-run). A bare `pip install .`
698             boots but dies
699             # on the first gs:// fetch (ModuleNotFoundError: google.cloud) or on dispatch
700             (google-cloud-run
701             # missing) ? real Cloud Run failures the mock-client tests can't see. Lock both
702             extras in structurally.
703             assert ".[storage-gcs,cloud-run]" in text
704
705         def test_dockerfile_uses_micromamba_not_anaconda_tos():

```

89. user (2026-07-04T09:10:47.221Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-dockerfile-auth\tests\test_compute_backend.py has been updated successfully. (file state is current in your context ? no need to Read it back)

90. user (2026-07-04T09:12:47.848Z)

-- Docs: <https://docs.pytest.org/en/stable/how-to/capture-warnings.html>
1626 passed, 10 skipped, 8 warnings in 110.40s (0:01:50)

91. user (2026-07-04T09:13:27.085Z)

All checks passed!
backend\providers__init__.py:9: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 147 source files

92. assistant (2026-07-04T09:13:36.352Z)

All gates green. Committing the contract-test update and pushing.

93. user (2026-07-04T09:13:50.308Z)

fix/dockerfile-auth-extra
9ed5d42 deploy(cloudrun): bake the auth extra (PyJWT) into the base image so an edge-auth

control plane needs no derived overlay
[fix/dockerfile-auth-extra c63d122] test: lock the auth extra into the Dockerfile structural guard
1 file changed, 6 insertions(+), 4 deletions(-)
To <https://github.com/Khelsutra/badminton-highlight-indexer.git>
9ed5d42..c63d122 fix/dockerfile-auth-extra -> fix/dockerfile-auth-extra
<https://github.com/Khelsutra/badminton-highlight-indexer/pull/470#issuecomment-4881427247>

94. user (2026-07-04T09:14:02.946Z)

```
{"message":"Successfully stopped task: b8qxd3ys (declare -A fp\nwhile true; do\n  for pr in 465 467 468 469; do\n    data=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr\" --jq '\"\\(.state)|\\(.merged)\"' 2>/dev/null) || continue\n    cm=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/issues/$pr/comments\" --jq 'length as $n | if $n==0 then \"0|none|none\" else \"\\($n)|\\(.[-1].created_at)|\\(.[-1].user.login):\\(.[-1].body[0:140] | gsub(\"\\n\"; \" \")\" end' 2>/dev/null) || continue\n    rv=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr/reviews\" --jq 'length as $n | if $n==0 then \"0\" else \"\\($n)|\\(.[-1].state)|\\(.[-1].user.login)\" end' 2>/dev/null) || continue\n    cur=\"$data|$cm|$rv\"\n    if [ -n \"${fp[$pr]}\" ] && [ \"${fp[$pr]}\" != \"$cur\" ]; then\n      echo \"PR #$pr CHANGED -> state/merged=$data | comments=$cm | reviews=$rv\"\n      fi\n      fp[$pr]=$cur\n      done\n      sleep 150\ndone\"),\"task_id\":\"b8qxd3ys\",\"task_type\":\"local_bash\",\"command\":\"declare -A fp\nwhile true; do\n  for pr in 465 467 468 469; do\n    data=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr\" --jq '\"\\(.state)|\\(.merged)\"' 2>/dev/null) || continue\n    cm=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/issues/$pr/comments\" --jq 'length as $n | if $n==0 then \"0|none|none\" else \"\\($n)|\\(.[-1].created_at)|\\(.[-1].user.login):\\(.[-1].body[0:140] | gsub(\"\\n\"; \" \")\" end' 2>/dev/null) || continue\n    rv=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr/reviews\" --jq 'length as $n | if $n==0 then \"0\" else \"\\($n)|\\(.[-1].state)|\\(.[-1].user.login)\" end' 2>/dev/null) || continue\n    cur=\"$data|$cm|$rv\"\n    if [ -n \"${fp[$pr]}\" ] && [ \"${fp[$pr]}\" != \"$cur\" ]; then\n      echo \"PR #$pr CHANGED -> state/merged=$data | comments=$cm | reviews=$rv\"\n      fi\n      fp[$pr]=$cur\n      done\n      sleep 150\ndone\"}
```

95. user (2026-07-04T09:14:07.671Z)

Monitor started (task bpriyemeyi, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

96. user (2026-07-04T09:14:19.132Z)

```
#467: pass=11 pending=0 fail=0
#468: pass=11 pending=0 fail=0
#469: pass=11 pending=0 fail=0
```

97. assistant (2026-07-04T09:14:33.119Z)

All three fix PRs are 11/11 green on their new heads. Let me arm a one-shot watcher for #470's re-run to complete, then summarize.

98. user (2026-07-04T09:14:35.814Z)

Command running in background with ID: b9ix3rveu. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\b9ix3rveu.output. You will be notified when it completes. To check interim output, use Read on that file path.

99. user (2026-07-04T09:15:42.033Z)

From <https://github.com/Khelsutra/badminton-highlight-indexer>
4847232..66d7d6f master -> origin/master
--- project id / aud / team domain occurrences in repo docs ---
c980da67: 1 hits
cloudflareaccess: 1 hits

gen-lang-client: 1 hits
khelsutra-rally-serving: 5 hits

100. assistant (2026-07-04T09:16:06.777Z)

All the identifiers are already public in committed runlogs ? safe as defaults. Master also just moved to the #465 squash (66d7d6f), pulled. Creating the generator branch now.

101. user (2026-07-04T09:16:10.102Z)

Preparing worktree (new branch 'feat/cloudrun-service-manifest')
branch 'feat/cloudrun-service-manifest' set up to track 'origin/master'.
HEAD is now at 66d7d6f fix(hosted): allowed_video_root jail must not reject storage URIs (gs://) (#465)
66d7d6f fix(hosted): allowed_video_root jail must not reject storage URIs (gs://) (#465)

102. assistant (2026-07-04T09:16:48.269Z)

```
"""Generate the control-plane Cloud Run *service manifest* (COMPUTE_DECOUPLED_SERVING M8/M9, #473).
```

The A1 alpha deploy drove `rally-control-plane` from a hand-generated YAML that lived in a session scratchpad ? this commits the generator so the deploy is reproducible from a clean clone. It emits a Knative service manifest for::

```
python deploy/cloudrun/gen_service_yaml.py --edge-auth off -o service.yaml  
gcloud run services replace service.yaml --region asia-southeast1
```

Why a MANIFEST (`services replace`) and not `gcloud run deploy` flags: the container command is a /bin/bash wrapper carrying shell metacharacters (pipes, redirects, &&) ? on Windows the gcloud cmd-shim mangles those through `--command/--args`, while a YAML round-trips them verbatim. Run gcloud from PowerShell, not git-bash (MSYS rewrites `/tmp`-style paths).

The app is configured through a config.json baked INTO the manifest: the wrapper base64-decodes it to `/tmp/apphome/config.json` and exports `RALLY\_APP\_HOME=/tmp/apphome` before exec'ing the server, so one image serves every config shape (config.json wins over the cloud-serving preset ? backend/config/loader.py). `--edge-auth` is REQUIRED and explicit: `off` is only for the identity-token smoke/e2e window; redeploy with `on` (the Cf-Access JWT verify, M9) before the service faces testers again.

Deliberate manifest choices (learned on the A1 deploy):

```
* `run.googleapis.com/cpu-throttling: "false"` + minScale 1 ? the in-process job worker drains  
  on a background thread; without always-allocated CPU a 202'd compile stalls (README §3a).  
* maxScale DEFAULTS TO 1 ? the jobs/videos DB is per-instance ephemeral SQLite, so a second  
  instance splits the queue and `/api/jobs/{id}` polls (issue #471, option 1). Raise it only  
  once state is shared.  
* `CLOUD_RUN_JOB` is NOT set ? it's a reserved runtime env name and the manifest is rejected  
  with it; the code default ("rally-worker") is already correct.  
* No secrets here: bucket/project/aud/team-domain are the same non-secret identifiers the  
  committed runlogs carry; the Gemini key stays in Secret Manager (the alpha runs  
  `skip_ai_handoff=true` and never needs it).
```

```
from __future__ import annotations
```

```
import argparse  
import base64  
import json  
import sys
```

```
DEFAULT_PROJECT = "gen-lang-client-0306026826"  
DEFAULT_REGION = "asia-southeast1"  
DEFAULT_SERVICE = "rally-control-plane"  
DEFAULT_SERVING_BUCKET = "gs://khelsutra-rally-serving"
```

```

DEFAULT_CF_TEAM_DOMAIN = "https://khehsutra.cloudflareaccess.com"
DEFAULT_CF_ACCESS_AUD = "c980da67ed888ffff935fcd913ae1eb4da63f9d65805d6e1c5ca76f92f517082b"

```

```

def build_config(args: argparse.Namespace) -> dict:
    """The app config the wrapper materializes at RALLY_APP_HOME/config.json."""
    bucket = args.serving_bucket.rstrip("/")
    return {
        "deployment": {"profile": "cloud-serving", "compute_target": "cloud_run"},
        # WASB-only alpha: candidate windows ARE the rallies; no Gemini key in this service.
        "indexing": {"skip_ai_handoff": True},
        "storage": {
            "backend": "gcs",
            "output_root": bucket,
            "input_backend": "gcs",
            "input_root": f"{bucket}/videos",
        },
        "security": {
            "edge_auth_enabled": args.edge_auth == "on",
            "cf_team_domain": args.cf_team_domain,
            "cf_access_aud": args.cf_access_aud,
            "allowed_video_root": "/tmp/apphome/output",
        },
    }

```

```

def build_yaml(args: argparse.Namespace) -> str:
    """Render the Knative service manifest for `gcloud run services replace`."""
    cfg_b64 = base64.b64encode(
        json.dumps(build_config(args), separators=(",", ":")).encode("utf-8")
    ).decode("ascii")
    # mkdir BEFORE the redirect (the redirect can't create the directory), then exec so the
    # server is PID 1 and Cloud Run's SIGTERM reaches it directly.
    wrapper = (
        "mkdir -p /tmp/apphome && "
        f"printf %s '{cfg_b64}' | base64 -d > /tmp/apphome/config.json && "
        "export RALLY_APP_HOME=/tmp/apphome && "
        "exec python3 -m backend.main --server --host 0.0.0.0 --port 8080"
    )
    return f"""\

```

```

apiVersion: serving.knative.dev/v1
kind: Service
metadata:
  name: {args.service}
  labels:
    cloud.googleapis.com/location: {args.region}
spec:
  template:
    metadata:
      annotations:
        autoscaling.knative.dev/minScale: "{args.min_instances}"
        autoscaling.knative.dev/maxScale: "{args.max_instances}"
        run.googleapis.com/cpu-throttling: "false"
    spec:
      timeoutSeconds: {args.timeout}
      containers:
        - image: {args.image}
          command: ["/bin/bash"]
          args: ["-c", "{wrapper}"]
          ports:
            - containerPort: 8080
          resources:
            limits:
              cpu: "{args.cpu}"
              memory: {args.memory}

```

```

env:
  - name: DEPLOYMENT_PROFILE
    value: cloud-serving
  - name: CLOUD_RUN_REGION
    value: {args.region}
  - name: GOOGLE_CLOUD_PROJECT
    value: {args.project}
  - name: RALLY_ALLOWED_HOSTS
    value: "*"
"""

def main(argv: "list[str] | None" = None) -> int:
    p = argparse.ArgumentParser(description=__doc__.splitlines()[0])
    p.add_argument(
        "--edge-auth",
        choices=("on", "off"),
        required=True,
        help="REQUIRED and explicit: 'off' only for the identity-token smoke/e2e window; "
        "redeploy 'on' (M9 Cf-Access verify) before testers touch it again.",
    )
    p.add_argument("--project", default=DEFAULT_PROJECT)
    p.add_argument("--region", default=DEFAULT_REGION)
    p.add_argument("--service", default=DEFAULT_SERVICE)
    p.add_argument(
        "--image",
        default=None,
        help="Container image; default = the project's rally-worker:latest (post-#470 it "
        "carries cloud-run + auth, so the worker image IS the control-plane image).",
    )
    p.add_argument("--serving-bucket", default=DEFAULT_SERVING_BUCKET)
    p.add_argument("--cf-team-domain", default=DEFAULT_CF_TEAM_DOMAIN)
    p.add_argument("--cf-access-aud", default=DEFAULT_CF_ACCESS_AUD)
    p.add_argument("--min-instances", type=int, default=1)
    p.add_argument(
        "--max-instances",
        type=int,
        default=1,
        help="Default 1: per-instance SQLite state splits across instances (issue #471).",
    )
    p.add_argument("--cpu", type=int, default=2)
    p.add_argument("--memory", default="4Gi")
    p.add_argument("--timeout", type=int, default=300)
    p.add_argument("-o", "--output", default="-", help="Output path, or '-' for stdout.")
    args = p.parse_args(argv)
    if args.image is None:
        args.image = f"{args.region}-docker.pkg.dev/{args.project}/rally/rally-worker:latest"

    text = build_yaml(args)
    if args.output == "-":
        sys.stdout.write(text)
    else:
        with open(args.output, "w", encoding="utf-8", newline="\n") as f:
            f.write(text)
    return 0

if __name__ == "__main__":
    raise SystemExit(main())

```

103. user (2026-07-04T09:16:48.301Z)

File created successfully at: C:\Users\avidu\Projects\kheIsutra-guru\wt-yaml-gen\deploy\cloudrun\gen_service_yaml.py (file state is current in your context ? no need to Read it back)

104. user (2026-07-04T09:16:54.232Z)

```
1          1 Syntax sweep + import smoke (no GPU stack): pass
2          1 Ruff lint: pass
3          1 Mypy (lenient baseline): pass
4          1 Markdown link check (live docs): pass
5          1 Leak scan (PII / attribution guard): pass
6          1 Golden fixtures cross-OS (windows-latest): pass
7          1 Golden fixtures cross-OS (ubuntu-latest): pass
8          1 Golden fixtures cross-OS (macos-latest): pass
9          1 Full test suite (offline, mocked): pass
10         1 Dependency license scan (no copyleft in runtime deps): pass
11         1 Built wheel smoke (install + launch): pass
12     PR470 CI: ALL GREEN
13
```

105. assistant (2026-07-04T09:17:17.133Z)

```
"""Structural guard for the control-plane service-manifest generator (#473).
```

CI can't `gcloud run services replace` anything, so ? like the Dockerfile guard in `test_compute_backend.py` ? lock the manifest's load-bearing shape structurally: the baked `config.json` must round-trip through the base64 wrapper byte-exactly (that's the piece a hand-edit silently corrupts), the always-allocated-CPU/scale annotations must survive (README §3a: without them a 202'd compile stalls), and the reserved ``CLOUD_RUN_JOB`` env name must never re-appear (a manifest carrying it is rejected at deploy time).

The generator is exercised through its real CLI (a subprocess), not an import ? deploy/ is not a package, and the CLI is exactly what a deployer runs.

```
"""
```

```
import base64
import json
import re
import subprocess
import sys
from pathlib import Path
```

```
_GEN = Path(__file__).resolve().parents[1] / "deploy" / "cloudrun" / "gen_service_yaml.py"
```

```
def _generate(*extra: str) -> str:
    out = subprocess.run(
        [sys.executable, str(_GEN), *extra],
        capture_output=True,
        text=True,
        check=True,
    )
    return out.stdout
```

```
def _baked_config(manifest: str) -> dict:
    """Decode the config.json the wrapper bakes into the manifest."""
    m = re.search(r"printf %s '([A-Za-z0-9+/=]+)' \\\| base64 -d", manifest)
    assert m, "wrapper must carry a base64 config payload"
    return json.loads(base64.b64decode(m.group(1)))
```

```
def test_manifest_bakes_the_cloud_serving_config():
    cfg = _baked_config(_generate("--edge-auth", "off"))
    assert cfg["deployment"] == {"profile": "cloud-serving", "compute_target": "cloud_run"}
    # WASB-only alpha: no Gemini key in this service, so the handoff must be off.
    assert cfg["indexing"]["skip_ai_handoff"] is True
    assert cfg["storage"]["backend"] == "gcs"
    assert cfg["storage"]["output_root"] == "gs://khehsutra-rally-serving"
```

```

assert cfg["storage"]["input_root"] == "gs://khehsutra-rally-serving/videos"
assert cfg["security"]["edge_auth_enabled"] is False
assert cfg["security"]["allowed_video_root"] == "/tmp/apphome/output"

def test_manifest_edge_auth_toggle_is_explicit():
    on = _baked_config(_generate("--edge-auth", "on"))
    assert on["security"]["edge_auth_enabled"] is True
    assert on["security"]["cf_team_domain"].startswith("https://")
    assert len(on["security"]["cf_access_aud"]) == 64 # the Access-app aud is a sha256 hex
    # --edge-auth is required: generating without choosing must fail, not default silently.
    rc = subprocess.run(
        [sys.executable, str(_GEN)], capture_output=True, text=True
    ).returncode
    assert rc != 0

def test_manifest_has_the_reliability_flags_and_no_reserved_env():
    text = _generate("--edge-auth", "off")
    # README §3a: the in-process drain needs always-allocated CPU + a warm instance.
    assert 'run.googleapis.com/cpu-throttling: "false"' in text
    assert 'autoscaling.knative.dev/minScale: "1"' in text
    # Issue #471: per-instance SQLite state ? a second instance splits the job queue.
    assert 'autoscaling.knative.dev/maxScale: "1"' in text
    # Reserved runtime env name ? a manifest carrying it is rejected at deploy time.
    assert "CLOUD_RUN_JOB" not in text
    assert "RALLY_ALLOWED_HOSTS" in text
    assert "containerPort: 8080" in text
    # The wrapper must create the config dir BEFORE redirecting into it, and exec the server.
    wrapper = re.search(r'args: \["-c", "(.+)"\]', text).group(1)
    assert wrapper.startswith("mkdir -p /tmp/apphome && ")
    assert "export RALLY_APP_HOME=/tmp/apphome" in wrapper
    assert wrapper.endswith("exec python3 -m backend.main --server --host 0.0.0.0 --port 8080")

```

106. user (2026-07-04T09:17:17.185Z)

File created successfully at: C:\Users\avidu\Projects\khehsutra-guru\wt-yaml-gen\tests\test_deploy_manifests.py (file state is current in your context ? no need to Read it back)

107. assistant (2026-07-04T09:17:29.566Z)

Now the README §3a pointer ? placed away from the region #470 edits, to avoid a cross-PR conflict.

108. user (2026-07-04T09:17:30.184Z)

```

66
67     The API server (`/api/process`, `/api/compile` ? #329) runs its **job worker IN-
PROCESS**: the
68     request returns `202` and a background thread drains the job (GPU detection dispatches
to the Job;
69     the reel stitch runs on the control plane itself). On Cloud Run that background thread
only makes
70     progress while the instance has CPU, so a scale-to-zero **service** deploy MUST set:
71     ``bash
72     gcloud run deploy rally-control-plane --image <image> --region $REGION \
73     --no-cpu-throttling \           # CPU stays allocated between requests ? the drain thread
finishes
74     --min-instances 1 \           # one always-warm instance drains reliably (cheap CPU;
NOT the D7 GPU pool)
75     --cpu 2 --memory 4Gi --port 8080 --allow-unauthenticated \
76     --set-env-vars DEPLOYMENT_PROFILE=cloud-serving,CLOUD_RUN_JOB=rally-
worker,CLOUD_RUN_REGION=$REGION,GOOGLE_CLOUD_PROJECT=$PROJECT

```

```
77      ```\n78      Without `--no-cpu-throttling`, a `/api/compile` returns 202 and then **stalls** ? the\nreel never\n79      finishes and startup crash-recovery would mark it terminal. `min-instances 1` does NOT\ncontradict
```

109. user (2026-07-04T09:17:45.827Z)

The file C:\\Users\\avidu\\Projects\\khelsutra-guru\\wt-yaml-gen\\deploy\\cloudrun\\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

110. assistant (2026-07-04T09:18:00.264Z)

Running the full gates in the generator worktree.

111. user (2026-07-04T09:18:03.941Z)

```
... [100%]\n3 passed in 0.39s
```

112. user (2026-07-04T09:20:39.685Z)

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html\n1637 passed, 10 skipped, 8 warnings in 110.38s (0:01:50)\nAll checks passed!\nSuccess: no issues found in 147 source files\n[check-md-links] clean ? no broken internal links in live docs.
```

113. user (2026-07-04T09:20:57.129Z)

```
feat/cloudrun-service-manifest\n66d7d6f fix(hosted): allowed_video_root jail must not reject storage URIs (gs://) (#465)\nwarning: in the working copy of 'deploy/cloudrun/gen_service_yaml.py', LF will be replaced by\nCRLF the next time Git touches it\nwarning: in the working copy of 'tests/test_deploy_manifests.py', LF will be replaced by CRLF\nthe next time Git touches it\n[feat/cloudrun-service-manifest 9de18d5] deploy(cloudrun): commit the control-plane service-\nmanifest generator (closes #473)\n 3 files changed, 262 insertions(+), 1 deletion(-)\n create mode 100644 deploy/cloudrun/gen_service_yaml.py\n create mode 100644 tests/test_deploy_manifests.py\n9de18d5 deploy(cloudrun): commit the control-plane service-manifest generator (closes #473)\nremote:\nremote: Create a pull request for 'feat/cloudrun-service-manifest' on GitHub by visiting:\nremote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/feat/cloudrun-\nservice-manifest\nremote:\nbranch 'feat/cloudrun-service-manifest' set up to track 'origin/feat/cloudrun-service-manifest'.\nTo https://github.com/Khelsutra/badminton-highlight-indexer.git\n * [new branch]      feat/cloudrun-service-manifest -> feat/cloudrun-service-manifest
```

114. user (2026-07-04T09:21:20.397Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/474>

115. user (2026-07-04T09:21:30.362Z)

```
{"message":"Successfully stopped task: bpryemeiy (declare -A fp\nwhile true; do\n  for pr in 467\n468 469 470; do\n    data=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr\"\n--jq '\"\\(.state)|\\(.merged)\"' 2>/dev/null) || continue\n    cm=$(gh api\n\"repos/Khelsutra/badminton-highlight-indexer/issues/$pr/comments\" --jq 'length as $n | if\n$==0 then \"0|none|none\" else \"\\($n)|\\(.[-1].created_at)|\\(.[-1].user.login):\n\\(.[-1].body[0:140] | gsub(\"\\n\"; \" \")\" end' 2>/dev/null) || continue\n    rv=$(gh api\n\"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr/reviews\" --jq 'length as $n | if $==0
```

```

then \"0\" else \"\\($n)|\\(. [-1].state)|\\(. [-1].user.login)\" end' 2>/dev/null) || continue\n
cur=\"$data|$cm|$rv\"\n    if [ -n \"${fp[$pr]}\" ] && [ \"${fp[$pr]}\" != \"$cur\" ]; then\n
echo \"PR #$pr CHANGED -> state/merged=$data | comments=$cm | reviews=$rv\"\n    fi\n
fp[$pr]=\"$cur\"\n done\n sleep
150\ndone)\", \"task_id\": \"bpryemeiy\", \"task_type\": \"local_bash\", \"command\": \"declare -A fp\nwhile true;
do\n    for pr in 467 468 469 470; do\n        data=$(gh api \"repos/Khelsutra/badminton-highlight-
indexer/pulls/$pr\" --jq '\"\\(.state)|\\(.merged)\"' 2>/dev/null) || continue\n        cm=$(gh api
\"repos/Khelsutra/badminton-highlight-indexer/issues/$pr/comments\" --jq 'length as $n | if
$n==0 then \"0|none|none\" else \"\\($n)|\\(. [-1].created_at)|\\(. [-1].user.login):
\\(. [-1].body[0:140] | gsub(\"\\n\"; \" \"))\" end' 2>/dev/null) || continue\n        rv=$(gh api
\"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr/reviews\" --jq 'length as $n | if $n==0
then \"0\" else \"\\($n)|\\(. [-1].state)|\\(. [-1].user.login)\" end' 2>/dev/null) || continue\n
cur=\"$data|$cm|$rv\"\n        if [ -n \"${fp[$pr]}\" ] && [ \"${fp[$pr]}\" != \"$cur\" ]; then\n
echo \"PR #$pr CHANGED -> state/merged=$data | comments=$cm | reviews=$rv\"\n        fi\n
fp[$pr]=\"$cur\"\n done\n sleep 150\ndone\"}

```

116. user (2026-07-04T09:21:35.286Z)

Monitor started (task betc1ot9n, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

117. user (2026-07-04T09:21:51.131Z)

The file c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

118. user (2026-07-04T09:22:02.043Z)

The file c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

119. user (2026-07-04T09:22:13.948Z)

```

#474 [MERGEABLE] deploy(cloudrun): commit the control-plane service-manifest generator (closes
#473)
#470 [MERGEABLE] deploy(cloudrun): bake the auth extra (PyJWT) into the base image (M9-ready
rebuilds)
#469 [MERGEABLE] fix(cloud-serving): serve large reels via a gs:// signed URL (#463)
#468 [MERGEABLE] fix(cloud-serving): parallelize the drain under cloud_run (#466)
#467 [MERGEABLE] fix(cloud-serving): stage browser-uploaded local files to gs:// before L4
dispatch (#461)

```

120. assistant (2026-07-04T09:22:40.225Z)

All caught up. Here's where things stand:

Summary

****Picked up the handoff and completed everything actionable; 5 PRs now await the owner's LGTM.****

****Merged:**** [#465](https://github.com/Khelsutra/badminton-highlight-indexer/pull/465) ? the owner LGTM'd it mid-session; I verified 11/11 CI checks + clean mergeability and squash-merged it. The other PRs stayed conflict-free after master moved.

****All three open review blockers fixed, pushed, and replied to**** (each fix ran through an adversarial verifier that proved the new regression test fails at the unfixed PR head, and the full suite + ruff + mypy were green on every branch):

- [#468](https://github.com/Khelsutra/badminton-highlight-indexer/pull/468) (`21ee141`) ? ``claim_due_job()`` now retries a lost CAS race instead of returning ``None``, so a loser can't masquerade as "queue empty". The 8-claimer barrier test failed 5/5 without the fix and passed 15/15 with it.

- [#467](https://github.com/Khelsutra/badminton-highlight-indexer/pull/467) (`c9bf37e`) ? the staging gate now classifies `input\_root` by its scheme via `parse\_uri` instead of trusting `input\_backend`; the reviewer's gs://-root-with-default-backend repro now stages and dispatches.
- [#469](https://github.com/Khelsutra/badminton-highlight-indexer/pull/469) (`ce450e5`) ? `storage.signed\_url()` defaults `ttl=None` so the operator's configured `signed\_url\_ttl` finally applies; tested at the helper (GCS + S3), end-to-end through `/api/highlights`, plus the explicit-override contract.

****Rough edges captured for the e2e north star:****

- [#470](https://github.com/Khelsutra/badminton-highlight-indexer/pull/470) ? bakes the `auth` extra (PyJWT) into the base image, so the e2e no longer needs a hand-built derived image. One honest stumble: my first push went CI-red on a Dockerfile structural-guard test I hadn't anticipated ? the exact "run the whole suite before pushing" lesson from #465. I owned it on the PR, fixed the guard, and it's now 11/11 green.
- [#474](https://github.com/Khelsutra/badminton-highlight-indexer/pull/474) ? commits the `gen\_service\_yaml.py` manifest generator (closes #473), with structural tests; the control-plane deploy is now reproducible from a clean clone.
- Issues filed: [#471](https://github.com/Khelsutra/badminton-highlight-indexer/issues/471) (per-instance SQLite splits control-plane state at maxScale>1 ? the generator defaults maxScale=1 as its option 1), [#472](https://github.com/Khelsutra/badminton-highlight-indexer/issues/472) (Google-managed cert renewal can stall behind the proxied CF CNAME ? check before ~Sept 30), [#473](https://github.com/Khelsutra/badminton-highlight-indexer/issues/473) (the manifest-generator gap, now addressed by #474).

****What's next (blocked on the owner):**** all five open PRs (#467/#468/#469/#470/#474) are CI-green and MERGEABLE, awaiting LGTM. A persistent monitor is polling them every 2.5 minutes ? when an LGTM lands I'll verify checks and squash-merge (watching for the #467?#469 / #468?#469 conflicts), and once all merge I'll run the full e2e deploy test: rebuild the worker image, deploy the control plane with `--edge-auth off`, run the CUJ verifying the >32 MiB signed-URL download (#463) and upload staging (#461), restore edge-auth, write the runlog, and flip A1 ? DONE. The memory handoff (`session-handoff.md`, `current-state.md`, `MEMORY.md`) is refreshed to this checkpoint.

121. user (2026-07-04T09:22:42.068Z)

#468 had a race condition is that resolved?

122. assistant (2026-07-04T09:23:12.212Z)

Yes ? resolved, pushed, and CI-green on the PR; it's awaiting the owner's LGTM to merge.

To be precise about which race: `claim\_due\_job()` was already CAS-safe ? two workers could never double-claim the same job. The actual bug (the reviewer's round-2 blocker) was a ****false-empty****: when several drain workers SELECTed the same head-of-queue row, only one won the CAS update, and the losers returned `None`. `\_drain\_due\_jobs()` treats `None` as "queue is empty" and exits ? so with 8 queued jobs and 8 workers, only 2 got claimed and the rest sat stranded until some later submit or wake-up.

The fix (commit `21ee141` on [#468](https://github.com/Khelsutra/badminton-highlight-indexer/pull/468)): a lost CAS now ****retries inside `claim\_due\_job()`**** ? re-SELECT the head runnable row, re-attempt the claim ? until it either wins a row or a fresh SELECT genuinely finds nothing runnable. So `None` now means "truly empty" by construction, and `\_drain\_due\_jobs()` needed no change. The retry loop is deliberately unbounded: every lost race means another worker successfully claimed that row, so the runnable set strictly shrinks and progress is guaranteed ? a retry cap would just reintroduce the false-empty.

Proof it's actually fixed, not just plausibly:

- New regression test (`test\_concurrent\_claimers\_claim\_all\_jobs\_distinctly`): 8 queued jobs, 8 claimer threads released simultaneously on a `threading.Barrier`; asserts all 8 claims return non-None, the claimed ids are exactly the 8 distinct jobs, and a 9th claim returns `None`.
- With the fix stashed, that test reproduces the reviewer's exact failure 5/5 runs (`['j0','j1',None,?]` ? same shape as their repro); with the fix it passed 15/15 consecutive runs, with no SQLite "database is locked" under the thread race.
- An independent adversarial verifier re-proved the test bites at the old head in a throwaway worktree before I pushed.

- Full suite 1634 passed / 10 skipped, ruff + mypy clean locally, and all 11 CI checks green on the new head.

The one related-but-separate item: the *park-vs-block* refinement (letting a compile job wait via `JobWaiting` instead of blocking a worker on the local-compute lane) was offered to the reviewer earlier and deliberately left out ? they haven't asked for it.

123. user (2026-07-04T09:25:17.511Z)

please keep watching out for the PR review comments and address them and once LGTM merge and move ahead towards the north star

124. assistant (2026-07-04T09:25:30.514Z)

The monitor is live and will re-invoke me on any PR activity. Let me do a manual sweep right now to catch anything that landed in the monitor's baseline gap, so nothing is missed.

125. user (2026-07-04T09:25:34.222Z)

```
Exit code 1
accepts 1 arg(s), received 2
accepts 1 arg(s), received 2
accepts 1 arg(s), received 2
accepts 1 arg(s), received 2
accepts 1 arg(s), received 2
```

126. user (2026-07-04T09:25:45.481Z)

```
#467: avidullu @ 2026-07-04T09:08:08Z: Fixed in c9bf37e ? good catch. The gate now classifies
the configured root itself: `storage.parse_ur
#468: avidullu @ 2026-07-04T09:08:06Z: Fixed in 21ee141, as suggested (first option):
`claim_due_job()` now retries the SELECT?CAS loop on
#469: avidullu @ 2026-07-04T09:08:09Z: Fixed in ce450e5 ? the helper was indeed forcing
`ttl=3600` into the backend, so the configured `sig
#470: avidullu @ 2026-07-04T09:13:50Z: CI red on the first push was
`tests/test_compute_backend.py::test_dockerfile_has_required_directives
#474: none
```

127. assistant (2026-07-04T09:26:10.122Z)

Sweep confirms nothing landed in the gap ? the latest comment on every PR is my own fix reply, and [#474](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/474>) has none yet. The watch posture is set:

- ****Monitor running**** (persistent, polls every 2.5 min) on #467, #468, #469, #470, #474 ? it re-invoices me on any new comment, review, or state change.
- ****On a new review comment:**** I'll address it ? fix, full suite + ruff + mypy locally, push, reply (the protocol that's now caught me out once and the reviewer twice).
- ****On LGTM:**** verify `gh pr checks` all green + MERGEABLE, then squash-merge with branch delete. After each merge I'll re-check the survivors for conflicts (#467?#469 share `orchestration.py`, #468?#469 share `main.py`) and resolve via merge-from-master + full suite if needed.
- ****Once #467/#468/#469 are in**** (plus ideally #470/#474, which remove the derived-image and scratchpad-YAML steps): I proceed to the north star ? the full e2e deploy test. Rebuild the worker image from merged master, deploy the control plane with `gen\_service\_yaml.py --edge-auth off`, run the CUJ (gs:// process ? compile ? verify the >32 MiB signed-URL download for #463 and the upload?gs:// staging for #461), restore edge-auth ON, verify the 401/302 posture at `api.khelsutra.guru`, write the DEPLOY_REPORT runlog, and flip A1 ? DONE in `docs/ALPHA\_LAUNCH\_READINESS.md`.

All quiet for now ? I'll act the moment the owner does.

128. user (2026-07-04T09:48:41.814Z)

you have logic review LGTM for all the PRs. please make sure the test coverage, ruff, mypy etc CI and all tests pass correctly and merge them

129. assistant (2026-07-04T09:49:40.647Z)

The owner's LGTM is given for all five PRs. Per the pre-merge gate, I'll verify each PR's CI is green, then ? because #467/#468/#469 share files ? validate each **merge result** locally (full suite with coverage vs a master baseline, plus ruff + mypy) in a temp worktree before each squash-merge, merging sequentially and re-checking the survivors after each merge. Starting with the CI status sweep and per-PR changed files.

130. user (2026-07-04T09:49:53.984Z)

```
#467: CI pass=11 pending=0 fail=0 | MERGEABLE/CLEAN
#468: CI pass=11 pending=0 fail=0 | MERGEABLE/CLEAN
#469: CI pass=11 pending=0 fail=0 | MERGEABLE/CLEAN
#470: CI pass=11 pending=0 fail=0 | MERGEABLE/CLEAN
#474: CI pass=11 pending=0 fail=0 | MERGEABLE/CLEAN
```

131. user (2026-07-04T09:49:58.226Z)

```
#467: backend/orchestration.py, tests/test_api_cloud_dispatch.py
#468: backend/api/job_worker.py, backend/config/models.py,
      backend/infrastructure/database_jobs.py, backend/main.py, backend/orchestration.py,
      tests/test_api_cloud_dispatch.py, tests/test_config_deployment.py,
      tests/test_drain_concurrency.py, tests/test_job_worker_loop.py
#469: backend/main.py, backend/orchestration.py, backend/storage/__init__.py,
      backend/storage/base.py, backend/storage/gcs.py, backend/storage/s3.py, tests/test_storage.py,
      tests/test_upload_download.py
#470: deploy/cloudrun/Dockerfile, deploy/cloudrun/README.md, tests/test_compute_backend.py
#474: deploy/cloudrun/README.md, deploy/cloudrun/gen_service_yaml.py,
      tests/test_deploy_manifests.py
```

132. user (2026-07-04T09:50:21.044Z)

```
True
66d7d6f fix(hosted): allowed_video_root jail must not reject storage URIs (gs://) (#465)
```

133. user (2026-07-04T09:52:57.409Z)

```
HEAD is now at 66d7d6f fix(hosted): allowed_video_root jail must not reject storage URIs (gs://)
(#465)
```

```
TOTAL                  11516   1681    85%
Required test coverage of 70% reached. Total coverage: 85.40%
1634 passed, 10 skipped, 15 warnings in 140.80s (0:02:20)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

134. user (2026-07-04T09:56:11.811Z)

```
HEAD is now at 66d7d6f fix(hosted): allowed_video_root jail must not reject storage URIs (gs://)
(#465)
```

```
tests/test_api_cloud_dispatch.py | 95 ++++++
2 files changed, 150 insertions(+), 4 deletions(-)
```

```
Required test coverage of 70% reached. Total coverage: 85.38%
1637 passed, 10 skipped, 15 warnings in 139.60s (0:02:19)
All checks passed!
Success: no issues found in 147 source files
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

135. user (2026-07-04T09:56:45.210Z)

```
#467 state=MERGED mergedAt=2026-07-04T09:56:41Z
```

From https://github.com/Khelsutra/badminton-highlight-indexer
66d7d6f..d7ebe9c master -> origin/master
d7ebe9c fix(cloud-serving): stage browser-uploaded local files to gs:// before L4 dispatch
(#461) (#467)

136. assistant (2026-07-04T09:56:54.093Z)

#467 is merged (master ? `d7ebe9c`). Now the #468 merge result against the new master.

137. user (2026-07-04T09:59:57.572Z)

HEAD is now at d7ebe9c fix(cloud-serving): stage browser-uploaded local files to gs:// before L4
dispatch (#461) (#467)
9 files changed, 292 insertions(+), 26 deletions(-)
create mode 100644 tests/test_drain_concurrency.py

Required test coverage of 70% reached. Total coverage: 85.44%
1645 passed, 10 skipped, 15 warnings in 140.31s (0:02:20)
All checks passed!
Success: no issues found in 147 source files
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

138. user (2026-07-04T10:00:17.130Z)

#468 state=MERGED mergedAt=2026-07-04T10:00:13Z
From https://github.com/Khelsutra/badminton-highlight-indexer
d7ebe9c..ffe68ed master -> origin/master
ffe68ed fix(cloud-serving): parallelize the drain under cloud_run (#466) (#468)
#469 now: UNKNOWN/UNKNOWN

139. user (2026-07-04T10:00:32.279Z)

HEAD is now at ffe68ed fix(cloud-serving): parallelize the drain under cloud_run (#466) (#468)
backend/storage/s3.py | 2 +-
tests/test_storage.py | 38 ++++++
tests/test_upload_download.py | 83 ++++++
8 files changed, 242 insertions(+), 9 deletions(-)

140. assistant (2026-07-04T10:00:39.479Z)

Clean auto-merge ? no conflicts. Running the gates on the #469 merge result.

141. user (2026-07-04T10:03:35.624Z)

Required test coverage of 70% reached. Total coverage: 85.39%
1652 passed, 10 skipped, 15 warnings in 136.55s (0:02:16)
All checks passed!
Success: no issues found in 147 source files
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

142. assistant (2026-07-04T10:03:44.139Z)

#469's merge result is green ? 1652 passed, coverage 85.39%, ruff + mypy clean. Merging.

143. user (2026-07-04T10:03:57.217Z)

#469 state=MERGED mergedAt=2026-07-04T10:03:53Z
From https://github.com/Khelsutra/badminton-highlight-indexer
ffe68ed..63d65f9 master -> origin/master
63d65f9 fix(cloud-serving): serve large reels via a gs:// signed URL (#463) (#469)

144. assistant (2026-07-04T10:04:04.037Z)

#469 merged (master ? `63d65f9`) ? all three serving fixes are in. Now #470.

145. user (2026-07-04T10:07:05.422Z)

HEAD is now at 63d65f9 fix(cloud-serving): serve large reels via a gs:// signed URL (#463) (#469)
tests/test_compute_backend.py | 10 ++++++----
3 files changed, 15 insertions(+), 7 deletions(-)

Required test coverage of 70% reached. Total coverage: 85.39%
1652 passed, 10 skipped, 15 warnings in 138.48s (0:02:18)
All checks passed!
Success: no issues found in 147 source files
[check-md-links] clean ? no broken internal links in live docs.
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

146. user (2026-07-04T10:07:21.393Z)

#470 state=MERGED mergedAt=2026-07-04T10:07:17Z
From https://github.com/Khelsutra/badminton-highlight-indexer
63d65f9..3b650df master -> origin/master
3b650df deploy(cloudrun): bake the auth extra (PyJWT) into the base image (M9-ready rebuilds) (#470)

147. assistant (2026-07-04T10:07:27.457Z)

#470 merged. Last one ? #474, which shares `README.md` with just-merged #470.

148. user (2026-07-04T10:10:26.466Z)

HEAD is now at 3b650df deploy(cloudrun): bake the auth extra (PyJWT) into the base image (M9-ready rebuilds) (#470)
3 files changed, 262 insertions(+), 1 deletion(-)
create mode 100644 deploy/cloudrun/gen_service_yaml.py
create mode 100644 tests/test_deploy_manifests.py

Required test coverage of 70% reached. Total coverage: 85.39%
1655 passed, 10 skipped, 15 warnings in 136.31s (0:02:16)
All checks passed!
Success: no issues found in 147 source files
[check-md-links] clean ? no broken internal links in live docs.
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

149. user (2026-07-04T10:10:47.374Z)

#474 state=MERGED mergedAt=2026-07-04T10:10:41Z
From https://github.com/Khelsutra/badminton-highlight-indexer
3b650df..195932f master -> origin/master
195932f deploy(cloudrun): commit the control-plane service-manifest generator (closes #473) (#474)
3b650df deploy(cloudrun): bake the auth extra (PyJWT) into the base image (M9-ready rebuilds) (#470)
63d65f9 fix(cloud-serving): serve large reels via a gs:// signed URL (#463) (#469)
--- issues auto-closed? ---
#461: CLOSED
#463: CLOSED
#466: CLOSED
#473: CLOSED

150. user (2026-07-04T10:11:16.052Z)

Updated task #2 status

151. user (2026-07-04T10:11:16.530Z)

Updated task #3 status

152. user (2026-07-04T10:11:27.030Z)

```
create mode 100644 tests/test_drain_concurrency.py
Deleted branch fix/allowed-video-root-uri (was cd4bc73).
Deleted branch fix/drain-concurrency (was 21ee141).
Deleted branch fix/upload-gcs-staging (was c9bf37e).
Deleted branch fix/highlights-signed-url (was ce450e5).
Deleted branch fix/dockerfile-auth-extra (was c63d122).
Deleted branch feat/cloudrun-service-manifest (was 9de18d5).
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer
195932f [master]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/cfg-wt      20b2d5a [claude/wasb-
config-validation]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/compile-wt    b09a1d4 [claude/compile-
reliability]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-pre-wt  355dad6 (detached HEAD)
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-wt      5de1ef2 (detached HEAD)
+ claude/compile-reliability
  claude/serving-loud-config-failure
+ claude/wasb-config-validation
  claude/wasb-gpu-batching
  codex/a2-golden-manifest-portability
  codex/a4-n15-nightly-baseline
  codex/a5-heuristic-n15-floor
  codex/a6-ablation-cli
  codex/a7-a9-quality-chain
  docs/a1-deploy-runlog
  fix/cloudrun-dispatch-client
* master
```

153. assistant (2026-07-04T10:11:37.517Z)

Cleanup done ? my worktrees and branches are gone (the `6e9d0958` scratchpad worktrees belong to a sibling session; left untouched). Now the e2e. GCP pre-flight first.

154. user (2026-07-04T10:11:50.639Z)

```
gen-lang-client-0306026826
avi.dullu@gmail.com
--- cloud run resources (asia-southeast1) ---
rally-worker
rally-control-plane      https://rally-control-plane-2bosgeukpa-as.a.run.app
--- serving bucket ---
gs://khelsutra-rally-serving/videos/MahadevpuraSingles.MP4
```

155. user (2026-07-04T10:12:00.282Z)

```
1      # Cloud Build config for the Cloud Run GPU worker image (COMPUTE_DECOUPLED_SERVING M7).
2      #
3      # `gcloud builds submit --tag ? -f ?` does NOT work ? `gcloud builds submit` has no `-f`
flag, so a
4      # non-root Dockerfile needs a build CONFIG like this one. Build with:
5      #
6      #   gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project <PROJECT> .
7      #
8      # Override the region/repo/image with --substitutions=_REGION=?,_REPO=?,_IMAGE=? if
needed.
9      # PROJECT_ID is provided by Cloud Build automatically.
```



```
serving","input_backend":"gcs","input_root":"gs://khelsutra-rally-serving/videos"},"security":{"edge_auth_enabled":false,"cf_team_domain":"https://khelsutra.cloudflareaccess.com","cf_access_aud":"c980da67ed888fff935fcd913ae1eb4da63f9d65805d6e1c5ca76f92f517082b","allowed_video_root":"/tmp/apphome/output"}}}
```

158. assistant (2026-07-04T10:12:30.568Z)

Manifests generated; the decoded config matches the handoff spec exactly. Updating the memory checkpoint with the merge milestone while the build runs.

159. user (2026-07-04T10:12:41.557Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

160. user (2026-07-04T10:12:59.572Z)

ffmpeg version 8.1.1-full_build-www.gyan.dev Copyright (c) 2000-2026 the FFmpeg developers
[aac @ 00000231923ebc80] Qavg: 489.570

Name	Length
----	-----
upload-e2e.mp4	70881

161. assistant (2026-07-04T10:13:06.768Z)

Test upload clip ready (70 KB, 4s). Checking build progress.

162. user (2026-07-04T10:13:08.763Z)

```
1      195932f deploy(cloudrun): commit the control-plane service-manifest generator (closes
#473) (#474)
2
3      Created [https://cloudbuild.googleapis.com/v1/projects/gen-lang-
client-0306026826/locations/global/builds/efe16581-f178
4      -4175-bf2c-efd56cf5142b].
5
```

163. user (2026-07-04T10:13:29.747Z)

```
Exit code 23
rally-control-plane-00004-tvr  asia-southeast1-docker.pkg.dev/gen-lang-
client-0306026826/rally/rally-control-plane:latest
--- health (edge-auth ON, before) ---
200
--- process sans CF JWT (expect 401) ---
422
```

164. user (2026-07-04T10:14:08.253Z)

```
--- POST /api/process, valid JSON, NO CF JWT (expect 401 if edge-auth enforces) ---
{"detail":"edge auth: missing Cf-Access-Jwt-Assertion header (request did not pass Cloudflare
Access)"}
HTTP 401
```

165. user (2026-07-04T10:14:27.174Z)

```
{"message":"Successfully stopped task: betc1ot9n (declare -A fp\nwhile true; do\n  for pr in 467
468 469 470 474; do\n    data=$(gh api \"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr\"
--jq '\"\\((.state)|\\((.merged)\"' 2>/dev/null) || continue\n    cm=$(gh api
\"repos/Khelsutra/badminton-highlight-indexer/issues/$pr/comments\" --jq 'length as $n | if
$n==0 then \"0|none|none\" else \"\\($n)|\\((.[ -1].created_at)|\\((.[ -1].user.login):
```

```

\\(. [-1].body[0:140] | gsub("\\\\n\\n"; \\n \\n"))\\n end' 2>/dev/null) || continue\\n    rv=$(gh api
\\"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr/reviews\\" --jq 'length as $n | if $n==0
then \\0\\" else \\\"\\n($n)\\\\(. [-1].state)|\\\\(. [-1].user.login)\\n' end' 2>/dev/null) || continue\\n
cur=\\\"$data|$cm|$rv\\n\\n    if [ -n \\\"${fp[$pr]}\\n' ] && [ \\\"${fp[$pr]}\\n' != \\\"$cur\\n' ]; then\\n
echo \\\"PR #$pr CHANGED -> state/merged=$data | comments=$cm | reviews=$rv\\n\\n    fi\\n
fp[$pr]=\\\"$cur\\n\\n done\\n sleep
150\\ndone\\n\\n\\\"task_id\\\":\\\"betc1ot9n\\\",\\\"task_type\\\":\\\"local_bash\\\",\\\"command\\\":\\\"declare -A fp\\nwhile true;
do\\n for pr in 467 468 469 470 474; do\\n    data=$(gh api \\\"repos/Khelsutra/badminton-
highlight-indexer/pulls/$pr\\n' --jq '\\\"\\\\(.state)|\\\\(.merged)\\n' 2>/dev/null) || continue\\n
cm=$(gh api \\\"repos/Khelsutra/badminton-highlight-indexer/issues/$pr/comments\\n' --jq 'length as
$n | if $n==0 then \\0|none|none\\n' else \\\"\\n($n)\\\\(. [-1].created_at)|\\\\(. [-1].user.login):
\\\\(. [-1].body[0:140] | gsub("\\\\n\\n"; \\n \\n'))\\n' end' 2>/dev/null) || continue\\n    rv=$(gh api
\\"repos/Khelsutra/badminton-highlight-indexer/pulls/$pr/reviews\\" --jq 'length as $n | if $n==0
then \\0\\" else \\\"\\n($n)\\\\(. [-1].state)|\\\\(. [-1].user.login)\\n' end' 2>/dev/null) || continue\\n
cur=\\\"$data|$cm|$rv\\n\\n    if [ -n \\\"${fp[$pr]}\\n' ] && [ \\\"${fp[$pr]}\\n' != \\\"$cur\\n' ]; then\\n
echo \\\"PR #$pr CHANGED -> state/merged=$data | comments=$cm | reviews=$rv\\n\\n    fi\\n
fp[$pr]=\\\"$cur\\n\\n done\\n sleep 150\\ndone\\n\\n\\\"}

```

166. user (2026-07-04T10:16:06.059Z)

<retrieval_status>success</retrieval_status>

<task_id>bfg5an95c</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>

195932f deploy(cloudrun): commit the control-plane service-manifest generator (closes #473) (#474)

Created [https://cloudbuild.googleapis.com/v1/projects/gen-lang-client-0306026826/locations/global/builds/efe16581-f178-4175-bf2c-efd56cf5142b].

libdrm-radeon1 libdrm2 libedit2 libelf1 liberror-perl libexpat1 libflac8

libdrm2 libedit2 libelf1 liberror-perl libexpat1 libflac8 libflite1

Get:254 http://archive.ubuntu.com/ubuntu jammy/main amd64 liberror-perl all 0.17029-1 [26.5 kB]

Selecting previously unselected package liberror-perl.

Preparing to unpack .../239-liberror-perl_0.17029-1_all.deb ...

</output>

167. user (2026-07-04T10:16:24.771Z)

WORKING

168. user (2026-07-04T10:16:44.262Z)

Command running in background with ID: blgl05jn5. Output is being written to:

C:\Users\avidu\AppData\Local\Temp\claudc\--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\blgl05jn5.output. You will be notified when it completes. To check interim output, use Read on that file path.

169. user (2026-07-04T10:19:17.663Z)

<retrieval_status>success</retrieval_status>

<task_id>blgl05jn5</task_id>

<task_type>local_bash</task_type>

```
<exit_code>0</exit_code>
```

BUILD TERMINAL: SUCCESS

</output>

Build ****SUCCESS**** ? new image digest `sha256:ed2fe448?` carries all five merged fixes plus the auth extra. Re-pinning the L4 job and deploying the control plane with the auth-off manifest.

Updating job...

asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest

[illegible]

```
Routing traffic....done
```

URL: <https://rally-control-plane-492532266377.asia-southeast1.run.app>

Control plane deployed from the new base image with edge-auth OFF. Verifying boot health and

capabilities.

174. user (2026-07-04T10:23:08.688Z)

```
rally-control-plane-00005-cgr
--- /api/health ---
{"status":"ok","sport":"badminton","default_segmenter":"yolo_hybrid"}
HTTP 200
--- /api/capabilities (first 400 chars) ---
{"api_version":1,"engine_version":"0.1.0","segmenters":["fusion_hybrid","gemini","motion","track
net_hybrid","wasb_hybrid","yolo_hybrid"],"default_segmenter":"yolo_hybrid","ffmpeg":{"ffmpeg":tr
ue,"ffprobe":true,"ffmpeg_path":"/usr/bin/ffmpeg","ffprobe_path":"/usr/bin/ffprobe"},"gpu":{"nvi
dia_smi_present":false},"gemini_key_present":false,"wasb_weights_present":false,"deployment":{"p
rofile":"cloud-s
```

175. user (2026-07-04T10:23:29.391Z)

```
{"job_id":"68d2ea67ca554567ab0676cfcc9f2125","status":"queued","poll":"/api/jobs/68d2ea67ca55456
7ab0676cfcc9f2125"}
HTTP 202
```

176. user (2026-07-04T10:23:41.654Z)

Command running in background with ID: bgz8gf7rh. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\bgz8gf7rh.output. You will be notified when
it completes. To check interim output, use Read on that file path.

177. user (2026-07-04T10:23:57.253Z)

No matches found

178. user (2026-07-04T10:24:02.197Z)

```
backend\api\uploads.py:86:@router.post("/api/upload/init")
backend\api\uploads.py:123:         "next_part": f"/api/upload/{upload_id}/part/{0}",
backend\api\uploads.py:127:@router.put("/api/upload/{upload_id}/part/{index}")
backend\api\uploads.py:173:@router.post("/api/upload/{upload_id}/complete")
backend\api\uploads.py:204:@router.delete("/api/upload/{upload_id}")
```

179. user (2026-07-04T10:24:08.949Z)

```
80         characters (for example a fullwidth dot) or trailing whitespace/dots. Keep the
existing
81         traversal/null-byte checks in ``safe_output_filename`` as the authority after this
cleanup.
82         """
83         return unicodedata.normalize("NFKC", filename).strip().rstrip(". ")
84
85
86     @router.post("/api/upload/init")
87     def upload_init(req: UploadInitRequest):
88         """Start a chunked upload. Returns upload_id; PUT sequential parts next."""
89         upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
90         try:
91             name = safe_output_filename(_normalize_upload_filename(req.filename))
92         except ValueError as e:
93             raise HTTPException(status_code=400, detail=str(e)) from e
94         ext = os.path.splitext(name)[1].lower().lstrip(".")
95         if ext not in exts:
96             raise HTTPException(
97                 status_code=400,
98                 detail=f"Extension '{ext}' not allowed. Allowed: {sorted(exts)}",
99             )
```

```

100     if req.total_size_bytes is not None and req.total_size_bytes > max_bytes:
101         raise HTTPException(
102             status_code=413,
103             detail=f"File exceeds the {max_bytes // (1024 * 1024)} MB upload limit.",
104         )
105     os.makedirs(upload_dir, exist_ok=True)
106     # P2 (D11): fail fast with 507 if the box can't hold the declared upload. Best-
107     # effort ? an
108     # undeclared size (total_size_bytes=None) or an unreadable disk skips the guard
109     # (never blocks a
110     # legit upload); the per-part total cap in upload_part is the hard backstop either
111     way.
112     require_free_bytes(upload_dir, req.total_size_bytes, stage="upload")
113     upload_id = uuid.uuid4().hex
114     tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")
115     open(tmp_path, "wb").close()
116     with _uploads_lock:
117         _uploads[upload_id] = {
118             "filename": name,
119             "tmp_path": tmp_path,
120             "parts": 0,
121             "bytes": 0,
122         }
123     return {
124         "upload_id": upload_id,
125         "max_part_mb": part_bytes // (1024 * 1024),
126         "next_part": f"/api/upload/{upload_id}/part/0",
127     }
128
129 @router.put("/api/upload/{upload_id}/part/{index}")
130 async def upload_part(upload_id: str, index: int, request: Request):
131     """Append one part (raw bytes body). Parts are strictly sequential (0,1,2,...)."""
132     _, max_bytes, part_bytes, _ = _upload_cfg()
133     with _uploads_lock:
134         st = _uploads.get(upload_id)
135         if st is None:
136             raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
137         if index != st["parts"]:
138             raise HTTPException(
139                 status_code=409,
140                 detail=f"Out-of-order part: expected {st['parts']}, got {index} (parts are
141                 sequential)",
142             )
143         received = 0
144         overflow = None
145         with open(st["tmp_path"], "ab") as fh:
146             async for chunk in request.stream():
147                 received += len(chunk)
148                 if received > part_bytes:
149                     overflow = (
150                         f"Part exceeds the {part_bytes // (1024 * 1024)} MB per-part limit."
151                     )
152                     break
153                 if st["bytes"] + received > max_bytes:
154                     overflow = (
155                         f"Upload exceeds the {max_bytes // (1024 * 1024)} MB total limit."
156                     )
157                     break
158                 fh.write(chunk)
159         if overflow:
160             _abort_upload(upload_id)
161             raise HTTPException(
162                 status_code=413, detail=overflow + " Upload aborted ? re-init to retry."
163             )

```

```

161         )
162
163     with _uploads_lock:
164         st["parts"] += 1
165         st["bytes"] += received
166     return {
167         "upload_id": upload_id,
168         "received_parts": st["parts"],
169         "received_bytes": st["bytes"],
170     }
171
172
173     @router.post("/api/upload/{upload_id}/complete")
174     def upload_complete(upload_id: str, req: UploadCompleteRequest):
175         """Finalize: verify part count, move into place, return {path, video_id} for
176 /api/process."""
177         upload_dir, _, _, _ = _upload_cfg()
178         with _uploads_lock:
179             st = _uploads.get(upload_id)
180             if st is None:
181                 raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
182             if st["parts"] == 0 or st["parts"] != req.total_parts:
183                 raise HTTPException(
184                     status_code=409,
185                     detail=f"Expected {req.total_parts} part(s), received {st['parts']}.",
186                 )
187             final_path = os.path.join(upload_dir, st["filename"])
188             if os.path.exists(
189                 final_path
190             ): # never overwrite an earlier upload with the same name
191                 stem, ext = os.path.splitext(st["filename"])
192                 final_path = os.path.join(upload_dir, f"{stem}-{upload_id[:8]}{ext}")
193             os.replace(st["tmp_path"], final_path)
194             with _uploads_lock:
195                 _uploads.pop(upload_id, None)
196             video_id = compute_video_id(final_path)
197             return {
198                 "path": os.path.abspath(final_path),
199                 "video_id": video_id,
200                 "size_bytes": os.path.getsize(final_path),
201                 "next": "POST /api/process with this path",
202             }
203
204     @router.delete("/api/upload/{upload_id}")

```

180. assistant (2026-07-04T10:24:16.705Z)

Upload flow confirmed (init ? PUT raw-bytes part 0 ? complete `{total\_parts:1}`). Checking detect-job progress.

181. user (2026-07-04T10:24:18.600Z)

```

1     JOB: running | validating
2

```

182. user (2026-07-04T10:34:29.817Z)

<retrieval_status>timeout</retrieval_status>

<task_id>bgz8gf7rh</task_id>

<task_type>local_bash</task_type>

```
<status>running</status>
```

```
<output>
```

```
JOB: running | validating
```

```
JOB: running | dispatching (cloud_run)
```

```
</output>
```

183. user (2026-07-04T10:38:00.429Z)

```
<retrieval_status>success</retrieval_status>
```

```
<task_id>bgz8gf7rh</task_id>
```

```
<task_type>local_bash</task_type>
```

```
<status>completed</status>
```

```
<exit_code>0</exit_code>
```

```
<output>
```

```
JOB: running | validating
```

```
JOB: running | dispatching (cloud_run)
```

```
JOB: done | finished
```

```
TERMINAL. Full: {"job_id":"68d2ea67ca554567ab0676cfcc9f2125","status":"done","progress":"finished","video_id":"915473794e0909c9f95a01e4511a5e98","error":null,"result":{"video_id":"915473794e0909c9f95a01e4511a5e98","status":"success","message":"Successfully indexed 48 play segments (cloud_run /
```

```
'wasb_hybrid')."},"results":[{"validator_name":"format_check","passed":true,"message":"Video format is valid MP4 container."},"details":{"format_name":"mov,mp4,m4a,3gp,3g2,mj2","duration":447.713933,"size_bytes":196182380,"codec_name":"hevc"}},{"validator_name":"resolution_fps_check","passed":true,"message":"Video resolution and frame rate meet minimum standards."},"details":{"width":1920,"height":1080,"fps":29.97}},{"validator_name":"pts_continuity_check","passed":true,"message":"PTS timeline is continuous with no abnormal gaps or backward jumps."},"details":{"keyframes_count":448,"max_keyframe_gap_seconds":1.0,"backward_jumps":0}}]>
```

184. user (2026-07-04T10:38:12.604Z)

```
backend\main.py:451:@app.get("/api/videos")
```

```
backend\main.py:997:@app.get("/api/videos/{video_id}/cost")
```

```
backend\main.py:1096:@app.post("/api/compile", status_code=202)
```

185. user (2026-07-04T10:38:26.866Z)

```
video: segments: 0
```

```
segment id range: ..
```

```
{"video_id":null,"output_filename":"e2e.mp4","segment_ids":[null]}
```

186. assistant (2026-07-04T10:38:32.297Z)

The response shape isn't what I assumed ? let me look at the raw payload.

187. user (2026-07-04T10:38:36.571Z)

```
{"videos":[{"id":"915473794e0909c9f95a01e4511a5e98","filepath":"gs://khehsutra-rally-serving/videos/MahadevpuraSingles.MP4","processed_at":"2026-07-04 10:24:10","status":"processed","validation_results":[{"validator_name":"format_check","passed":true,"message":"Video format is valid MP4 container."},"details":{"format_name":"mov,mp4,m4a,3gp,3g2,mj2","duration":447.713933,"size_bytes":196182380,"codec_name":"hevc"}},{"validator_name":"resolution_fps_check","passed":true,"message":"Video resolution and frame rate meet minimum standards."},"details":{"width":1920,"height":1080,"fps":29.97}},{"validator_name":"pts_continuity_check","passed":true,"message":"PTS timeline is continuous with no abnormal gaps or backward jumps."},"details":{"keyframes_count":448,"max_keyframe_gap_seconds":1.0,"backward_jumps":0}}]}
```

188. user (2026-07-04T10:38:48.286Z)

```
id, filepath, processed_at, status, validation_results, user_id, video_id, match_name,
```

local_path, created_at, fps, resolution

189. user (2026-07-04T10:38:59.295Z)

```
236: _ingest_remote_segments,
266: # play_segments is returned from cache so refresh/Try again cannot re-bill paid
segmenters.
1085: def query_play_segments(req: QueryRequest, request: Request):
1092:     segments = request.app.state.db.query_play_segments(req.video_id, filters)
1093:     return {"play_segments": segments}
1103: got the URL). The cheap checks (video exists, segments match + survive the floor, safe
filename)
1544:         print(f"[CLI] Indexing play segments using segmenter: {seg_name}...")
1556:         segments = []
1558:         segments = result.value if hasattr(result, "value") else result
1560:         f"[SUCCESS] Process complete. Indexed {len(segments)} play segments
inside database: {db_path}"
1563:         _finalize_reingest(db_instance, video_id, segments, play_wm, cand_wm)
```

190. user (2026-07-04T10:39:05.212Z)

```
1078         detail=f"No compiled file named '{name}'. Compile first via /api/compile.",
1079     )
1080     media = "video/quicktime" if name.lower().endswith(".mov") else "video/mp4"
1081     return FileResponse(path, media_type=media, filename=name)
1082
1083
1084 @app.post("/api/query")
1085 def query_play_segments(req: QueryRequest, request: Request):
1086     filters = {
1087         "server": req.server,
1088         "receiver": req.receiver,
1089         "duration_min": req.duration_min,
1090         "event_type": req.event_type,
1091     }
1092     segments = request.app.state.db.query_play_segments(req.video_id, filters)
1093     return {"play_segments": segments}
1094
1095
```

191. user (2026-07-04T10:39:20.755Z)

```
segments: 48
id range: 1..48
{"video_id": "915473794e0909c9f95a01e4511a5e98", "output_filename": "e2e.mp4", "segment_ids": [1, 2, 3,
4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23,
```

192. assistant (2026-07-04T10:39:25.525Z)

48 segments (ids 1?48). Firing the compile.

193. user (2026-07-04T10:39:31.608Z)

```
{"job_id": "06eb0372ba294cd7bd6a19c3446c687b", "status": "queued", "poll": "/api/jobs/06eb0372ba294cd
7bd6a19c3446c687b"}
HTTP 202
```

194. user (2026-07-04T10:39:49.128Z)

Command running in background with ID: b83z43dtp. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\b83z43dtp.output. You will be notified when
it completes. To check interim output, use Read on that file path.

195. user (2026-07-04T10:40:05.611Z)

```
init: upload_id=afea8aa687164f4c854001d0255a6484
part0: parts=1 bytes=70881
complete: path=/tmp/apphome/output/uploads/upload-e2e.mp4
video_id=1e3d610cbff4e4d8a22985d003d81d0b size=70881
```

196. assistant (2026-07-04T10:40:13.249Z)

Upload landed on the CP's local disk. Now the #461 moment of truth: process that local path with cloud compute ? it must stage to gs:// and dispatch.

197. user (2026-07-04T10:40:20.825Z)

```
{"job_id":"75d4de634aa84ab9ba5c99a26cc74c8e","status":"queued","poll":"/api/jobs/75d4de634aa84ab9ba5c99a26cc74c8e"}
HTTP 202
```

198. user (2026-07-04T10:40:30.637Z)

Command running in background with ID: bbgcntng6. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\bbgcntng6.output. You will be notified when it completes. To check interim output, use Read on that file path.

199. assistant (2026-07-04T10:40:38.726Z)

Both jobs are polling in the background. Quick early check: did the staged object land in the bucket?

200. user (2026-07-04T10:40:43.036Z)

```
Exit code 1
ERROR: (gcloud.storage.ls) One or more URLs matched no objects.
```

201. assistant (2026-07-04T10:40:50.431Z)

The upload job already terminated ? suspiciously fast, and no staged object visible yet. Reading the result.

202. user (2026-07-04T10:40:51.174Z)

```
1      UPLOADJOB: done | finished
2      TERMINAL. Full: {"job_id":"75d4de634aa84ab9ba5c99a26cc74c8e","status":"done","progress":
"finished","video_id":"1e3d610cbff4e4d8a22985d003d81d0b","error":null,"result":{"video_id":"1e3d
610cbff4e4d8a22985d003d81d0b","status":"failed","message":"Video failed validation:
resolution_fps_check: Low resolution: Found 640x360, but minimum is
1280x720.","results":[{"validator_name":"format_check","passed":true,"message":"Video format is
valid MP4 container.","details":{"format_name":"mov,mp4,m4a,3gp,3g2,mj2","duration":4.0,"size_by
tes":70881,"codec_name":"h264"}},{ "validator_name":"resolution_fps_check","passed":false
3
```

203. user (2026-07-04T10:41:13.109Z)

```
size: 105345
complete: path=/tmp/apphome/output/uploads/upload-e2e-720p.mp4
video_id=d8335f51acd9b50c9f15bff3f5fcf4e7 size=105345
{"job_id":"6a9cb088133b45b69fdb685f620f35a7","status":"queued","poll":"/api/jobs/6a9cb088133b45b69fdb685f620f35a7"}
HTTP 202
```

204. user (2026-07-04T10:41:26.444Z)

Command running in background with ID: btq1cko74. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\btq1cko74.output. You will be notified when it completes. To check interim output, use Read on that file path.

205. user (2026-07-04T10:41:34.800Z)

```
1      COMPILE: running | stitching
2
```

206. assistant (2026-07-04T10:41:42.942Z)

Compile is stitching. Checking the staging bucket for the new upload's video id ? the staged object should appear seconds into the job.

207. user (2026-07-04T10:42:07.697Z)

```
105345 2026-07-04T10:41:30Z gs://khehsutra-rally-
serving/videos/d8335f51acd9b50c9f15bff3f5fcf4e7/upload-e2e-720p.mp4
TOTAL: 1 objects, 105345 bytes (102.88kiB)
```

208. user (2026-07-04T10:44:44.631Z)

```
<retrieval_status>success</retrieval_status>

<task_id>b83z43dtp</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>
COMPILE: running | stitching
COMPILE: done | finished
TERMINAL. Full: {"job_id":"06eb0372ba294cd7bd6a19c3446c687b","status":"done","progress":"finishe
d","video_id":"915473794e0909c9f95a01e4511a5e98","error":null,"result":{"status":"success","file
path":"/tmp/apphome/output/e2e.mp4","download_url":"/api/highlights/915473794e0909c9f95a01e4511a
5e98/e2e.mp4","video_id":"915473794e0909c9f95a01e4511a5e98"},"reports":null,"created_at":"2026-
07-04 10:39:32","started_at":"2026-07-04 10:39:32","finished_at":"2026-07-04 10:44:26"}
</output>
```

209. assistant (2026-07-04T10:44:50.367Z)

Compile done (`e2e.mp4` ready, ~5 min stitch). Reading the upload job's terminal state.

210. user (2026-07-04T10:44:51.452Z)

```
1      UPLOADJOB2: running | validating
2      UPLOADJOB2: running | dispatching (cloud_run)
3      UPLOADJOB2: done | finished
4      TERMINAL. Full: {"job_id":"6a9cb088133b45b69fdb685f620f35a7","status":"done","progress":
"finished","video_id":"d8335f51acd9b50c9f15bff3f5fcf4e7","error":null,"result":{"video_id":"d833
5f51acd9b50c9f15bff3f5fcf4e7","status":"success","message":"Successfully indexed 0 play segments
(cloud_run /
'wasb_hybrid').","results":[{"validator_name":"format_check","passed":true,"message":"Video
format is valid MP4 container.","details":{"format_name":"mov,mp4,m4a,3gp,3g2,mj2","duration":4.
0,"size_bytes":105345,"codec_name":"h264"}},{ "validator_name":"resolution_fps_check","passed":tr
ue,"message":"Video resolution and frame rate meet minimum
standards.","details":{"width":1280,"height":720,"fps":30.0}},{ "validator_name"
5
```

211. user (2026-07-04T10:45:12.564Z)

```
--- redirect check (no -L): expect 307 + storage.googleapis.com location ---
HTTP 500
location:
--- gs:// mirror check ---
174046228 2026-07-04T10:44:26Z gs://khehsutra-rally-
serving/highlights/915473794e0909c9f95a01e4511a5e98/e2e.mp4
TOTAL: 1 objects, 174046228 bytes (165.98MiB)
```

212. assistant (2026-07-04T10:45:26.844Z)

The reel ****is**** mirrored (165.98 MiB ? well over the 32 MiB cap), but ``/api/highlights`` returned ****500** instead of the 307 redirect******. Pulling the control-plane logs for the exception.

213. user (2026-07-04T10:45:32.718Z)

```
2026-07-04T10:45:10.993747Z      Response size was too large. Please consider reducing response
size.
2026-07-04T10:45:10.434707Z
```

214. user (2026-07-04T10:45:49.245Z)

```
INFO:      169.254.169.126:31240 - "GET /api/highlights/915473794e0909c9f95a01e4511a5e98/e2e.mp4
HTTP/1.1" 200 OK
10:45:10 [WARNING] backend.main: [highlights] signed-url path failed for
915473794e0909c9f95a01e4511a5e98; serving local: you need a private key to sign credentials.the
credentials you are currently using <class 'google.auth.compute_engine.credentials.Credentials'>
just contains a token. see https://googleapis.dev/python/google-api-
core/latest/auth.html#setting-up-a-service-account for more details.
```

215. user (2026-07-04T10:46:07.886Z)

```
60         return StorageStat(
61             size=int(b.size or 0),
62             modified=b.updated,
63             etag=b.etag,
64             content_type=b.content_type,
65         )
66
67     def list(self, prefix) -> Iterator[StorageRef]:
68         for blob in self._client.list_blobs(prefix.bucket, prefix=prefix.key):
69             yield StorageRef(
70                 "gcs", prefix.bucket, blob.name, f"gcs://{prefix.bucket}/{blob.name}"
71             )
72
73     def delete(self, ref) -> None:
74         self._blob(ref).delete()
75
76     def signed_url(self, ref, *, ttl=None, method="GET") -> str:
77         blob = self._blob(ref)
78         exp = timedelta(seconds=ttl or self._ttl)
79         creds = getattr(self._client, "_credentials", None)
80         # Local-key signing (a service-account JSON key exposes a `signer`) works out of
the box. But
81         # on Cloud Run / GCE the ADC credentials come from the metadata server and have
NO private key,
82         # so a plain generate_signed_url raises "you need a private key". Fall back to
the IAM signBlob
83         # API by passing the SA email + a fresh access token ? which requires the
runtime SA to hold
84         # roles/iam.serviceAccountTokenCreator on ITSELF. Best-effort: if we can't build
that, defer to
85         # the default call (which will raise a clear error rather than silently mis-
sign).
86         if creds is not None and not hasattr(creds, "signer"):
```

```

87         try:
88             from google.auth.transport.requests import Request as _AuthRequest
89
90             if not getattr(creds, "token", None):
91                 creds.refresh(_AuthRequest())
92             sa_email = getattr(creds, "service_account_email", None)
93             if sa_email and sa_email != "default":
94                 return blob.generate_signed_url(
95                     version="v4",
96                     expiration=exp,
97                     method=method,
98                     service_account_email=sa_email,
99                     access_token=creds.token,
100                 )
101         except Exception: # pragma: no cover - exercised only on real Cloud Run
102             pass
103         return blob.generate_signed_url(version="v4", expiration=exp, method=method)
104

```

216. user (2026-07-04T10:46:45.715Z)

```

{
  "bindings": [
    {
      "members": [
        "serviceAccount:492532266377-compute@developer.gserviceaccount.com"
      ],
      "role": "roles/iam.serviceAccountTokenCreator"
    }
  ],
  "etag": "BwZVwYeVtgA=",
  "version": 1
}

```

217. user (2026-07-04T10:47:17.173Z)

```

iam.googleapis.com
iamcredentials.googleapis.com
--- also try a direct signBlob as the SA would (impersonation smoke) ---
WARNING: This command is using service account impersonation. All API calls will be
executed as [492532266377-compute@developer.gserviceaccount.com].
ERROR: (gcloud.storage.sign-url) PERMISSION_DENIED: Failed to impersonate
[492532266377-compute@developer.gserviceaccount.com]. Make sure the account that's trying to
impersonate it has access to the service account itself and the
"roles/iam.serviceAccountTokenCreator" role. Permission 'iam.serviceAccounts.getAccessToken'
denied on resource (or it may not exist). Remediate access with this Troubleshooter URL or share
it with your administrator - https://console.cloud.google.com/iam-admin/troubleshooter/summary;e
rrorId=CiQwMTlmMmNiZC02NjQ1LTczM2MtOTM5Ni1hYjY4Y2E1MGI0N2MSAA%3D%3D . This command is
authenticated as avi.dullu@gmail.com which is the active account specified by the [core/account]
property. Impersonation is used to impersonate
492532266377-compute@developer.gserviceaccount.com.

```

218. user (2026-07-04T10:48:17.658Z)

```

Client.SCOPE = ('https://www.googleapis.com/auth/devstorage.full_control',
'https://www.googleapis.com/auth/devstorage.read_only',
'https://www.googleapis.com/auth/devstorage.read_write')
gcs 3.12.0 | google-auth 2.53.0

```

219. user (2026-07-04T10:48:54.675Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/issues/475>

220. user (2026-07-04T10:49:01.164Z)

```
1      """Google Cloud Storage backend ? first-class object buckets.
2
3      Uses ``google-cloud-storage``, imported lazily so ``import backend.storage`` never pulls
it;
4      install with the ``storage-gcs`` extra. Credentials come from Application Default
Credentials
5      (``GOOGLE_APPLICATION_CREDENTIALS`` pointing at a service-account JSON, or workload
identity) ?
6      never from config. A ``client`` may be injected for tests.
7      """
8
9      from __future__ import annotations
10
11     from datetime import timedelta
12     from importlib import import_module
13     from typing import Any, Iterator, Optional
14
15     from backend.storage.base import StorageBackend
16     from backend.storage.ref import StorageRef, StorageStat
17
18
19     class GCSStorage(StorageBackend):
20         def __init__(
21             self,
22             *,
23             client: Any = None,
24             project: Optional[str] = None,
25             signed_url_ttl: int = 3600,
26             **_ignored,
27         ) -> None:
28             self._ttl = timedelta(seconds=signed_url_ttl)
29             if client is not None:
30                 self._client = client
31             return
32
33             storage_module: Any = import_module("google.cloud.storage")
34             self._client = storage_module.Client(project=project)
35
36         def _blob(self, ref: StorageRef):
37             return self._client.bucket(ref.bucket).blob(ref.key)
38
39         def fetch_to_local(self, ref, dst=None, *, overwrite=False) -> str:
40             if dst is None:
41                 raise ValueError("GCSStorage.fetch_to_local requires a dst path")
42             import os
43
44             dst = str(dst)
45             if os.path.exists(dst) and not overwrite:
46                 return dst
```

221. user (2026-07-04T10:49:13.645Z)

Preparing worktree (new branch 'fix/gcs-keyless-signblob-scope')
branch 'fix/gcs-keyless-signblob-scope' set up to track 'origin/master'.
HEAD is now at 195932f deploy(cloudrun): commit the control-plane service-manifest generator
(closes #473) (#474)
195932f deploy(cloudrun): commit the control-plane service-manifest generator (closes #473)
(#474)

222. user (2026-07-04T10:49:21.718Z)

```
76     def signed_url(self, ref, *, ttl=None, method="GET") -> str:
77         blob = self._blob(ref)
78         exp = timedelta(seconds=ttl or self._ttl)
```

```

79         creds = getattr(self._client, "_credentials", None)
80         # Local-key signing (a service-account JSON key exposes a `signer`) works out of
the box. But
81         # on Cloud Run / GCE the ADC credentials come from the metadata server and have
NO private key,
82         # so a plain generate_signed_url raises "you need a private key". Fall back to
the IAM signBlob
83         # API by passing the SA email + a fresh access token ? which requires the
runtime SA to hold
84         # roles/iam.serviceAccountTokenCreator on ITSELF. Best-effort: if we can't build
that, defer to
85         # the default call (which will raise a clear error rather than silently mis-
sign).
86         if creds is not None and not hasattr(creds, "signer"):
87             try:
88                 from google.auth.transport.requests import Request as _AuthRequest
89
90                 if not getattr(creds, "token", None):
91                     creds.refresh(_AuthRequest())
92                 sa_email = getattr(creds, "service_account_email", None)
93                 if sa_email and sa_email != "default":
94                     return blob.generate_signed_url(
95                         version="v4",
96                         expiration=exp,
97                         method=method,
98                         service_account_email=sa_email,
99                         access_token=creds.token,
100                     )
101             except Exception: # pragma: no cover - exercised only on real Cloud Run
creds
102                 pass
103         return blob.generate_signed_url(version="v4", expiration=exp, method=method)
104

```

223. user (2026-07-04T10:49:42.637Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-signblob\backend\storage\gcs.py has been updated successfully. (file state is current in your context ? no need to Read it back)

224. user (2026-07-04T10:49:51.846Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-signblob\backend\storage\gcs.py has been updated successfully. (file state is current in your context ? no need to Read it back)

225. assistant (2026-07-04T10:50:13.791Z)

Now the regression tests, locking the exact root-cause contract.

226. user (2026-07-04T10:50:16.251Z)

```

class _FakeGCSClient:
    def __init__(self):
        self.store = {}

    def bucket(self, name):
        return type(
            "_B", (), {"blob": lambda _self, key: _FakeBlob(self.store, name, key)}
        )()

    def list_blobs(self, bucket, prefix=""):
        return [
            _FakeBlob(self.store, bucket, k)
            for (b, k) in self.store

```

227. user (2026-07-04T10:50:24.648Z)

339
349

228. user (2026-07-04T10:50:30.271Z)

```
330
331     def test_module_signed_url_honors_configured_s3_ttl():
332         url = storage.signed_url(
333             "s3://bucket/highlights/vid/reel.mp4",
334             config={"s3": {"client": _FakeS3Client(), "signed_url_ttl": 456}},
335         )
336         assert "X-Amz-Expires=456" in url
337
338
339     def test_module_signed_url_explicit_ttl_overrides_config():
340         url = storage.signed_url(
341             "gcs://bucket/x",
342             config={"gcs": {"client": _FakeGCSClient(), "signed_url_ttl": 123}},
343             ttl=999,
344         )
345         assert "ttl=999" in url
346
347
348     # ----- Google Drive (local
mount)
349     # The backend reads a locally MOUNTED Google Drive (Google Drive for Desktop). A
tmp_path
```

229. user (2026-07-04T10:50:56.003Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-signblob\tests\test_storage.py has been updated successfully. (file state is current in your context ? no need to Read it back)

230. user (2026-07-04T10:51:06.080Z)

```
..... [100%]
38 passed in 0.68s
```

231. user (2026-07-04T10:54:10.429Z)

Required test coverage of 70% reached. Total coverage: 85.46%
1659 passed, 10 skipped, 15 warnings in 138.82s (0:02:18)
All checks passed!
Success: no issues found in 147 source files

232. user (2026-07-04T10:54:28.960Z)

```
fix/gcs-keyless-signblob-scope
195932f deploy(cloudrun): commit the control-plane service-manifest generator (closes #473)
(#474)
[fix/gcs-keyless-signblob-scope 5b9929f] fix(storage): GCS keyless signing mints a cloud-
platform-scoped token for signBlob (closes #475)
 2 files changed, 154 insertions(+), 12 deletions(-)
remote:
remote: Create a pull request for 'fix/gcs-keyless-signblob-scope' on GitHub by visiting:
remote:     https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/fix/gcs-keyless-
signblob-scope
remote:
branch 'fix/gcs-keyless-signblob-scope' set up to track 'origin/fix/gcs-keyless-signblob-scope'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      fix/gcs-keyless-signblob-scope -> fix/gcs-keyless-signblob-scope
```

233. user (2026-07-04T10:54:48.845Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/476>

234. user (2026-07-04T10:54:56.832Z)

Command running in background with ID: baky7rsqf. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\baky7rsqf.output. You will be notified when it completes. To check interim output, use Read on that file path.

235. user (2026-07-04T10:58:03.551Z)

<retrieval_status>success</retrieval_status>

<task_id>baky7rsqf</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>

11
PR476 CI: ALL GREEN
</output>

236. user (2026-07-04T10:58:20.556Z)

MERGEABLE/CLEAN
#476 state=MERGED mergedAt=2026-07-04T10:58:16Z
error: failed to delete 'C:/Users/avidu/Projects/khelsutra-guru/wt-signblob': Permission denied
Deleted branch fix/gcs-keyless-signblob-scope (was 5b9929f).
2 files changed, 154 insertions(+), 12 deletions(-)
3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob
(closes #475) (#476)

237. user (2026-07-04T10:58:34.083Z)

fatal: 'C:\Users\avidu\Projects\khelsutra-guru\wt-signblob' is not a working tree
3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob
(closes #475) (#476)

238. user (2026-07-04T10:58:51.423Z)

False

239. user (2026-07-04T10:58:54.067Z)

Command running in background with ID: bq510ilzr. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\bq510ilzr.output. You will be notified when it completes. To check interim output, use Read on that file path.

240. assistant (2026-07-04T10:59:08.527Z)

Rebuild is running. Checkpointing the e2e progress in memory while it builds.

241. user (2026-07-04T10:59:19.568Z)

The file c:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra-guru\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read

it back)

242. user (2026-07-04T11:09:24.947Z)

```
<retrieval_status>timeout</retrieval_status>
```

<task_id>bq510ilzr</task_id>

```
<task_type>local_bash</task_type>
```

```
<status>running</status>
```

```
<output>
```

BUILD ID: 7a7bd329-8345-4f35-b749-552c64199229

</output>

243. user (2026-07-04T11:09:36.282Z)

```
<retrieval_status>success</retrieval_status>
```

<task_id>bq510ilzr</task_id>

<task_type>local_bash</task_type>

```
<status>completed</status>
```

```
<exit_code>0</exit_code>
```

```
<output>
```

BUILD ID: 7a7bd329-8345-4f35-b749-552c64199229

BUILD TERMINAL: SUCCESS

SUCCESS sha256:31475f27dd1b2d2544e7fa9feb03c73913bd670448cd2e0fc3748d7b499d1843

</output>

244. user (2026-07-04T11:09:59.100Z)

Done.

Done .

URL: <https://rally-control-plane-492532266377.asia-southeast1.run.app>

rally-control-plane-00005-cgr

245. user (2026-07-04T11:12:23.113Z)

```
Creating Revision.....done
Routing traffic.....done
```

Done.
rally-control-plane-00006-qh4
--- deployed digest check ---
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker@sha256:31475f27dd1b2d2544e7fa9feb03c73913bd670448cd2e0fc3748d7b499d1843

246. user (2026-07-04T11:12:42.803Z)

Exit code 23
--- redirect check (no -L): expect 307 + storage.googleapis.com ---
HTTP 404
location:

247. assistant (2026-07-04T11:12:51.504Z)

404 ? the redeploy wiped the ephemeral instance state and something in the route depends on it before the bucket check. Reading the actual route.

248. user (2026-07-04T11:12:52.517Z)

```
1040 # --- Highlight download (B3, PR-2) -----
1041
1042
1043 @app.get("/api/highlights/{video_id}/{filename}")
1044 def download_highlight(video_id: str, filename: str, request: Request):
1045     """Stream a compiled highlight reel ? `filename` is the bare name given to
/api/compile
1046     (which only writes into output_dir; the browser previously got back an unreachable
1047     server-local path)."""
1048     try:
1049         name = safe_output_filename(filename)
1050     except ValueError as e:
1051         raise HTTPException(status_code=400, detail=str(e)) from e
1052     if not request.app.state.db.get_video(video_id):
1053         raise HTTPException(status_code=404, detail="Indexed video record not found.")
1054     # #463: when the reel was mirrored to the output bucket, 307 to a short-lived signed
URL (bytes flow
1055     # client<->store, bypassing Cloud Run's ~32 MiB response cap + the ephemeral local
/tmp). The
1056     # exists/sign logic (with the active storage config) lives in
orchestration.signed_reel_url to keep
1057     # this module lean; best-effort ? any error falls back to the local file below.
1058     try:
1059         signed = signed_reel_url(request.app.state.config, video_id, name)
1060         if signed:
1061             return RedirectResponse(signed, status_code=307)
1062     except Exception as e: # noqa: BLE001 ? signing/exists hiccup ? serve the local
file instead
1063         logger.warning("[highlights] signed-url path failed for %s; serving local: %s",
video_id, e)
1064     output_dir = (
1065         getattr(
1066             getattr(request.app.state.config, "output", None), "output_dir", "output"
1067         )
1068         or "output"
1069     )
1070     path = os.path.join(output_dir, name)
1071     try:
1072         path = resolve_under_root(path, output_dir)
1073     except ValueError as e:
1074         raise HTTPException(status_code=400, detail=str(e)) from e
1075     if not os.path.isfile(path):
1076         raise HTTPException(
1077             status_code=404,
```

```

1078         detail=f"No compiled file named '{name}'. Compile first via /api/compile.",
1079     )
1080     media = "video/quicktime" if name.lower().endswith(".mov") else "video/mp4"
1081     return FileResponse(path, media_type=media, filename=name)
1082
1083
1084     @app.post("/api/query")

```

249. user (2026-07-04T11:13:27.937Z)

```

--- process gs:// video ---
{"job_id":"18dc6502661148fa9ee1b1e7ccaf5526","status":"queued","poll":"/api/jobs/18dc6502661148fa9ee1b1e7ccaf5526"}
--- upload CUJ (720p) ---
upload complete: /tmp/apphome/output/uploads/upload-e2e-720p.mp4
(d8335f51acd9b50c9f15bff3f5fcf4e7)
{"job_id":"19bb8304824b4b85acdd293245a90f60","status":"queued","poll":"/api/jobs/19bb8304824b4b85acdd293245a90f60"}

```

250. user (2026-07-04T11:13:39.329Z)

Command running in background with ID: bfu2g9t4l. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\bfu2g9t4l.output. You will be notified when it completes. To check interim output, use Read on that file path.

251. assistant (2026-07-04T11:13:50.396Z)

Both jobs are polling. Meanwhile I'll draft the e2e runlog, following the structure of the merged #464 report.

252. user (2026-07-04T11:13:52.941Z)

A1 alpha serving deploy ? hosted CUJ run report (cloud-serving)

****Date:**** 2026-07-04 · ****Region:**** asia-southeast1 · ****Project:**** gen-lang-client-0306026826
****Scope:**** A1 (ALPHA_LAUNCH_READINESS §7) ? stand up the alpha serving endpoint and verify the hosted `process ? jobs ? compile ? download` CUJ against the WASB-only owned detector.

> Companion to `DEPLOY\_REPORT\_cloud-serving\_2026-07-04.md` (PR #457, the GPU-batching runlog).
That
> report proved the L4 worker + measured the batching win via the M6 dispatch ****shim****; THIS report
> proves the deployed ****control-plane service**** end-to-end. They do not overlap.

1. What was deployed

Resource	Value
--- ---	
L4 worker **Job**	`rally-worker` ? nvidia-l4, 8 vCPU / 32Gi, `--no-gpu-zonal-redundancy`, `--max-retries 0 --task-timeout 3600`, GCS-FUSE mount `kheisutra-rally-serving`?`/gcs`, `WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar`
Control-plane **Service**	`rally-control-plane` ? rev 00002, `--no-cpu-throttling`, `min/maxScale 1/3`, cpu 2 / mem 4Gi, port 8080, **private (IAM-gated, no allUsers)**
Service image	`asia-southeast1-docker.pkg.dev/?/rally/rally-control-plane:latest` (= `rally-worker:latest` + `google-cloud-run`; see §5 bug 1)
Service URL	`https://rally-control-plane-2bosgeukpa-as.a.run.app`
Config (via base64 startup wrapper ? `RALLY_APP_HOME=/tmp/apphome/config.json`)	 `deployment.profile=cloud-serving`, `compute_target=cloud_run`, `indexing.skip_ai_handoff=true`, `storage.backend=gcs`, `output_root=gs://kheisutra-rally-serving`, `input_backend=gcs`, `input_root=?/videos`
Env	`DEPLOYMENT_PROFILE=cloud-serving`, `CLOUD_RUN_REGION=asia-southeast1`, `GOOGLE_CLOUD_PROJECT=?`, `RALLY_ALLOWED_HOSTS=*` (CLOUD_RUN_JOB is reserved ? code default

`rally-worker` is correct) |

****Boot-crash trap avoided:**** the `cloud-serving` preset forces `indexing.skip\_ai\_handoff=False` (Gemini ON), which would trip the `AI\_HANDOFF\_KEY\_MISSING` startup check (no Gemini key) and crash-loop. Verified `loader.py:194` deep-merges `defaults < preset < config.json`, so the explicit `skip\_ai\_handoff=true` wins. Also pinned `storage.backend=gcs` explicitly to satisfy the `STORAGE\_INCOHERENT` check. Boot log confirmed: `deployment profile ACTIVE: cloud-serving (compute\_target=cloud\_run)` + `Uvicorn running on http://0.0.0.0:8080`.

2. Health / capabilities (authenticated via `gcloud auth print-identity-token`)

```
- `GET /api/health` ? `200`  
{ "status": "ok", "sport": "badminton", "default_segmenter": "yolo_hybrid" }`  
- `GET /api/capabilities` ? `segmenters` incl. `wasb_hybrid`; `deployment.cloud_available=true`;  
  `gemini_key_present=false` (no crash ? skip_ai_handoff honored); `gcs` backend usable.
```

3. gs:// bucket-path CUJ ? PROVEN end-to-end

```
1. `POST /api/process {"video_path": "gs://khelsutra-rally-serving/videos/MahadevpuraSingles.MP4", "segmenter": "wasb_hybrid", "compute": "cloud"}` ? `202`  
{ "job_id": "d16654d1" }`  
2. Poll `GET /api/jobs/{id}`: `validating ? dispatching (cloud_run) ? done`.  
   - CP pulled + validated the video (format/resolution/pts/scene/blur/relevance all pass).  
   - Dispatched L4 Job execution `rally-worker-k49kg`; worker log: `WASB in FULLY-LOCAL mode`  
     (skip_ai_handoff=True) ? no gemini handoff, no API key needed, `torch 1.11.0+cu113 cuda`  
     True`.  
   - **Result: `success` ? "Successfully indexed 48 play segments (cloud_run /`  
     'wasb_hybrid')"***;  
     outputs at `gs://khelsutra-rally-serving/outputs/915473?/`.  
   - Wall: created 02:12:42 ? finished 02:26:27 (**~13m45s**, incl. L4 cold image pull +  
     inference).  
3. `POST /api/compile {video_id, segment_ids:[1..48],`  
   output_filename:"mahadevpura_wasb_highlights.mp4"}` ? `202 { "job_id": "d947a9e6" }` ? poll ?  
   **success**. In-process ffmpeg stitch (trim 48 clips w/ padding ? concat ? watermark) on the  
   CP; **Dynamic highlight reel saved ? (166.0 MB)**; wall ~3m26s (02:27:47?02:31:13, incl. re-  
   fetch of the gs:// source).  
4. `GET /api/highlights/{video_id}/mahadevpura_wasb_highlights.mp4` ? the reel is served  
   (correct `206`/`video/mp4`/etag/size headers), and the **full 174,038,655-byte reel was  
   retrieved via <32 MiB Range requests and fprobes as a VALID video: H.264 1920x1080 @ 29.97fps +  
   AAC, duration 244.3s** (48 rallies). ? **A single full `GET` returns edge-500** ? Cloud Run's  
   ~32 MiB response cap vs. the `FileResponse` fixed `Content-Length` (see §5 bug 3).
```

4. Notes / observations

```
- **In-process compile serializes behind process jobs** (drain  
  `ThreadPoolExecutor(max_workers=1)`):  
  a queued compile waits while a process job holds the worker in its bucket-poll. Fine for the  
  alpha  
  (low concurrency); worth noting for load.  
- **WASB batch tuning is NOT forwarded on the dispatch path.** `_maybe_dispatch_remote` forwards  
  only  
  `skip_ai_handoff` (orchestration.py:274-283); the L4 job runs its baseline batch (not the  
  preset's  
  batch64/prefetch4), so the measured #457 ?30.5% win is not applied via the deployed CP. Minor  
  (correctness unaffected); follow-up = forward `indexing.wasb.*` overrides like  
  `skip_ai_handoff`.  
- `player_pool` / `max_frames` are dropped on the cloud path (already noted at  
  orchestration.py:273).
```

5. Bugs surfaced + handled

253. user (2026-07-04T11:14:01.271Z)

```
1      JOB 18dc6502: running | validating
2      JOB 19bb8304: running | dispatching (cloud_run)
3
```

254. assistant (2026-07-04T11:14:08.193Z)

Both jobs progressing. Drafting the runlog while they run.

255. assistant (2026-07-04T11:14:49.618Z)

A1 alpha serving ? post-merge e2e deploy test (cloud-serving)

```
**Date:** 2026-07-04 . **Region:** asia-southeast1 . **Project:** gen-lang-client-0306026826
**Scope:** A1 (ALPHA_LAUNCH_READINESS $7) closure ? prove the five merged serving fixes
(#465 input jail, #467 upload staging, #468 drain concurrency, #469 signed-URL download,
#470 auth-in-base-image) plus the committed manifest generator (#474) on a REAL deployed build,
then restore the live edge-auth posture.
```

```
> Companion to `DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md` (PR #464): that report stood
the
> stack up and FILED #461/#463/#466; THIS report proves their fixes end-to-end on merged master
and
> records the one new bug the run surfaced (#475, fixed by #476 mid-run).
```

1. What was deployed

```
| Resource | Value |
|---|---|
| Source | merged `master` ? `195932f` (first build) ? `3c97117` (final build incl. #476) |
| Worker image | `~/rally/rally-worker:latest` ? build 1 `sha256:ed2fe448?` (195932f), build 2
**`sha256:31475f27?`** (3c97117); job re-pinned via `gcloud run jobs update` after each build |
| L4 worker **Job** | `rally-worker` ? unchanged shape (nvidia-l4, 8vCPU/32Gi, FUSE `kheIsutra-
rally-serving`~/gcs) |
| Control-plane **Service** | `rally-control-plane` ? rev **00005** (build 1) ? rev **00006**
(final digest `31475f27?`), deployed FROM THE BASE IMAGE directly (no derived PyJWT overlay ?
#470 proven), `maxScale=1` (#471 option 1) |
| Manifest | generated by the **committed** `deploy/cloudrun/gen_service_yaml.py` (#474) ?
`--edge-auth off` for the test window, `--edge-auth on` to restore |
```

2. Boot / health (rev 00005, edge-auth OFF, identity-token auth)

```
- `GET /api/health` ? `200 {"status":"ok",?}`; `GET /api/capabilities` ? `wasb_hybrid` present,
`gemini_key_present=false` (skip_ai_handoff honored), ffmpeg/ffprobe present.
- Before-state also captured on the OLD rev 00004: `api/process` without a CF JWT ? **401**
`edge auth: missing Cf-Access-Jwt-Assertion header` (the M9 posture was live as documented).
```

3. CUJ on the FINAL image (rev 00006) ? all green

```
1. **gs:// process** ? `POST /api/process {gs://?/MahadevpuraSingles.MP4, wasb_hybrid,
compute=cloud}`
? 202 ? `validating ? dispatching (cloud_run) ? done` ? **48 play segments** (same count as
the
#464 run; job `18dc6502?`). <!-- TODO: wall time -->
2. **#461 upload staging** ? chunked upload (`init ? part/0 ? complete`) of a 105,345-byte 720p
clip
? `POST /api/process` with the returned **local** path + `compute=cloud` ? job staged it to
`gs://kheIsutra-rally-serving/videos/d8335f51?/upload-e2e-720p.mp4` (byte-exact) and
dispatched;
**success** (0 segments ? correct for a synthetic test pattern). NOT the pre-fix
"cloud-serving needs a cloud input" hard-fail.
- ? Validation floor: a 640x360 test clip is rejected (`resolution_fps_check`, min 1280x720)
?
```

first attempt failed for that reason; harmless, but alpha docs should mention the floor.

3. ****#468 drain concurrency, live**** ? the upload job staged + dispatched ****while**** the compile of the big reel was stitching on the CP (rev 00005 run) ? remote dispatch overlapped the local compile lane exactly as designed; no serialization, no false-empty stall.

4. ****compile**** ? ``POST /api/compile {video_id, segment_ids:[1..48], "e2e.mp4"}`` ? 202 ? done (~4m54s on rev 00005); reel ****mirrored to** ``gs://khelsutra-rally-serving/highlights/915473?/e2e.mp4`` (174,046,228 bytes ? 166 MiB)******.
 <!-- TODO: rev 00006 compile time -->

5. ****#463 signed-URL download**** ? ``GET /api/highlights/915473?/e2e.mp4`` ?
 <!-- TODO: 307 + location + full download size + ffprobe -->

4. Bug surfaced mid-run: #475 ? fixed by #476 (merged, redeployed)

On the FIRST attempt (rev 00005, build `ed2fe448?` = pre-#476), the #463 check ****failed****:
``GET /api/highlights/...`` ? ****500****, log:
``[highlights] signed-url path failed ?: you need a private key to sign credentials``.

- The reel WAS mirrored correctly; the ****keyless signBlob path**** failed and the handler fell back to serving the 166 MiB file locally ? Cloud Run's ~32 MiB cap ? edge-500 (the original #463 symptom, resurrected by a different root cause).
- ****Root cause (verified live):**** ``GCSStorage.signed_url()`` reused the storage client's own token for ``iamcredentials.signBlob``; ``google.cloud.storage.Client.SCOPE`` is ****devstorage-only****, which ``signBlob`` rejects (``ACCESS_TOKEN_SCOPE_INSUFFICIENT``) ? and the branch's bare ``except: pass`` swallowed it, so the fallback raised the misleading private-key error. Environment was clean:
 - the SA holds self-``serviceAccountTokenCreator`` and ``iamcredentials.googleapis.com`` is enabled.
- ****Why tests never saw it:**** the fakes carry no ``_credentials``; this was the FIRST run of #469's path on real compute credentials. Filed ****#475****, fixed by ****#476**** (``_signblob_identity()`` mints a fresh ``cloud-platform``-scoped ADC credential; failures now LOGGED), verified 1659?/85.46%, merged on green CI, rebuilt (``31475f27?``), redeployed as rev 00006.

5. Notes / observations (rough edges captured)

- ****`services replace` with an unchanged manifest is a NO-OP**** ? it does not re-resolve ``:latest``. After a rebuild, force a new revision (``gcloud run services update --image ? :latest``) or stamp a revision annotation. Follow-up candidate: a ``--revision-suffix`` in ``gen_service_yaml.py``.
- ****A redeploy wipes the per-instance SQLite state**** ? ``/api/highlights`` 404s on the DB record check (``main.py:1052``) even though the reel mirror exists in ``gs://``. Known + filed as ****#471****; the CUJ was re-run on rev 00006 to regenerate state. Worth noting for alpha operators: a deploy invalidates in-flight jobs and past download URLs (the SPA library repopulates on re-process).
- Validation floor for uploads (?1280x720) ? see §3.2.
- The 422-before-401 quirk: a ****malformed-JSON**** body gets FastAPI's 422 before the edge-auth dependency runs; any valid request without a CF JWT still 401s. No exposure; noted for triage literacy.

6. Edge-auth restore + live posture

<!-- TODO: services replace auth-on (new revision), 401 verify, api.khelsutra.guru 302 verify -->

7. A1 DoD checklist

DoD item	Status
---	---

```
| Serving endpoint deployed + healthy | <!-- TODO --> |  
| Upload ? process ? jobs ? compile ? download CUJ | <!-- TODO --> |  
| Edge-auth (M9) restored + verified | <!-- TODO --> |  
| >32 MiB download via signed URL (#463) | <!-- TODO --> |  
| Browser-upload staging to gs:// (#461) | ? |  
| Parallel drain (#466/#468) observed live | ? |
```

256. user (2026-07-04T11:14:49.650Z)

File created successfully at: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\scratchpad\DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md (file state is current in your context ? no need to Read it back)

257. user (2026-07-04T11:24:56.625Z)

```
<retrieval_status>timeout</retrieval_status>
```

```
<task_id>bfu2g9t4l</task_id>
```

```
<task_type>local_bash</task_type>
```

```
<status>running</status>
```

```
<output>
```

```
JOB 18dc6502: running | validating  
JOB 19bb8304: running | dispatching (cloud_run)  
JOB 18dc6502: running | dispatching (cloud_run)  
JOB 19bb8304: done | finished  
JOB 19bb8304 TERMINAL: success - Successfully indexed 0 play segments (cloud_run /  
'wasb_hybrid').  
</output>
```

258. user (2026-07-04T11:27:57.253Z)

```
<retrieval_status>success</retrieval_status>
```

```
<task_id>bfu2g9t4l</task_id>
```

```
<task_type>local_bash</task_type>
```

```
<status>completed</status>
```

```
<exit_code>0</exit_code>
```

```
<output>
```

```
JOB 18dc6502: running | validating  
JOB 19bb8304: running | dispatching (cloud_run)  
JOB 18dc6502: running | dispatching (cloud_run)  
JOB 19bb8304: done | finished  
JOB 19bb8304 TERMINAL: success - Successfully indexed 0 play segments (cloud_run /  
'wasb_hybrid').  
JOB 18dc6502: done | finished  
JOB 18dc6502 TERMINAL: success - Successfully indexed 48 play segments (cloud_run /  
'wasb_hybrid').  
BOTH JOBS TERMINAL  
</output>
```

259. user (2026-07-04T11:28:08.046Z)

```
{"job_id":"2289271edd8e44ac81fd4b78c8dbe70a","status":"queued","poll":"/api/jobs/2289271edd8e44a  
c81fd4b78c8dbe70a"}  
HTTP 202
```

260. user (2026-07-04T11:28:18.477Z)

Command running in background with ID: b59imjubi. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\b59imjubi.output. You will be notified when it completes. To check interim output, use Read on that file path.

261. user (2026-07-04T11:31:36.277Z)

```
<retrieval_status>success</retrieval_status>

<task_id>b59imjubi</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>
COMPILE2: running | stitching
COMPILE2: done | finished
TERMINAL: {"job_id":"2289271edd8e44ac81fd4b78c8dbe70a","status":"done","progress":"finished","video_id":"915473794e0909c9f95a01e4511a5e98","error":null,"result":{"status":"success","filepath":"/tmp/apphome/output/e2e.mp4","download_url":"/api/highlights/915473794e0909c9f95a01e4511a5e98/e2e.mp4","video_id":"915473794e0909c9f95a01e4511a5e98"},"reports":null,"created_at":"2026-07-04 11:28:08","started_at":"202
</output>
```

262. assistant (2026-07-04T11:31:43.922Z)

Compile done. The decisive check: does `/api/highlights` now 307-redirect and deliver the full 166 MiB reel?

263. user (2026-07-04T11:31:50.715Z)

```
Exit code 23
--- redirect (no -L) ---
HTTP 307
location: https://storage.googleapis.com/khelsutra-rally-serving/highlights/915473794e0909c9f95a01e4511a5e98/e2e.mp4?X-Goog-Algorithm=G00G4-RSA-SHA256&X-Goog-Credential=492532266377-compute%40developer.gserviceaccount.com%2F20260704%2Fauto%2Fstorage%2Fgoog4_request&X-Goog-Date=20260704T113151Z&X-Goog-Expires=3600&X-Goog-SignedHeaders=host&X-Goog-Signature=<redacted>
```

264. user (2026-07-04T11:32:16.006Z)

```
final HTTP 200, 174046228 bytes in 8.103722s
codec_name=h264
width=1920
height=1080
avg_frame_rate=1464000/48853
codec_name=aac
avg_frame_rate=0/0
codec_name=unknown
avg_frame_rate=30000/1001
duration=244.286000
size=174046228
```

265. user (2026-07-04T11:32:59.691Z)

```
Creating Revision.....
.....done
Routing traffic.....done
revision: rally-control-plane-00007-8hs
```

False

fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
fatal: not a git repository (or any of the parent directories): .git
--- dir timestamps ---

Saturday, July 4, 2026 8:11:40 AM

--- branch tip ---

3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob
(closes #475) (#476)

272. user (2026-07-04T11:34:46.624Z)

HEAD is now at 3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob (closes #475) (#476)

3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob
(closes #475) (#476)

docs/a1-e2e-runlog

273. assistant (2026-07-04T11:34:52.319Z)

Now finalizing the runlog with all measured results and writing it into the repo, plus the A1 flip.

274. user (2026-07-04T11:34:52.950Z)

```
92
93     ## 7. Deliverables & progress tracker
94
95     Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
96
97     | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
98     |----|-----|-----|-----|-----|-----|----|
99     | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | DONE | #451 |
100    | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
101    | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | DONE | #452 |
102    | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | DONE | collector #62 |
103    | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | DONE | #453 |
104    | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
105    | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
```

275. user (2026-07-04T11:35:03.177Z)

The file C:\Users\avidu\Projects\khelsutra-guru\wt-a1-runlog\docs\ALPHA_LAUNCH_READINESS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

276. assistant (2026-07-04T11:35:51.881Z)

A1 alpha serving ? post-merge e2e deploy test (cloud-serving)

Date: 2026-07-04 . **Region:** asia-southeast1 . **Project:** gen-lang-client-0306026826

****Scope:**** A1 (ALPHA_LAUNCH_READINESS \$7) closure ? prove the merged serving fixes (#465 input jail, #467 upload staging, #468 drain concurrency, #469 signed-URL download, #470 auth-in-base-image) plus the committed manifest generator (#474) on a REAL deployed build, then restore the live edge-auth posture.

> Companion to `DEPLOY\_REPORT\_cloud-serving\_a1-cuj\_2026-07-04.md` (PR #464): that report stood the
> stack up and FILED #461/#463/#466; THIS report proves their fixes end-to-end on merged master and
> records the one new bug the run surfaced (#475, fixed by #476 mid-run).

1. What was deployed

Resource	Value
Source	merged `master` ? `195932f` (first build) ? **`3c97117`** (final build, incl. #476)
Worker image	`?/rally/rally-worker:latest` ? build 1 `sha256:ed2fe448?` (195932f), build 2 **`sha256:31475f27?`** (3c97117); the Job re-pinned via `gcloud run jobs update` after each build
L4 worker ****Job****	`rally-worker` ? unchanged shape (nvidia-l4, 8 vCPU/32Gi, FUSE `khelsutra-rally-serving`?)`gcs`)
Control-plane ****Service****	`rally-control-plane` ? rev **00005** (auth-off, build 1) ? rev **00006** (auth-off, final digest) ? rev **00007** (auth-ON restore, final digest). Deployed FROM THE BASE IMAGE directly ? no derived PyJWT overlay (**#470 proven**); `maxScale=1` (#471 option 1)
Manifest	generated by the ****committed**** `deploy/cloudrun/gen\_service\_yaml.py` (#474) ? `--edge-auth off` for the test window, `--edge-auth on` to restore. The scratchpad-artifact gap (#473) is closed: this deploy used only repo-committed tooling

2. Boot / health (rev 00005+, identity-token auth)

- `GET /api/health` ? `200 {"status":"ok",?}`; `GET /api/capabilities` ? `wasb\_hybrid` present, `gemini\_key\_present=false` (skip_ai_handoff honored), ffmpeg/ffprobe present.
- Before-state captured on the outgoing rev 00004: `/api/process` without a CF JWT ? ****401**** `edge auth: missing Cf-Access-Jwt-Assertion header` ? the M9 posture was live as documented.

3. CUJ on the FINAL image (rev 00006, digest `31475f27?`) ? all green

- **gs:// process**** ? `POST /api/process {gs://?/MahadevpuraSingles.MP4, wasb\_hybrid, compute=cloud}` ? 202 ? `validating` ? dispatching (cloud_run) ? done` ? **\*\*48 play segments\*\*** (same count as the #464 run; job `18dc6502?`).
- **#461 upload staging**** ? chunked upload (`init` ? part/0 ? complete`) of a 105,345-byte 720p clip ? `POST /api/process` with the returned **\*\*local\*\*** path + `compute=cloud` ? the job staged
it to `gs://khelsutra-rally-serving/videos/d8335f51?/upload-e2e-720p.mp4` (byte-exact size verified in the bucket) and dispatched to the L4 ? ****success**** (0 segments ? correct for a synthetic test pattern). NOT the pre-fix "cloud-serving needs a cloud input" hard-fail. Verified on BOTH builds (`ed2fe448?` and the final `31475f27?`).
- ? Upload validation floor: a 640x360 test clip is rejected (`resolution\_fps\_check`, min 1280x720) ? the first attempt failed on this; alpha docs should mention the floor.
- **#468 drain concurrency, observed live**** ? the upload job staged + dispatched ****while**** the 48-segment reel compile was stitching on the CP: remote dispatch overlapped the local compile lane exactly as designed (no serialization, no false-empty stall). On rev 00006 the two process jobs (gs:// + upload) also ran the drain concurrently.
- **compile**** ? `POST /api/compile {video\_id, segment\_ids:[1..48], "e2e.mp4"}` ? 202 ? done; measured stitch wall on rev 00005: created 10:39:32 ? finished 10:44:26 (****~4m54s****, incl. the
gs:// source re-fetch); rev 00006 re-run equally green. Reel ****mirrored to**
`gs://khelsutra-rally-serving/highlights/915473?/e2e.mp4` ? 174,046,228 bytes (~166 MiB).
- **#463 signed-URL download ? VERIFIED****:
- `GET /api/highlights/915473?/e2e.mp4` ? ****307`**** ?
`https://storage.googleapis.com/khelsutra-rally-serving/highlights/?/e2e.mp4?X-Goog-Algorithm=G00G4-RSA-SHA256&X-Goog-Credential=492532266377-compute@?&X-Goog-Expires=3600&X-Goog-

Signature=?`

(v4, signed AS the runtime SA via keyless signBlob ? no key file anywhere).

- `curl -L` through the redirect: ****full 174,046,228 bytes in 8.1s**** (single GET ? no 32 MiB edge-500, no Range workaround), and the file ****ffprobes valid****: H.264 1920x1080 @ 29.97 fps
- + AAC, duration 244.29s (the 48-rally reel).

4. Bug surfaced mid-run: #475 ? fixed by #476 (merged, redeployed)

On the FIRST attempt (rev 00005, build `ed2fe448?` = pre-#476), the #463 check ****failed****:

`GET /api/highlights/...` ? ****500****; log:

`[highlights] signed-url path failed ?: you need a private key to sign credentials`.

- The reel WAS mirrored correctly; the ****keyless signBlob path**** failed and the handler fell back to serving the 166 MiB file locally ? Cloud Run's ~32 MiB cap ? edge-500 (the original #463 symptom, resurrected by a different root cause).
- ****Root cause (verified live):**** `GCSStorage.signed\_url()` reused the storage client's own token for `iamcredentials.signBlob`; `google.cloud.storage.Client.SCOPE` is ****devstorage-only****, which signBlob rejects (`ACCESS\_TOKEN\_SCOPE\_INSUFFICIENT`) ? and the branch's bare `except: pass` swallowed the real error, so the fallback raised the misleading private-key message. Environment was clean: the SA holds self-`serviceAccountTokenCreator` and `iamcredentials.googleapis.com` is enabled ? code bug, not config.
- ****Why tests never saw it:**** the fakes carry no `\_credentials`; this was the FIRST run of #469's path on real compute credentials. Filed ****#475**** ? fixed by ****#476**** (`\_signblob\_identity`) mints a fresh `cloud-platform`-scoped ADC credential; failures now LOGGED before falling back; regression tests reproduce the credential shape). Suite 1659? / cov 85.46% / ruff+mypy clean; merged on green CI per the standing agreement; rebuilt (`31475f27?`) and redeployed (rev 00006);
- \$3.5 is the pass on the fixed build.

5. Notes / observations (rough edges captured)

- ****`services replace` with an unchanged manifest is a NO-OP**** ? it does not re-resolve `:latest` after a rebuild. Force a new revision (`gcloud run services update --image ?:latest`) or stamp a revision suffix. Follow-up candidate: `--revision-suffix` in `gen\_service\_yaml.py`.
- ****A redeploy wipes the per-instance SQLite state**** ? `/api/highlights` then 404s on the DB record check (`backend/main.py:1052`) even though the reel mirror exists in gs://. Known, filed as ****#471****; the CUJ was re-run on rev 00006 to regenerate state. Alpha-operator implication: a deploy invalidates in-flight jobs and past download URLs (re-process regenerates them).
- Upload validation floor (?1280x720) ? \$3.2.
- 422-before-401 quirk: a ****malformed-JSON**** body gets FastAPI's 422 before the edge-auth dependency runs; any well-formed request without a CF JWT still 401s. No exposure; noted so a 422 in edge-auth triage isn't misread.

6. Edge-auth restore + live posture (rev 00007, digest `31475f27?`)

- Redeployed with the generator's `--edge-auth on` manifest ? rev ****00007****.
- `GET /api/health` ? ****200**** (deliberately exempt).
- `POST /api/process` (valid JSON, no CF JWT) ? ****401**** `edge auth: missing Cf-Access-Jwt-Assertion header`.
- `https://api.khelsutra.guru/api/health` (no CF session) ? ****302**** ? `https://khelsutra.cloudflareaccess.com/cdn-cgi/access/login/api.khelsutra.guru??` with the Access-app aud `c980da67?082b` in the login meta ? the exact pre-test posture, restored.

7. A1 DoD checklist

```
| DoD item | Status |
|---|---|
| Serving endpoint deployed + healthy | ? rev 00007, final digest `31475f27?` |
| Upload ? process ? jobs ? compile ? download CUJ | ? end-to-end on the final image |
| Edge-auth (M9) restored + verified | ? 401 sans JWT; api.khelsutra.guru ? CF login 302 |
| >32 MiB download via signed URL (#463) | ? 307 + full 174 MB single-GET + ffprobe-valid |
| Browser-upload staging to gs:// (#461) | ? staged byte-exact + dispatched + success |
| Parallel drain (#466/#468) observed live | ? dispatch overlapped compile; 2 process jobs
drained concurrently |
```

****A1 ? DONE**** (flipped in `docs/ALPHA\_LAUNCH\_READINESS.md` §7 in this PR).

277. user (2026-07-04T11:35:51.913Z)

File created successfully at: C:\Users\avidu\Projects\khelsutra-guru\wt-a1-runlog\docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md (file state is current in your context ? no need to Read it back)

278. user (2026-07-04T11:37:48.049Z)

[check-md-links] clean ? no broken internal links in live docs.
1659 passed, 10 skipped, 8 warnings in 105.36s (0:01:45)
All checks passed!

279. user (2026-07-04T11:38:02.413Z)

```
docs/a1-e2e-runlog
3c97117 fix(storage): GCS keyless signing mints a cloud-platform-scoped token for signBlob
(closes #475) (#476)
warning: in the working copy of 'docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md', LF will be replaced by CRLF the next time Git touches it
[docs/a1-e2e-runlog d80790d] docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE
 2 files changed, 117 insertions(+), 1 deletion(-)
 create mode 100644 docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md
d80790d docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE
remote:
remote: Create a pull request for 'docs/a1-e2e-runlog' on GitHub by visiting:
remote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/docs/a1-e2e-runlog
remote:
branch 'docs/a1-e2e-runlog' set up to track 'origin/docs/a1-e2e-runlog'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]      docs/a1-e2e-runlog -> docs/a1-e2e-runlog
```

280. user (2026-07-04T11:38:23.154Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/477>

281. user (2026-07-04T11:38:34.094Z)

Command running in background with ID: b0jys7ujb. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\b0jys7ujb.output. You will be notified when it completes. To check interim output, use Read on that file path.

282. user (2026-07-04T11:41:40.723Z)

<retrieval_status>success</retrieval_status>

<task_id>b0jys7ujb</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>

PR477 CI: ALL GREEN (11 checks)

</output>

283. user (2026-07-04T11:41:56.882Z)

#477 state=MERGED mergedAt=2026-07-04T11:41:53Z

Deleted branch docs/a1-e2e-runlog (was d80790d).

create mode 100644 docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md

122f174 docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE (#477)

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer

122f174 [master]

C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/cfg-wt 20b2d5a [claude/wasb-config-validation]

C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/compile-wt b09a1d4 [claude/compile-reliability]

C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-pre-wt 355dad6 (detached HEAD)

C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-wt 5de1ef2 (detached HEAD)

284. assistant (2026-07-04T11:42:09.160Z)

[#477](https://github.com/Khelsutra/badminton-highlight-indexer/pull/477) is merged ? **A1 is officially DONE** (master `122f174`). Final housekeeping: memory refresh and task closure.

285. user (2026-07-04T11:42:09.781Z)

Updated task #3 status

286. user (2026-07-04T11:42:10.109Z)

Updated task #4 status

287. user (2026-07-04T11:42:12.865Z)

(PowerShell completed with no output)

288. user (2026-07-04T11:42:35.781Z)

<tool_use_error>String to replace not found in file.

String: > Last refreshed: 2026-07-04 (pickup session, post-fix checkpoint). **A1 alpha serving is DEPLOYED +

> LIVE behind Cloudflare Access. #465 is MERGED (owner LGTM). All open review blockers are FIXED,

> pushed, and replied-to; #467/#468/#469/#470/#474 await LGTM.** Remaining: merge on LGTM ? (B) full e2e

> deploy test. See [[current-state]].

? YOU ARE HERE ? watch LGTMs ? merge ? (B) full e2e test

The A1 stack is live (see "Deployed state" below). Of the original five bug-fix PRs, **#462, #464,

#465 are MERGED**. **#467, #468, #469 are OPEN with every review comment addressed (incl. the round-2

blockers) on heads c9bf37e / 21ee141 / ce450e5; plus new #470** (bake `auth` extra into the

```

cloudrun
base image). All MERGEABLE vs post-#465 master. Do NOT merge until each has an LGTM. After all
merge ?
run the e2e. Rough-edge issues filed this session: **#471** (per-instance SQLite CP state at
maxScale>1),
**#472** (cert-renewal watch-item ~2026-09-30), **#473** (commit the service-YAML generator).

**FIRST STEPS for the next session:** `git -C C:\Users\avidu\Projects\khelsutra-guru\badminton-
highlight-indexer
pull --ff-only`; `gh auth status` (should be `avidullu`); read this whole file; then start
watching the
open PRs (protocol below). Any review watcher loop from a prior session does NOT carry over;
just poll
the PRs yourself (loop or periodic check).
(note: Edit also tried swapping \uxxxx escapes and their characters; neither form matched, so
the mismatch is likely elsewhere in old_string. Re-read the file and copy the exact surrounding
text.)</tool_use_error>

```

289. assistant (2026-07-04T11:42:42.060Z)

The handoff file changed under me ? possibly a sibling session refreshed it. Re-reading before editing.

290. user (2026-07-04T11:42:42.989Z)

```

1      ---
2      name: session-handoff
3      description: "Living handoff: current state, next steps, ramp-up kit, and key
decisions."
4      metadata:
5          node_type: memory
6          type: project
7      ---
8
9      # Session handoff ? khelsutra-guru
10
11      > Last refreshed: 2026-07-04 (pickup session, post-fix checkpoint). **A1 alpha serving
is DEPLOYED +
12      > LIVE behind Cloudflare Access. #465 is MERGED (owner LGTM). All open review blockers
are FIXED,
13      > pushed, and replied-to; #467/#468/#469/#470 await LGTM.** Remaining: merge on LGTM ?
(B) full e2e
14      > deploy test. See [[current-state]].
15
16      ## ? YOU ARE HERE ? watch LGTMs ? merge ? (B) full e2e test
17
18      The A1 stack is live (see "Deployed state" below). Of the original five bug-fix PRs,
**#462, #464,
19      #465 are MERGED**. **#467, #468, #469 are OPEN with every review comment addressed
(incl. the round-2
20      blockers) on heads c9bf37e / 21ee141 / ce450e5; plus new #470** (bake `auth` extra into
the cloudrun
21      base image). All MERGEABLE vs post-#465 master. Do NOT merge until each has an LGTM.
After all merge ?
22      run the e2e. Rough-edge issues filed this session: **#471** (per-instance SQLite CP
state at maxScale>1),
23      **#472** (cert-renewal watch-item ~2026-09-30), **#473** (commit the service-YAML
generator).
24
25      **FIRST STEPS for the next session:** `git -C C:\Users\avidu\Projects\khelsutra-
guru\badminton-highlight-indexer
26      pull --ff-only`; `gh auth status` (should be `avidullu`); read this whole file; then
start watching the
27      open PRs (protocol below). Any review watcher loop from a prior session does NOT carry
over; just poll

```

```

28     the PRs yourself (loop or periodic check).
29
30     ---
31
32     ## (A) PR-REVIEW RESOLUTION ? remaining open PRs
33
34     Repo `Khelsutra/badminton-highlight-indexer`. Reviewer = **avidullu** (the owner).
    "LGTM" arrives as a
35     PR comment or a formal approval. Merge convention = **squash** (`gh pr merge <n>
    --squash --delete-branch`).
36
37     | PR | branch | fixes | head | review state |
38     |---|---|---|---|---|
39     | ~~#465~~ | ~~`fix/allowed-video-root-uri`~~ | hosted-mode input jail vs gs:// | ? |
**MERGED 2026-07-04 09:03Z** (owner LGTM + 11/11 CI) |
40     | **#467** | `fix/upload-gcs-staging` | closes #461 (upload?gs:// staging) | c9bf37e |
    round-2 blocker (input_root-scheme gate) ADDRESSED; awaiting LGTM |
41     | **#468** | `fix/drain-concurrency` | closes #466 (parallel videos) | 21ee141 | round-2
    blocker (claim_due_job lost-race false-empty) ADDRESSED; awaiting LGTM |
42     | **#469** | `fix/highlights-signed-url` | closes #463 (32 MiB download ? signed URL) |
    ce450e5 | round-2 blocker (configured signed_url_ttl ignored) ADDRESSED; awaiting LGTM |
43     | **#470** | `fix/dockerfile-auth-extra` | base image M9-ready (auth extra) | c63d122 |
    NEW this session; 11/11 green (first push tripped the Dockerfile structural guard ? guard
    updated). Once merged, the e2e SKIPS the derived-image step |
44     | **#474** | `feat/cloudrun-service-manifest` | closes #473 (gen_service_yaml.py + tests
    + README §3a) | 9de18d5 | NEW this session; encodes maxScale=1 (#471 opt-1), no CLOUD_RUN_JOB,
    explicit --edge-auth. Once merged, the e2e regenerates the YAML from the repo |
45

```

291. user (2026-07-04T11:43:06.962Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

292. user (2026-07-04T11:43:23.486Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

293. user (2026-07-04T11:43:41.082Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

294. assistant (2026-07-04T11:44:07.421Z)

****A1 is done ? all PRs merged after full verification, and the end-to-end service run passed on the final deployed build with edge-auth restored.****

What happened

****Merges (each verified before merging).**** For every PR I ran the ***merge result*** against master locally in a temp worktree ? full suite with coverage, ruff, mypy ? before squash-merging, per your pre-LGTM gate. Baseline master was 85.40% coverage; the results:

[#467](https://github.com/Khelsutra/badminton-highlight-indexer/pull/467) 1637 passed / 85.38% (?0.02pp, the new defensive `except` branch ? noise),

[#468](https://github.com/Khelsutra/badminton-highlight-indexer/pull/468) 1645 / ****85.44%****,

[#469](https://github.com/Khelsutra/badminton-highlight-indexer/pull/469) 1652 / 85.39%,

[#470](https://github.com/Khelsutra/badminton-highlight-indexer/pull/470) 1652 + md-links clean,

[#474](https://github.com/Khelsutra/badminton-highlight-indexer/pull/474) 1655 / 85.39%. All 11 CI checks were green on every head; the feared `orchestration.py`/`main.py` overlaps auto-merged cleanly and the suites proved no semantic conflicts. Issues #461/#463/#466/#473 auto-closed.

****The e2e (north star).**** Rebuilt the worker image from merged master, re-pinned the L4 job, deployed the control plane from the base image via the newly committed ``gen_service_yaml.py`` ? zero scratchpad artifacts, proving #470 and #474. Then ran the full CUJ:

- ****gs:// video ? 48 rallies**** on the L4 (same count as the prior run), compile ? reel mirrored to gs:// (166 MiB).
- ****#461 verified:**** chunked upload staged byte-exact to ``videos/<id>/?`` and dispatched ? on both builds.
- ****#468 verified live:**** the upload job dispatched **while** the compile was stitching; two process jobs later drained concurrently.
- ****#463 found broken, then fixed and verified.**** The first redirect attempt ****500'd** on real credentials^{**}: the keyless signBlob path rode the storage client's devstorage-scoped token (rejected by ``iamcredentials``), a bare ``except`` swallowed the real error, and the handler fell back to local serving ? the 32 MiB cap. Filed [#475](https://github.com/Khelsutra/badminton-highlight-indexer/issues/475), fixed in [#476](https://github.com/Khelsutra/badminton-highlight-indexer/pull/476) (fresh ``cloud-platform``-scoped credential + loud logging; 1659 passed / 85.46%; merged on green CI per the standing agreement), rebuilt, redeployed. Result: ****307 ? signed `storage.googleapis.com` URL, full 174,046,228 bytes in 8.1s single GET, fprobe-valid**** 1080p29.97+AAC, 244.3s.
- ****Edge-auth restored**** (rev 00007, same fixed digest): health 200, process-without-JWT 401, ``api.khelsutra.guru`` ? CF Access login 302 with the correct app aud. The stack is left running per your standing instruction.

****Record:**** runlog + A1?DONE flip merged as [#477](https://github.com/Khelsutra/badminton-highlight-indexer/pull/477); master is ``122f174``.

Rough edges captured along the way

Fixed: #475 (signBlob scope). Filed and open: [#471](https://github.com/Khelsutra/badminton-highlight-indexer/issues/471) ? per-instance SQLite state (it bit during the e2e: a redeploy wipes the DB so ``/api/highlights`` 404s despite the gs:// mirror; alpha runs `maxScale=1`) and [#472](https://github.com/Khelsutra/badminton-highlight-indexer/issues/472) ? cert-renewal watch before ~Sept 30. Documented in the runlog: ``services replace`` with an unchanged manifest silently doesn't re-resolve ``:latest`` (a ``--revision-suffix`` for the generator is a good follow-up), and uploads below 1280x720 fail validation ? worth a line in the alpha docs.

All worktrees and branches from this session are cleaned up, and the memory handoff points at the next tracks: the A13 launch review, the khelsutra UX backlog (async ``/api/compile`` #329), and the #471 durable-state fix.

295. user (2026-07-04T11:44:09.799Z)

Post that pls proceed to the next stage and lmk how it goes! The \$300 budget to use GCloud and Gemini API key is still available to you so ensure thorough testing

296. assistant (2026-07-04T11:45:34.174Z)

Picking up from A1 ? the next stage per the tracker is ****A13: the alpha launch review**** (runbook, observability/job-failure triage, known limits, GO/NO-GO recommendation) ? and with the budget + Gemini key available, I'll back it with a thorough live test battery on the deployed stack, including the Gemini handoff leg that the WASB-only alpha e2e deliberately skipped. First, grounding in the tracker's remaining rows.

297. user (2026-07-04T11:45:38.555Z)

Already up to date.

ID	Deliverable	Repo / owner	Depends on	Gated?	Status	PR
A0	Create this alpha readiness tracker and index it in docs	`badminton-highlight-indexer`				
-	No	DONE	#451			
A1	Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth					
ON	upload -> process -> jobs -> compile -> download verified	`khelsutra` + `badminton-highlight-indexer`				
A0	Owner spend / CF config	DONE				
#462/#464/#465/#467/#468/#469/#470/#474/#476;	e2e runlog					

```
`docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md` |
| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that
fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No |
DONE | #452 |
| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile
collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` +
vault | A2 | Owner upload/storage scope | DONE | collector #62 |
| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | DONE | #453 |
| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for
future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the
n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | DONE | #458 |
| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | DONE | #458 |
| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-
serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | DONE | #458 |
| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold,
significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5,
A8 | Owner decision | GATED | - |
| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion,
`train/reeval` job, delta-F1 surface, and alignment with `COMPUTE_DECOUPLED_SERVING` M11 |
`khealsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen
(`khealsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
| A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure triage,
known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner
launch decision | TODO | - |
```

8. Concrete next PR sequence

1. ****Docs PR:**** land this tracker.
2. ****Corpus portability PR:**** rebase/portable manifest tooling plus missing-file smoke test.
3. ****Hosted alpha PR/batch:**** health/auth/upload/process/job/compile/download verification and runbook updates.

298. user (2026-07-04T11:46:31.097Z)

Task #5 created successfully: Live thorough-test battery on deployed stack (failure modes + concurrency + Gemini leg)

299. user (2026-07-04T11:46:33.472Z)

Task #6 created successfully: A13 alpha launch review doc + PR (GO/NO-GO recommendation for owner)

300. user (2026-07-04T11:46:40.851Z)

```
gemini-api-key
--- cert status (#472 check while here) ---
Ready;CertificateProvisioned;DomainRoutable      True;True;True
```

301. user (2026-07-04T11:47:45.394Z)

```
<tool_use_error>InputValidationError: Workflow failed due to the following issue:
An unexpected parameter `run_in_background` was provided</tool_use_error>
```

302. user (2026-07-04T11:48:25.904Z)

Workflow launched in background. Task ID: w3atf3i6j
Summary: Fan out 4 research agents over the repos to draft the A13 alpha launch review sections
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\subagents\workflows\wf_d2c10282-601
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\workflows\scripts\A13-launch-review-inputs-wf_d2c10282-601.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\workflows\scripts\A13-launch-review-inputs-wf_d2c10282-601.js"} to iterate without resending the script.)
Run ID: wf_d2c10282-601
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\workflows\scripts\A13-launch-review-inputs-wf_d2c10282-601.js", resumeFromRunId: "wf_d2c10282-601"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.

303. user (2026-07-04T11:48:33.542Z)

Updated task #5 status

304. user (2026-07-04T11:49:04.007Z)

Exit code 23
Updated IAM policy for service [rally-control-plane].
Creating Revision.....
.....done
rally-control-plane-00008-wnc
--- public reachability check (expect 403, IAM blocks) ---
HTTP 200

305. user (2026-07-04T11:50:04.880Z)

Exit code 23
{
 "etag": "BwZVx59_Brs=",
 "version": 1
}
--- retry public check ---
HTTP 403
HTTP 403

306. user (2026-07-04T11:50:39.647Z)

Exit code 23
1.jail-foreign-bucket : HTTP 400 :: {"detail":"hosted mode: 'gs://evil-bucket/x.mp4' is not under the allowed input root 'gs://khelsutra-rally-serving/videos'"}
2.jail-sep-boundary : HTTP 400 :: {"detail":"hosted mode: 'gs://khelsutra-rally-serving/videos-evil/x.mp4' is not under the allowed input root 'gs://khelsutra-rally-serving/v"}
3.nonexistent-object : HTTP 202 :: {"job_id":"032c60109b1445d2be951537acf86faf","status":"queued","poll":"/api/jobs/032c60109b1445d2be951537acf86faf"}
4.bad-segmenter : HTTP 400 :: {"detail":"Segmenter 'not_a_segmenter' not found. Available segmenters: ['fusion_hybrid', 'gemini', 'motion', 'tracknet_hybrid', 'wasb_hybri"}
6.compile-bad-video : HTTP 404 :: {"detail":"Indexed video record not found."}
5.jobs-404 : HTTP 404
8a.highlights-traversal: HTTP 404
8b.highlights-missing : HTTP 404

307. assistant (2026-07-04T11:50:49.019Z)

Jail, boundary, and 4xx shapes are all crisp. Now the async-failure shape for the nonexistent object, plus the upload error probes.

308. user (2026-07-04T11:51:26.805Z)

Exit code 23

```
--- 3.nonexistent-object job outcome ---
{"job_id":"032c60109b1445d2be951537acf86faf","status":"failed","progress":"hashing","video_id":null,"error":"NotFound: 404 GET https://storage.googleapis.com/download/storage/v1/b/khelsutra-rally-serving/o/videos%2Fdefinitely-not-here.mp4?alt=media: No such object: khelsutra-rally-serving/videos/definitely-not-here.mp4: ('Request failed with status code', 404, 'Expected one of', <HTTPStatus.OK: 20

--- 9a. upload bad extension ---
{"detail":"Extension '.exe' not allowed. Allowed: ['mov', 'mp4']}|400
--- 9b. upload oversize declare ---
{"detail":"File exceeds the 4096 MB upload limit."}|413
--- 9c. out-of-order part ---
{"detail":"Out-of-order part: expected 0, got 5 (parts are sequential)}|409
cleanup delete: 200
```

309. user (2026-07-04T11:51:58.028Z)

```
jobs: 3ad560c8eb05487e94274701b4f8c656 e7981cb3b93441a182b40069e4340348
350773b904ae408bbe8cb6248bc5cad2
```

310. user (2026-07-04T11:52:09.770Z)

Command running in background with ID: b31c5ap32. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\b31c5ap32.output. You will be notified when it completes. To check interim output, use Read on that file path.

311. user (2026-07-04T11:52:31.063Z)

```
{
  "etag": "BwZVuzykA0o="
}
```

312. user (2026-07-04T11:52:35.376Z)

Exit code 1

```
wrote C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\scratchpad\service-gemini-off.yaml
- name: GEMINI_API_KEY
secretKeyRef:
name: gemini-api-key
  File "<string>", line 1
    import base64,re,json; s=open(r'C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\scratchpad\service-gemini-off.yaml',encoding='utf-8').read(); m=re.search(r\
                                                                    ^
SyntaxError: '(' was never closed
```

313. user (2026-07-04T11:53:01.413Z)

```
Updated IAM policy for secret [gemini-api-key].
{'skip_ai_handoff': False}
False
```

314. assistant (2026-07-04T11:53:09.785Z)

Gemini manifest verified. Waiting for the three concurrency jobs before the Gemini redeploy (a redeploy would wipe their job rows).

315. user (2026-07-04T11:53:41.004Z)

<retrieval_status>success</retrieval_status>

<task_id>b31c5ap32</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>

CONC 3ad560c8: done - success Successfully indexed 0 play segments (cloud_run / 'wasb_hybrid').
CONC e7981cb3: done - success Successfully indexed 0 play segments (cloud_run / 'wasb_hybrid').
CONC 350773b9: done - success Successfully indexed 0 play segments (cloud_run / 'wasb_hybrid').
ALL 3 CONCURRENT JOBS TERMINAL
</output>

316. user (2026-07-04T11:54:27.379Z)

Done.

Creating Revision.....
.....done

revision: rally-control-plane-00009-58c

--- health + capabilities on the Gemini config ---

{"status":"ok","sport":"badminton","default_segmenter":"yolo_hybrid"}{"gemini_key_present":true}

317. assistant (2026-07-04T11:54:36.276Z)

Gemini config booted clean ? rev 00009, `gemini\_key\_present: true`, the startup key-check passed. Dispatching the one paid Gemini run on the real video.

318. user (2026-07-04T11:54:43.575Z)

{"job_id":"35151328570d44f28d8df995efe9ff15","status":"queued","poll":"/api/jobs/35151328570d44f28d8df995efe9ff15"}

HTTP 202

319. user (2026-07-04T11:54:54.533Z)

Command running in background with ID: bu9bh6s5g. Output is being written to:

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\bu9bh6s5g.output. You will be notified when it completes. To check interim output, use Read on that file path.

320. user (2026-07-04T12:03:09.102Z)

<retrieval_status>success</retrieval_status>

<task_id>w3atf3i6j</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>

[Truncated. Full output: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\w3atf3i6j.output]

```

r/owner access to bucket contents (stated in the tester checklist) follows from GCP project
ownership; no IAM policy was read [unverified]"
    ]
  },
  {
    "key": "limits",
    "markdown": "### Known limits + rollback/disable steps\n\n### K1. Locked alpha shape (the
limits that are *by design*)\n\nThese are deliberate, config-pinned constraints of the live A1
stack ? launch copy and tester expectations must match them.\n\n| Locked decision | Evidence
|\n|---|---|\n| **WASB-only detection ? no Gemini anywhere in serving.**
`indexing.skip_ai_handoff: true` is baked into the control-plane manifest (`candidate windows
ARE the rallies; no Gemini key in this service`) | `deploy/cloudrun/gen_service_yaml.py:55-56`;
live-verified `gemini_key_present=false` (e2e runlog $2,
`docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:25-26`)
|\n| **Single region, single GPU.** Everything is in asia-southeast1 (`gen_service_yaml.py:43`);
the project's GPU quota is `GPUS_ALL_REGIONS=1` ? one L4 execution at a time, globally |
`docs/archives/decisions/DECISIONS.md:675` (`quota granted (`GPUS_ALL_REGIONS=1`)`); worker
Job shape e2e runlog $1 |\n| **One control-plane instance (`maxScale=1`).** Deliberate #471
option-1 mitigation ? per-instance SQLite state means >1 instance splits the job queue |
`gen_service_yaml.py:25-27,97,145-150`; e2e runlog $1 (rev 00007 deployed with maxScale=1) |\n|
**Upload size cap: 4096 MB whole-file, 95 MB parts.** `SecurityConfig.max_upload_mb: int =
4096`, `max_part_mb: int = 95` (`backend/config/models.py:370-371`), enforced at
`backend/api/uploads.py:57-58` with the 413-style rejection at `uploads.py:153` (`Upload
exceeds the {N} MB total limit`). Parts stay ~95 MB because free-tier edge proxies cap bodies
~100 MB (`models.py:368-369`) | files cited |\n| ? [analysis] the 4 GB cap is theoretical on
this deploy: uploads land under `/tmp/apphome/output/uploads` (a1-cuj runlog $5.2) on a **4Gi-
memory** instance (`gen_service_yaml.py:152`) whose `/tmp` is RAM-backed
(`deploy/cloudrun/README.md:47-48`) ? a multi-GB upload would exhaust instance memory well
before the configured cap. Not observed live; treat effective cap as ~4 GB | [analysis] |\n|
**Upload validation floor: ~1280x720 and ~29.9 fps.** A 640x360 clip was rejected live by
`resolution_fps_check` | e2e runlog $3.2 + $5 (lines 41-42, 90); defaults at
`backend/config/models.py:25-27` |\n| **Auth posture:** Cloudflare Access at
`api.khelsutra.guru`, same `KhelSutra Studio (alpha)` Access app, **4-email allowlist**, plus
in-app CF-JWT verification (401 without `Cf-Access-Jwt-Assertion`) | a1-cuj runlog $6
(`DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md:78-90`); e2e runlog $6 (restore verified, rev
00007) |\n\n### K2. Known-limits table (open issues + live rough edges)\n\n| Limit | Impact on
the alpha | Source |\n|---|---|---|\n| **Control-plane state is per-instance ephemeral SQLite**
? a redeploy/restart wipes jobs/videos/segments; `/api/highlights` then 404s on the DB check
(`backend/main.py:1052`) even though the reel still exists in gs://. A deploy invalidates in-
flight jobs and past download URLs | Any redeploy mid-tester-session breaks their job polls and
download links; re-processing regenerates them | #471 (open, bug); e2e runlog $5 (lines 86-89)
|\n| **Google-managed cert on `api.khelsutra.guru` can stall at ~90-day renewal** behind the
proxied CF CNAME ? verify before ~2026-09-30 (`gcloud beta run domain-mappings describe --domain
api.khelsutra.guru --region asia-southeast1`) | Silent future outage risk; calendar item for the
owner | #472 (open) |\n| **CORS default is `allow_origins=['*']`** ? and the deployed manifest
sets no `RALLY_CORS_ALLOW_ORIGINS` (env block `gen_service_yaml.py:111-119`), so the wildcard is
active on the hosted service too [analysis]. Edge auth + no-credentials CORS blunt it, but it is
an unresolved audit residual (R1-12) | Residual cross-origin exposure, mainly on non-PNA
browsers | #443 (open) |\n| **No content-safety/NSFW pre-flight** before processing. The WASB-
only alpha removes the send-to-Gemini ToS risk (no Gemini in serving), but abusive uploads still
burn paid L4 time and produce garbage | `Some users will try` ? accepted for a 4-email
allowlist, not for wider invites | #332 (open) |\n| **Path-jail vs storage schemes**: gs://
ingest through the jail was unblocked by PR #465 (verified live: gs:// ? 202 with jail enforced,
a1-cuj $6), but the general scheme-exempt design (s3://, gdrive-api://) remains open | gs://
works today; other cloud schemes 400 | #327 (open); a1-cuj runlog $6 |\n| **No automated upload
retention** ? the T11 sweep tool exists but nothing schedules it, and it targets the *local*
upload dir. Staged tester uploads persist in `gs://khelsutra-rally-serving/videos/<video_id>/?`
indefinitely; no bucket lifecycle policy found in-repo [unverified] | Tester footage accumulates
in the bucket until manually deleted (see R5) | #301 (open); staging path
`backend/orchestration.py:284` |\n| **Observability gaps**: no user-facing per-video trace
endpoint (#328); no request capture/replay, and CF-edge failures (302s at Access) never reach
engine logs so they're invisible in-app (#326) | Tester-reported failures need manual CF-
dashboard + Cloud Logging triage | #328, #326 (open) |\n| **Performance**: hosted processing is
CPU-bound (ffmpeg/cv2) ? measured ~65% of wait is software encode (#331); WASB decode is single-
threaded in streaming mode so the paid L4 sits ~80% idle, ~2x realtime detection (#349). Live A1

```

numbers: ~13m45s process (incl. cold pull) + ~3m26s?4m54s compile for a ~7-min match | Set tester latency expectations in minutes, not seconds; GPU cost per match is inflated | #331, #349 (open); a1-cuj \$3; e2e \$3.4 | \n| **Golden eval data is machine-split** (GPU box ? dev sessions, OneDrive sync incomplete) | Slows quality-lane reproduction; no direct tester impact | #175 (open) | \n| **`gcloud run services replace` with an unchanged manifest is a NO-OP** ? it does not re-resolve `:latest` after a rebuild; force a new revision with `gcloud run services update --image` | Operator footgun: a `\"redeploy\"` can silently ship the old image | e2e runlog \$5 (lines 83-85) | \n| **422-before-401 quirk**: malformed-JSON bodies get FastAPI's 422 before edge-auth runs; well-formed requests without a CF JWT still 401 | No exposure; don't misread a 422 during auth triage | e2e runlog \$5 (lines 91-93) | \n| **No video-delete API** ? no supported way to remove a video/job record; only manual DB/bucket surgery | See R5 for the actual deletion procedure | #447 (open) | \n\n### K3. GATED items that constrain launch copy (A10 / A11 ? both OWNER-GATED, still open) | \n\n**A11 ? WASB-C3 provenance (weights NOT commercially cleared).** The worker image itself carries the warning: *\"the pretrained badminton weights are academic-data-trained ? provenance NOT cleared for commercial sharing? Resolve WASB-C3 before any public, non-owner serving\"* (`deploy/cloudrun/Dockerfile:14-16`). The tracker keeps A11 GATED on owner/legal (`docs/ALPHA\_LAUNCH\_READINESS.md:110`) and names C3 a **hard model-release gate** that must not be hidden inside product-launch tasks (\$5 risk table, `ALPHA\_LAUNCH\_READINESS.md:85`). Consequences for the alpha: \n\n- **Who may use it:** only the current 4-email invite-only allowlist (a1-cuj \$6). Note honestly for the GO/NO-GO: those testers are already *non-owner* users, which sits in tension with the Dockerfile's \"before any public, non-owner serving\" ? A13 needs the owner to either resolve C3 or **explicitly defer it with the documented \"no commercial model sharing\" copy guardrail** (that's exactly what the A11 row requires). No wider invite wave until then. \n- **What copy must NOT claim:** an \"owned\" or commercially cleared rally-detection model, any model sharing/download, or any commercial license posture. Tracker guardrail: *\"Keep all alpha copy scoped to invite-only processing, not owned-model claims\"* (`ALPHA\_LAUNCH\_READINESS.md:80`; also \$0 TL;DR line 15 and D1, line 31). Mitigating fact for the record: the weights are **not baked into the image or distributed** ? they're staged at runtime from the bucket and used server-side only (`Dockerfile:15-16`; `deploy/cloudrun/README.md:15,29-30`). \n\n- **A10 ? ship-gate target not owner-ratified (#172).** The F1@tIoU targets in `gate\_verdict` are explicitly provisional (\"the owner has explicitly stated they did not commit to that figure\" ? #172), the A10 row is GATED on the owner decision (`ALPHA\_LAUNCH\_READINESS.md:109`), and the latest Gen-0 shadow run is `ship=False` (sign test 9/15, p=0.3036; D6 + \$8 A9 record, `ALPHA\_LAUNCH\_READINESS.md:37,214`). Gate-A served default was **rejected** (served mean ?F1 ?0.0328, `ALPHA\_LAUNCH\_READINESS.md:205`). Consequence: **the alpha can make no quality claims** ? no accuracy numbers, no \"best-in-class\", not even the internal R2/tolF1 contrast as a headline (its strict per-video guard still blocks blanket promotion language, `ALPHA\_LAUNCH\_READINESS.md:206`). Allowed framing until A10 is ratified: invite-only alpha that detects candidate rally segments and compiles reels, quality under active measurement. \n\n### K4. Rollback / disable inventory | \n\nAll commands run from PowerShell (not git-bash ? MSYS mangles paths/flags, `deploy/cloudrun/README.md:129-133`), region `asia-southeast1`, project `gen-lang-client-0306026826` (`gen\_service\_yaml.py:42-43`). \n\n- **R1. Disable the whole alpha (fast, reversible, keeps GCS data).** Two independent switches ? throwing either one cuts testers off: \n\n1. *Edge off:* disable (or delete the policy of) the \"KhelSutra Studio (alpha)\" CF Access application in Zero Trust ? Access ? Applications ? the login path dies at the edge (app identified at a1-cuj \$6, lines 81-82; \"disable the Access application\" is the documented teardown lever, `kheisutra/deploy/DEPLOY\_ALPHA.md:156` ? that runbook is RETIRED but kept as the CF-mechanics reference, line 3). Dashboard click-op; exact click path [unverified]. \n\n2. *Origin off:* remove the public invoker that CF-proxied traffic rides in on (the inverse of the documented enable step, a1-cuj \$6 remaining-(a), lines 93-95): \n\n```\ngcloud run services remove-iam-policy-binding rally-control-plane --region asia-southeast1 --member=allUsers --role=roles/run.invoker\n```\nRequests then 403 at Google Front End; `min-instances 1` CPU cost continues until you also scale down or delete. \n\n3. *Full teardown* (documented in the runlog; remember state is ephemeral anyway ? but staged videos/reels in gs:// survive): \n\n```\ngcloud run services delete rally-control-plane --region asia-southeast1\n\ngcloud run jobs delete rally-worker --region asia-southeast1\n```\n(a1-cuj \$7, lines 107-108: bucket/secret/repo retained.) \n\n- **R2. Roll back to a previous image digest.** Do NOT use `services replace` with an unchanged manifest ? it's a no-op (e2e \$5). Pin the digest explicitly: \n\n```\ngcloud run services update rally-control-plane --region asia-southeast1 --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker@sha256:<digest>\ngcloud run jobs update rally-worker --region asia-southeast1 --image asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker@sha256:<digest>\n```\n\nThe `jobs update` re-pin is the exact mechanism used in the e2e run (e2e \$1); the service image path/name from `gen\_service\_yaml.py:157` (one image serves both roles, #470). **Last-known-good digest:

```

`sha256:31475f27?`** (rev 00007, edge-auth ON ? e2e §1/§6). Do not roll back to
`sha256:ed2fe448?`: it predates #476 and reintroduces the broken >32 MiB signed-URL download
(#475, e2e §4). Caveats: any new revision **wipes the per-instance SQLite state** ? in-flight
jobs and existing download URLs die (#471; e2e §5); and if config (not code) is what regressed,
regenerate the manifest instead: `python deploy/cloudrun/gen_service_yaml.py --edge-auth on
--image <pin-digest-ref> -o service.yaml` then `gcloud run services replace service.yaml
--region asia-southeast1` (`gen_service_yaml.py:6-8,136-140`).\n\n**R3. Revoke a tester.** Zero
Trust ? Access ? Applications ? \"Khelsutra Studio (alpha)\" ? edit the Allow policy's *Emails*
rule and remove the address (policy mechanics per `khelsutra/deploy/DEPLOY_ALPHA.md:113`; same-
app/allowlist confirmed live at a1-cuj §6). Also revoke the user's active Access session in Zero
Trust so an existing cookie doesn't outlive the policy edit ? [unverified: exact dashboard path;
not in repo docs]. Defense-in-depth already holds: every `/api/process` re-verifies the CF JWT
in-app (`backend/config/models.py:374-379`; e2e §6 shows 401 without it), so a revoked user
can't fall back to hitting the run.app URL directly. There is no app-side account/data scoping
to clean up (per-user subdirs are a future PR-3 item, `models.py:364-366`).\n\n**R4. Emergency-
stop a running L4 execution.**\n\n``\ngcloud run jobs executions list --job rally-worker
--region asia-southeast1\ngcloud run jobs executions cancel <execution-name> --region asia-
southeast1\n``\n\nExecution names look like `rally-worker-k49kg` (a1-cuj §3.2). The `cancel`
verb is standard gcloud but not exercised in any repo runlog [unverified in-repo]. Known
consequence chain, from repo sources: a killed container writes **no completion/failure
marker**, so the control plane's bucket-marker poll runs to its bounded timeout ?
`ExecutionPolicy.max_wait_sec` default **1800 s**
(`backend/infrastructure/execution_policy.py:54`; `backend/compute/cloud_run.py:245`) ? then
surfaces a clean `cloud dispatch error` job failure, not a hang
(`deploy/cloudrun/README.md:112-115`). GPU billing stops with the execution; the job row on the
CP resolves ~30 min later or dies with the instance on the next redeploy (#471).\n\n**R5. Delete
a user's data for a given `video_id`.** Bucket is the durable store; the DB is ephemeral.\n\n-
*Bucket objects* (all three prefixes, plus the user's own source object if they supplied a raw
gs:// path outside `videos/<video_id>/`):\n\n``\n gcloud storage rm -r gs://khelsutra-rally-
serving/videos/<video_id>/ # staged upload ? orchestration.py:284 (input_root
gen_service_yaml.py:61)\n gcloud storage rm -r gs://khelsutra-rally-serving/outputs/<video_id>/
# worker outputs ? a1-cuj §3.2 (\n\"outputs at gs://?/outputs/915473?/\n\")\n gcloud storage rm -r
gs://khelsutra-rally-serving/highlights/<video_id>/ # mirrored reels ?
orchestration.py:748-755\n\n``\n Also sweep the dispatch done-markers
(`<out_uri>/_dispatch/<token>.done`, `deploy/cloudrun/README.md:109-110`) ? exact prefix in the
serving bucket [unverified].\n- *DB reality:* rows live in per-instance ephemeral SQLite (#471)
with **no delete API** (#447 open ? the only known method is manual row surgery, impractical
inside a Cloud Run instance). Practically: after the bucket `rm`, any surviving DB record is
inert (`/api/highlights` 404s once the mirror check fails / on the `main.py:1052` record check
after the next restart), and **any redeploy wipes all DB rows anyway** (e2e §5) ? a blunt but
real \"forget everything\" lever. This is adequate for a 4-email alpha but is not a real GDPR-
style deletion story; flag alongside #447/#301 in the launch review.\n\n**Cross-cutting caution
for every lever above:** rollback ? data rollback. `gs://khelsutra-rally-serving` (weights,
staged videos, outputs, reels) is untouched by service/job changes; state loss is confined to
the CP's ephemeral SQLite (#471).\",
    \"evidence_gaps\": [
        \"Exact Cloudflare Zero Trust click-path for disabling the Access application and for
        revoking an individual tester's active Access session ? dashboard operations, not captured in
        any repo doc (the policy/app mechanics are cited from the retired DEPLOY_ALPHA.md and the a1-cuj
        runlog).\",
        \"The `gcloud run jobs executions cancel` command is standard gcloud but is not
        exercised or documented anywhere in the repo; its consequence chain (no marker -> bounded poll
        timeout -> clean dispatch-error failure) IS repo-cited.\",
        \"Whether gs://khelsutra-rally-serving has any bucket lifecycle/retention policy
        configured ? nothing found in-repo; #301 covers only the local upload-dir sweep tool.\",
        \"Exact object prefix of the `_dispatch/<token>.done` markers inside the serving bucket
        (the pattern is documented in deploy/cloudrun/README.md but the concrete out_uri layout was not
        confirmed).\",
        \"The four allowlisted email addresses themselves (CF dashboard state; only the count
        '4-email allowlist' is repo-cited from the a1-cuj runlog).\",
        \"The claim that a multi-GB upload would OOM the 4Gi control-plane instance is
        [analysis] from config values (max_upload_mb=4096, memory 4Gi, RAM-backed /tmp), not an observed
        failure.\",
        \"Wildcard CORS being active on the hosted control plane is [analysis] (no
        RALLY_CORS_ALLOW_ORIGINS in the generated manifest env block) ? not verified against the live

```

service's response headers."

```
    ]
  }
],
"critique": {
  "missing": [
    "BIGGEST GAP - browser/SPA CUJ never verified, and it is the tracker's own DoD #1: docs/ALPHA_LAUNCH_READINESS.md:226 '[ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH' is UNCHECKED, and D2 (:32) mandates a page-driven alpha. Both runlogs (a1-cuj, a1-e2e) drove the CUJ via curl with identity tokens/JWTs only. No section audits the SPA's live deployment state: is Cloudflare Pages still serving app.khelsutra.guru with VITE_API_BASE=https://api.khelsutra.guru/api (khelsutra/deploy/DEPLOY_ALPHA.md:102-103 - a RETIRED doc), is the CF Access app's 'Bypass OPTIONS/CORS preflight' setting on (DEPLOY_ALPHA.md:115 says without it EVERY cross-origin API call fails), does the SPA's cloud compute toggle send compute=cloud against the new Cloud Run backend, does the wizard's /api/capabilities fetch work behind CF? The review must either include a browser-CUJ verification or state explicitly that the alpha is currently curl-only.",
    "No GO/NO-GO scorecard or recommendation: the tracker has an explicit Definition-of-done checklist (ALPHA_LAUNCH_READINESS.md:224-235, 6 of 10 boxes unchecked) and open owner questions (:216-222, incl. 'acceptable alpha wording while C3 unresolved'). No section maps the review to that checklist or states the actual decision shape the owner must sign (e.g. GO for the current 4-email allowlist conditional on fixing SetupWizard copy; NO-GO for any wider invite wave until A10/A11 close). A12 (GATED, :111 - flywheel/feedback loop deferred) is never mentioned in any section.",
    "No cost/budget projection for the standing stack: idle cost of the always-on control plane (minScale=1 + cpu-throttling off, 2 vCPU/4Gi - gen_service_yaml.py:96-98,151-152), L4 cost precedent ~$1-2/run (docs/archives/decisions/DECISIONS.md:675), expected monthly figure for the 4-tester wave, and the [unverified] budget-alert threshold ($100 per deploy/gcp/README.md:18 vs the claimed $300 spend gate) - that unverified is a single console check and should be resolved before the owner decision.",
    "No expected-quality statement for testers: the sections say what may NOT be claimed (A10) but never what testers WILL experience from the WASB-only path (skip_ai_handoff=true means candidate windows ARE the rallies). The measured numbers exist: served milestone toLF1 0.364 / F1@0.5 0.472 vs baseline-off 0.138/0.237 with the strict split-count guard still failing (ALPHA_LAUNCH_READINESS.md:142-153,206), 48 segments on the ~7-min e2e match. Owner needs 'testers will see roughly-right rally boundaries with over-splitting' framing to judge alpha-burn risk.",
    "Runtime service-account blast radius unaudited: the compute SA (492532266377-compute@) holds project-wide roles/editor plus self-serviceAccountTokenCreator (memory session-handoff.md:159-160) on a service with allUsers run.invoker. No section inventories IAM or recommends a least-privilege SA before the allowlist grows - launch-review-grade security gap beyond the CORS residual already covered.",
    "No tester onboarding/support content: what an invited tester is actually sent (app URL, CF email-OTP flow, the >=720p/29.9fps floor, 4GB/mp4|mov caps, expect ~15-20 min per match) collected in one place, and where testers report failures (hello@khelsutra.guru?) given A12 (feedback loop) is GATED and observability gap 2 means they cannot self-serve diagnostics.",
    "Watch items have no owner/calendar mechanism: the #472 cert check before ~2026-09-30 and the budget watch are documented as commands but nobody is assigned; an A13 review should name the owner and the reminder mechanism (the repo owner runs multiple parallel sessions - a calendar entry or issue-due-date is the difference between documented and done).",
    "Privacy audit misses two site-copy items: (a) policy line locales/en/common.json:528 'We process your video on our own infrastructure' is arguably FALSE on rented Google Cloud infrastructure - same class as the SetupWizard issue but on the public policy page, stronger than the draft's 'not disclosed' framing; (b) the site footer trust line :488 'Run from Bengaluru, India' sits in tension with all footage stored/processed in Singapore.",
    "Doc-hygiene decision: the runbook reproduces the operational secret location G:\\My Drive\\cloudflare.token.txt (and memory session-handoff.md:169's tester emails are one hop away). Decide whether the assembled A13 review is an owner-only doc or must be scrubbed before any wider sharing."
  ],
  "contradictions": [
    "Runbook Gotcha A overstates its provenance: 'that exact fallback was used live' - the cited e2e runlog section (DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:83-85) records the no-op and the remedy as an observation, not as an executed live command, and the full flag string is sourced only from memory current-state.md:64 which the runbook itself marks
```

[unverified]. Downgrade 'used live' to [analysis] or cite the runlog line that actually shows the command ran.",

"Limits R4 states 'GPU billing stops with the execution' unconditionally, but cancel is best-effort (backend/compute/cloud_run.py:268-277) and R4 itself marks 'executions cancel' as [unverified in-repo]; the observability timeout row correctly warns the opposite ('Confirm the cancel landed - otherwise L4 bills to --task-timeout 3600s'). The billing claim should carry the same [unverified] tag.",

"Privacy section applies inconsistent severity: SetupWizard.tsx:58 copy is called 'factually wrong... must be fixed before any tester uploads' (correct - verified at locales/en/common.json:612), but policy :528 'our own infrastructure' - false by the same standard on the hosted alpha - is graded only as a nondisclosure in the section 2 table ('Disclosed today? No').",

"Runbook section 5 attributes 'gcloud beta run jobs executions logs read' to 'the committed pattern' at deploy/gcp/nightly_regression/README.md:111 - the committed line has no 'beta' release track and targets nightly-regression-launcher in asia-south1; the adaptation is fine but the citation implies more than the source carries.",

"Trivial: runbook section 6 quotes deploy/cloudrun/README.md:86,90-93 while saying '~2-vCPU' - the quoted README line 92 actually says '~1-vCPU CPU instance'; the deployed manifest is cpu=2 (gen_service_yaml.py:151), so the runbook's number is right but does not match the source it quotes.",

"(Verification note, not a defect) All other spot-checked claims held: 18 issue/PR states re-confirmed via gh on 2026-07-04 (#471/#472/#301/#332/#326/#328/#447/#443/#327/#175/#331/#349/#172 OPEN; #465/#476/#470 merged; #461/#463 closed); ~40 file:line cites verified across gen_service_yaml.py, backend/main.py, uploads.py, job_worker.py, cloud_run.py, execution_policy.py, orchestration.py, flywheel.py (consent hard-deny confirmed at 44-52), config/models.py, Dockerfile:14-16, both runlogs, the tracker, khelsutra locales/SetupWizard/App.tsx, and the web/src privacy-string zero-match claim (grep reproduces: only two code comments, no user-facing privacy copy)."

```
    ],
    "verdict": "needs_additions"
  }
},
"workflowProgress": [
  {
    "type": "workflow_phase",
    "index": 1,
    "title": "Research"
  },
  {
    "type": "workflow_phase",
    "index": 2,
    "title": "Critique"
  },
  {
    "type": "workflow_agent",
    "index": 1,
    "label": "draft:runbook",
    "phaseIndex": 1,
    "phaseTitle": "Research",
    "agentId": "a5fffddeaf184a077",
    "model": "claude-fable-5",
    "state": "done",
    "startedAt": 1783165705934,
    "queuedAt": 1783165705919,
    "attempt": 1,
    "lastToolName": "StructuredOutput",
    "lastToolSummary": "## Operator runbook ? live A1 alpha (Cloud Run asia-southea?",
    "promptPreview": "You are drafting ONE section of the A13 \"alpha launch review\" for the KhelSutra badminton-highlight alpha\n(repo Khelsutra/badminton-highlight-indexer, engine checkout at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at C:/Users/avidu/Projects/khelsutra-guru/khelsutra).\nContext: the A1 serving stack is LIVE on Cloud Run asia-southeast1 ? control plane service\n'ra?",
    "lastProgressAt": 1783165983146,
    "tokens": 108196,
    "toolCalls": 26,
```

```

    "durationMs": 277212,
    "resultPreview": "{\n\"markdown\\\":\n\"## Operator runbook ? live A1 alpha (Cloud Run asia-southeast1)\n\n> Scope note: this is an **operations** runbook for the stack that is already live. It does not imply launch GO ? A10 (ship-gate target) and A11 (WASB-C3 provenance) are still **GATED** owner rows (`docs/ALPHA_LAUNCH_READINESS.md:109-110`).\n\n**Live topology** (e2e runlog `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY?\"
  },
  {
    "type": "workflow_agent",
    "index": 2,
    "label": "draft:observability",
    "phaseIndex": 1,
    "phaseTitle": "Research",
    "agentId": "a4b697c820517aa72",
    "model": "claude-fable-5",
    "state": "done",
    "startedAt": 1783165705936,
    "queuedAt": 1783165705919,
    "attempt": 1,
    "lastToolName": "StructuredOutput",
    "lastToolSummary": "## Observability & job-failure triage Scope: the live A1 st?",
    "promptPreview": "You are drafting ONE section of the A13 `alpha launch review` for the KhelSutra badminton-highlight alpha\n(repo Khelsutra/badminton-highlight-indexer, engine checkout at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at C:/Users/avidu/Projects/khelsutra-guru/khelsutra).\nContext: the A1 serving stack is LIVE on Cloud Run asia-southeast1 ? control plane service\n'ra'",
    "lastProgressAt": 1783166043563,
    "tokens": 140244,
    "toolCalls": 24,
    "durationMs": 337627,
    "resultPreview": "{\n\"markdown\\\":\n\"## Observability & job-failure triage\n\nScope: the live A1 stack ? Cloud Run service `rally-control-plane` (rev 00007, edge-auth ON) + L4 Job `rally-worker`, bucket `gs://khelsutra-rally-serving` (DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md:1-22). All engine paths below are relative to `badminton-highlight-indexer`.\n\n**Reading a job's state.** `GET /api/jobs/{job_id}` returns ?\"
  },
  {
    "type": "workflow_agent",
    "index": 3,
    "label": "draft:privacy",
    "phaseIndex": 1,
    "phaseTitle": "Research",
    "agentId": "afc8517f62c5cd0c9",
    "model": "claude-fable-5",
    "state": "done",
    "startedAt": 1783165705937,
    "queuedAt": 1783165705920,
    "attempt": 1,
    "lastToolName": "StructuredOutput",
    "lastToolSummary": "## Privacy & consent copy audit (A13) **Verdict:** the mark?",
    "promptPreview": "You are drafting ONE section of the A13 `alpha launch review` for the KhelSutra badminton-highlight alpha\n(repo Khelsutra/badminton-highlight-indexer, engine checkout at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at C:/Users/avidu/Projects/khelsutra-guru/khelsutra).\nContext: the A1 serving stack is LIVE on Cloud Run asia-southeast1 ? control plane service\n'ra'",
    "lastProgressAt": 1783166051749,
    "tokens": 85348,
    "toolCalls": 34,
    "durationMs": 345812,
    "resultPreview": "{\n\"markdown\\\":\n\"## Privacy & consent copy audit (A13)\n\n**Verdict:** the marketing site has a real, decent privacy policy in all 6 locales ? but the Studio SPA (the surface testers actually upload through) contains **zero privacy/consent copy**, and its first-run wizard affirmatively tells users their videos never leave the device, which is false on the hosted alpha. One code-level guarantee (traini?\"
  },

```

```

{
  "type": "workflow_agent",
  "index": 4,
  "label": "draft:limits",
  "phaseIndex": 1,
  "phaseTitle": "Research",
  "agentId": "af5800ff62c0b9c44",
  "model": "claude-fable-5",
  "state": "done",
  "startedAt": 1783165705938,
  "queuedAt": 1783165705920,
  "attempt": 1,
  "lastToolName": "StructuredOutput",
  "lastToolSummary": "## Known limits + rollback/disable steps ### K1. Locked alp?",
  "promptPreview": "You are drafting ONE section of the A13 \"alpha launch review\" for the
KhelSutra badminton-highlight alpha\n(repo Khelsutra/badminton-highlight-indexer, engine
checkout at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at
C:/Users/avidu/Projects/khelsutra-guru/khelsutra).\nContext: the A1 serving stack is LIVE on
Cloud Run asia-southeast1 ? control plane service\n'ra?",
  "lastProgressAt": 1783166106075,
  "tokens": 115615,
  "toolCalls": 30,
  "durationMs": 400137,
  "resultPreview": "{\n\"markdown\": \"## Known limits + rollback/disable steps\\n\\n### K1.
Locked alpha shape (the limits that are *by design*)\\n\\nThese are deliberate, config-pinned
constraints of the live A1 stack ? launch copy and tester expectations must match them.\\n\\n|
Locked decision | Evidence |\\n|---|---|\\n| **WASB-only detection ? no Gemini anywhere in
serving.* `indexing.skip_ai_handoff: true` is baked into the?\"
},
{
  "type": "workflow_agent",
  "index": 5,
  "label": "critic",
  "phaseIndex": 2,
  "phaseTitle": "Critique",
  "agentId": "a67e3e151830df5e2",
  "model": "claude-fable-5",
  "state": "done",
  "startedAt": 1783166106092,
  "queuedAt": 1783166106076,
  "attempt": 1,
  "lastToolName": "StructuredOutput",
  "lastToolSummary": "needs_additions",
  "promptPreview": "You are drafting ONE section of the A13 \"alpha launch review\" for the
KhelSutra badminton-highlight alpha\n(repo Khelsutra/badminton-highlight-indexer, engine
checkout at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at
C:/Users/avidu/Projects/khelsutra-guru/khelsutra).\nContext: the A1 serving stack is LIVE on
Cloud Run asia-southeast1 ? control plane service\n'ra?",
  "lastProgressAt": 1783166589025,
  "tokens": 136224,
  "toolCalls": 37,
  "durationMs": 482933,
  "resultPreview": "{\n\"missing\": [\"BIGGEST GAP - browser/SPA CUJ never verified, and it is
the tracker's own DoD #1: docs/ALPHA_LAUNCH_READINESS.md:226 '[ ] Invited tester can complete
the hosted record/upload -> reel download CUJ without SSH' is UNCHECKED, and D2 (:32) mandates a
page-driven alpha. Both runlogs (a1-cuj, a1-e2e) drove the CUJ via curl with identity
tokens/JWTs only. No section audits the SPA's live dep?\"
}
],
"totalTokens": 585627,
"totalToolCalls": 151
}
</output>

```

```
root: HTTP 200 type=text/html; charset=utf-8
<!doctype html>
```

322. user (2026-07-04T12:03:40.063Z)

```
1      GEMINI-JOB: running | validating
2      GEMINI-JOB: running | dispatching (cloud_run)
3
```

323. user (2026-07-04T12:03:57.150Z)

<https://github.com/avidullu/khelsutra.git>

324. user (2026-07-04T12:04:15.293Z)

<https://github.com/avidullu/khelsutra/issues/154>

325. user (2026-07-04T12:13:56.837Z)

```
<retrieval_status>success</retrieval_status>

<task_id>bu9bh6s5g</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>
GEMINI-JOB: running | validating
GEMINI-JOB: running | dispatching (cloud_run)
GEMINI-JOB: done | finished
TERMINAL: done :: success - Successfully indexed 43 play segments (cloud_run / 'wasb_hybrid').
err: None
</output>
```

326. user (2026-07-04T12:14:21.118Z)

latest execution: rally-worker-khx58

327. assistant (2026-07-04T12:14:35.164Z)

No matches ? let me look at the raw execution logs to see what's actually logged.

328. user (2026-07-04T12:14:40.772Z)

```
12:13:20 [INFO] backend.worker: [worker] wrote completion marker -> gs://khelsutra-rally-
serving/_dispatch/eda6f43be6054152824e59fe2244be90.done
  gs://khelsutra-rally-serving/outputs/915473794e0909c9f95a01e4511a5e98/report.json
  gs://khelsutra-rally-serving/outputs/915473794e0909c9f95a01e4511a5e98/intervals.csv
[worker] infer done: 43 segment(s); pushed 2 artifact(s):
12:13:19 [INFO] backend.pipeline.segmenters.trajectory_hybrid: Phase 3 completed. AI confirmed
43 semantic play segments.
12:12:49 [INFO] backend.providers.gemini: [Gemini Handoff 47] Response received in 4.9s:
12:12:49 [INFO] httpx: HTTP Request: POST
https://generativelanguage.googleapis.com/v1beta/models/gemini-flash-latest:generateContent
"HTTP/1.1 200 OK"
12:12:49 [INFO] backend.providers.gemini: [Gemini Handoff 48] Response received in 3.7s:
12:12:49 [INFO] httpx: HTTP Request: POST
https://generativelanguage.googleapis.com/v1beta/models/gemini-flash-latest:generateContent
"HTTP/1.1 200 OK"
12:12:45 [INFO] backend.providers.gemini: [Gemini Handoff 48] Submitting multimodal request to
gemini-flash-latest...
```

```
12:12:45 [INFO] backend.providers.gemini: [Gemini Handoff 48] Processing finished in 10.6s.
12:12:45 [INFO] backend.providers.gemini: Gemini Handoff 48 processing: ready after 1 poll(s) /
11s
12:12:44 [INFO] backend.providers.gemini: [Gemini Handoff 47] Submitting multimodal request to
gemini-flash-latest...
12:12:44 [INFO] backend.providers.gemini: [Gemini Handoff 47] Processing finished in 10.8s.
12:12:44 [INFO] backend.providers.gemini: Gemini Handoff 47 processing: ready after 1 poll(s) /
11s
```

329. user (2026-07-04T12:15:35.278Z)

```
Done.
Creating Revision.....
.....
.....done
Updated IAM policy for service [rally-control-plane].
revision: rally-control-plane-00010-qbw
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker@sha256:31475f27dd1b2d2544e7fa9feb03c73913bd670448cd2e0fc3748d7b499d1843
```

330. user (2026-07-04T12:15:51.003Z)

```
--- posture verification (rev 00010, restored) ---
health (token): HTTP 200
health (public): HTTP 200
process sans JWT: 401 :: {"detail":"edge auth: missing Cf-Access-Jwt-Assertion header (request
did not pa
api.khelsutra.guru: HTTP 302
gemini_key_present: False
```

331. user (2026-07-04T12:16:27.334Z)

Updated task #5 status

332. user (2026-07-04T12:16:27.805Z)

Updated task #6 status

333. assistant (2026-07-04T12:16:55.138Z)

A13 live test-battery evidence (2026-07-04, this session ? all measured, none inferred)

Run against the deployed stack during a controlled test window: `allUsers` invoker REMOVED first (public probes 403 ? zero public exposure; NB this also takes api.khelsutra.guru down for testers during the window since CF rides that binding), then edge-auth-off revisions; restored to rev 00010 (digest `31475f27?`, auth ON, allUsers re-added) and posture re-verified at the end: health(token)=200, health(public)=200 (exempt), process-sans-JWT=401, api.khelsutra.guru=302, gemini_key_present=false.

Failure-mode battery (all via identity token, rev 00008)

```
| Probe | Result |
|---|---|
| process gs://evil-bucket/x.mp4 (foreign bucket) | **400** `hosted mode: '?' is not under the
allowed input root 'gs://khelsutra-rally-serving/videos'` (#465 live at the API surface) |
| process gs://khelsutra-rally-serving/videos-evil/x.mp4 (separator-boundary attack) | **400**
same jail message (the boundary check holds) |
| process nonexistent object under input_root | **202** job queued ? job `failed`, error
`NotFound: 404 ? No such object: khelsutra-rally-serving/videos/definitely-not-here.mp4` (clean
async failure, client-visible via poll) |
```

```
| process bad segmenter | **400** `Segmenter 'not_a_segmenter' not found. Available segmenters:
[?]|` |
| GET /api/jobs/deadbeef | **404** |
| compile bad video_id | **404** `Indexed video record not found.` |
| GET /api/highlights/x/..%2F..%2Fconfig.json (traversal) | **404** (rejected; no file
disclosure) |
| GET /api/highlights/<missing>/reel.mp4 | **404** |
| upload init filename=malware.exe | **400** `Extension '.exe' not allowed. Allowed: ['mov',
'mp4']` |
| upload init total_size 9999999999 | **413** `File exceeds the 4096 MB upload limit.` |
| upload PUT part/5 first (out-of-order) | **409** `Out-of-order part: expected 0, got 5` ;
DELETE cleanup ? 200 |
```

Concurrency smoke (#468 live, rev 00008)

3 distinct 720p clips (4s/5s/6s) uploaded + submitted `compute=cloud` back-to-back ? ALL THREE jobs claimed and completed `success` (0 segments each ? synthetic content). No false-empty, no stranded job, GPU-quota=1 serialized the L4 executions but the drain claimed everything.

Gemini handoff leg (rev 00009 ? one-off test config, NOT the alpha posture; restored after)

- Prep finding: the `gemini-api-key` secret existed but had ****NO accessor grant**** ? the documented secret-mount path had never been exercised. Fixed live:
 - `gcloud secrets add-iam-policy-binding gemini-api-key`
 - member=serviceAccount:492532266377-compute@developer.gserviceaccount.com
 - role=roles/secretmanager.secretAccessor`
 - (grant left in place; the secret is only mounted when a config asks for it).
- Job: `gcloud run jobs update rally-worker --set-secrets GEMINI\_API\_KEY=gemini-api-key:latest` ? worked.
- CP: manifest with `skip\_ai\_handoff=false` + secret env ? booted clean
- (`AI\_HANDOFF\_KEY\_MISSING`
 - startup check passed; `gemini\_key\_present=true` in /api/capabilities).
- Run: MahadevpuraSingles, wasb_hybrid, compute=cloud ? ****success, 43 play segments**** (vs 48 in WASB-only mode ? the AI pass genuinely refined/filtered).
- Worker logs prove real Gemini traffic: `backend.providers.gemini [Gemini Handoff 1..48]

Submitting

```
multimodal request to gemini-flash-latest` ? `POST
https://generativelanguage.googleapis.com/v1beta/
models/gemini-flash-latest:generateContent "HTTP/1.1 200 OK"` for all 48 windows; `Phase 3
completed. AI confirmed 43 semantic play segments.`; execution `rally-worker-khx58`, wall
~similar
to WASB-only + ~4-11s per handoff window.
```

- Restored after: secret unmounted from the job (`--remove-secrets`), alpha config back (skip_ai_handoff=true), `gemini\_key\_present=false` re-verified.

Other live facts gathered today

- The CP serves the bundled SPA at its root: `GET /` ? 200 `text/html` (single-origin ? the tester path has NO cross-origin CORS dependency; the CF-Access "bypass OPTIONS" concern applies only to a split-origin deploy).
- Domain-mapping cert (issue #472): `Ready;CertificateProvisioned;DomainRoutable = True;True;True` today (2026-07-04) ? healthy; the ~90-day renewal watch date stands (~2026-09-30).
- Better auth-off window practice (used today, recommend for the runbook): REMOVE the allUsers invoker for the duration of any edge-auth-off window (public 403; identity tokens still work), re-add on restore. Cost: api.khelsutra.guru is down for testers during the window.
- khelsutra copy launch-blockers filed as khelsutra issue #154 (SetupWizard "never leaves your device" false on hosted; policy "our own infrastructure" on GCP; SPA has zero consent copy).

334. user (2026-07-04T12:16:55.177Z)

File created successfully at: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\scratchpad\ai3-live-evidence.md (file state is current in your context ? no need to Read it back)

335. user (2026-07-04T12:17:06.081Z)

HEAD is now at 122f174 docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE (#477)

122f174 docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE (#477)

336. user (2026-07-04T12:17:45.613Z)

Async agent launched successfully.

agentId: a705e7a7d538f83b1 (internal ID - do not mention to user. Use SendMessage with to: 'a705e7a7d538f83b1', summary: '<5-10 word recap>' to continue this agent.)

The agent is working in the background. You will be notified automatically when it completes.

Do not duplicate this agent's work ? avoid working with the same files or topics it is using.

output_file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\a705e7a7d538f83b1.output

Do NOT Read or tail this file via the shell tool ? it is the full subagent JSONL transcript and reading it will overflow your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.

337. user (2026-07-04T12:18:05.688Z)

The file c:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra-guru\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

338. user (2026-07-04T12:28:11.388Z)

<retrieval_status>timeout</retrieval_status>

<task_id>a705e7a7d538f83b1</task_id>

<task_type>local_agent</task_type>

<status>running</status>

339. user (2026-07-04T12:33:12.972Z)

<retrieval_status>success</retrieval_status>

<task_id>a705e7a7d538f83b1</task_id>

<task_type>local_agent</task_type>

<status>completed</status>

<output>

All clean ? the only matches are the owner's public GitHub handle (`@avidullu`), which the tracker already uses. Work complete; nothing committed.

Report

****Files changed**** (worktree `C:\Users\avidu\Projects\khelsutra-guru\wt-a13`, branch `docs/ai3-launch-review` at `122f174`, no commits made):

- ****NEW**** `C:\Users\avidu\Projects\khelsutra-guru\wt-a13\docs\ALPHA\_LAUNCH\_REVIEW\_2026-07-04.md`
? the A13 owner-facing launch review (~6 sections: TL;DR + recommendation, DoD scorecard + 5-minute browser-CUJ checklist, runbook incl. cost + watch-items, observability/triage incl. the live failure-mode battery, privacy/consent audit, known limits + rollback incl. the Gemini-leg

validation, SA blast radius, tester one-pager, open gates). Registered via `git add -N` (intent-to-add, content unstaged) so the `git ls-files`-based link gate actually scans it.

- `docs\ALPHA\_LAUNCH\_READINESS.md` ? A13 row ? `GATED`, PR column ? `review DELIVERED 2026-07-04` ? docs\ALPHA_LAUNCH_REVIEW_2026-07-04.md; GO/NO-GO awaiting owner`; §10 auth-posture/rollback box ? `[x]` with citations (e2e runlog §6 + review §5.6); browser-CUJ box left `[]`; changelog line added. All edits were in-place so the review's `ALPHA_LAUNCH_READINESS.md:<line>` citations stay valid.
- `docs\DOC\_STATUS.md` ? new §3c REPORT row for the review; §3a ALPHA_LAUNCH_READINESS row's status/next-action refreshed (A0?A9 done, A13 delivered, owner actions listed).

****Gate outputs (verbatim final lines, run from the worktree root):****

1. `python -m tools.check\_md\_links` ? `[check-md-links] clean ? no broken internal links in live docs.`
2. `python -m pytest tests/ -q -p no:cacheprovider` ? `1659 passed, 10 skipped, 8 warnings in 103.81s (0:01:43)`
3. `python -m ruff check .` ? `All checks passed!`

****Review verdict/emphases (5 bullets):****

- ****Conditional GO for the current 4-email allowlist**** ? conditions: fix khelsutra #154 copy blockers (SetupWizard "never leaves this device" + policy "our own infrastructure" are **false** on the hosted alpha; SPA has zero consent copy) and owner completes the 5-minute browser-CUJ checklist; ****NO-GO for any wider wave**** until A10/A11 close plus least-privilege SA and retention fixes. Recording GO also requires the owner to explicitly defer A11 (current testers are already non-owner users vs the Dockerfile's C3 warning).
- ****DoD scorecard: 5 of 10 boxes done**** ? the browser CUJ has **never** been run through a real browser (all evidence is curl+token); the live SPA-served-at-root fact makes the tester path single-origin (kills the CF-Access CORS-preflight worry), but only the owner checklist closes the box.
- ****Live battery folded in as authoritative****: 11-probe failure-mode table (jail holds against foreign-bucket/boundary attacks, clean 4xx/async failures), 3-job concurrency smoke all-green, cert healthy today with a dated ~2026-09-30 watch item + owner/mechanism, and the improved auth-off-window practice (remove `allUsers` for the window) now the runbook standard.
- ****Gemini leg recorded as "validated beyond the alpha config"****: real gemini-flash-latest calls, 48?43 segments, secret-accessor grant gap found+fixed live, fully restored to WASB-only ? deployable in hours if ever wanted, but flagged as a privacy-copy change, not just config.
- ****Every critique fix applied and the doc scrubbed for repo publication****: cost subsection (idle 2vCPU/4Gi CP, ~\$1?2/L4-run precedent, monthly \$ `[unverified ? owner billing check]`), honest tester-expectation numbers (tolF1 0.364/F1@0.5 0.472 vs 0.138/0.237, over-splitting persists, no public claims until A10), SA `roles/editor` blast-radius paragraph, tester onboarding one-pager, all five contradiction fixes, and zero secret paths / tester emails / session-memory citations.

</output>

340. assistant (2026-07-04T12:33:20.372Z)

Assembly is done with all gates green. Spot-reviewing the deliverable myself before committing ? the TL;DR and a scrub-violation sweep.

341. user (2026-07-04T12:33:25.596Z)

lines: 351
 --- scrub sweep ---
 (empty = clean)

Alpha Launch Review ? A13 (2026-07-04)

> ****Status:**** `REPORT` ? review ****DELIVERED 2026-07-04****; ****GO/NO-GO awaiting owner**** .
****Owner:**** @avidullu . ****Created:**** 2026-07-04
 > ****Lifecycle:**** terminal record ? archive with the alpha-launch cluster once the owner records GO/NO-GO on the tracker's Definition of done.
 > ****Tracking anchors:**** `docs\ALPHA\_LAUNCH\_READINESS.md` §7 row ****A13****; registered in `docs\DOC\_STATUS.md` §3c.
 > ****Relation to existing docs:**** the A13 deliverable of ALPHA_LAUNCH_READINESS.md; builds on the two A1 runlogs ([a1-cuj](COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md),

```
[a1-e2e](COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md)).
> Honesty note: `[verified]` = checked against the repos or the live stack on 2026-07-04.
"live battery, 2026-07-04" = measured this session against the deployed service in a
controlled test window (`allUsers` invoker removed first ? public probes 403 during the window ?
then edge-auth-off revisions; restored to `rev 00010`, digest `31475f27?`, edge-auth ON,
`allUsers` re-added; restore posture re-verified: health(token)=200, health(public)=200
(deliberately exempt), process-sans-JWT=401, `api.khelsutra.guru?302 CF login,
`gemini_key_present=false`). `[analysis]` = derived, not observed. `[unverified]` = stated but
not checked ? every one is tagged. "Session records" = non-repo operational notes from the
deploy sessions. Issue states (#471/#472/#301/#332/#326/#328/#447/#443/#327/#175/#331/#349/#172
open; #465/#476/#470 merged) were re-confirmed via `gh` on 2026-07-04. This is not legal advice.

---
```

0. TL;DR + recommendation

The A1 stack is live and end-to-end green on the final image (48-segment CUJ, signed-URL download, edge auth restored ? e2e runlog §3/§6), and this session's live battery adds: the hosted-mode input jail holds at the API surface against foreign-bucket and separator-boundary attacks, upload/compile misuse fails with clean 4xx, a missing gs:// object fails cleanly and client-visibly, three concurrent cloud jobs all completed with no stranded state, and the domain-mapping cert is healthy today. Auth posture, rollback levers, and data-deletion mechanics are documented below. Training capture is hard-denied in code ? stronger than the privacy policy promises.

What is **not** done: nobody has ever completed the CUJ **in** a browser through Cloudflare Access[®] (both runlogs and the battery drove it with curl + tokens/JWTs ? the tracker's own DoD #1 is unchecked); the Studio SPA's first-run wizard tells testers **"your videos never leave this device"**, which is **false** on the hosted alpha, and the public privacy policy says footage is processed on **"our own infrastructure"** while it actually runs on Google Cloud in Singapore (both filed as **khelsutra #154**); A10 (ship-gate target) and A11 (WASB-C3 provenance) remain owner-gated, so **no** quality or owned-model claims are allowed in any copy[®]; the runtime service account is over-privileged for a public-invoker service; and retention/deletion is manual-only (#301, #447).

Recommendation (the decision is the owner's ? this is the evidence-supported shape):

- **Conditional GO** for the CURRENT 4-email allowlist[®], with two conditions to clear **before** testers touch it:

1. Fix the khelsutra copy blockers (**khelsutra #154**): suppress/fix the SetupWizard "never leaves this device" claim for the hosted build, correct the policy's "our own infrastructure" line, and add minimum upload-moment disclosure (§4.4 draft copy is ready to ship).

2. Owner completes the **5-minute browser-CUJ checklist** (§1.1) ? closing the tracker's DoD #1 with a real browser pass.

Rationale: the blast radius is 4 known, owner-invited people; the stack, auth, failure modes, and rollback are verified; the two conditions are the only tester-facing falsehoods/unknowns left. Recording GO also requires the owner to **explicitly defer A11** with the documented "no commercial model sharing" copy guardrail (``docs/ALPHA_LAUNCH_READINESS.md:110``) ? the current testers are already non-owner users, which sits in tension with the Dockerfile's "no public, non-owner serving" warning (``deploy/cloudrun/Dockerfile:14-16``); that tension must be owned, not implied.

- **NO-GO** for any wider invite wave[®] until: the owner closes **A10** (ship-gate target + allowed claims) and **A11** (WASB-C3 resolve-or-defer), the runtime SA is replaced with a **least-privilege SA** (§5.5 ? today it holds project ``roles/editor`` on a service with an ``allUsers`` invoker), and the **retention/deletion over-promise** is addressed (automation via #301 or softened policy wording). The no-content-screening gap (#332) is acceptable at 4 trusted users and not beyond.

1. Scorecard ? the tracker's Definition of done, mapped to evidence

Against ``docs/ALPHA_LAUNCH_READINESS.md`` §10 (lines 224?235):

#	DoD item	Verdict	Evidence
---	---	---	---

| 1 | Invited tester completes the hosted upload?reel CUJ without SSH | ****NOT DONE ? browser leg never run**** | Both runlogs drove the CUJ via curl with identity tokens/CF-JWT checks, not a browser (a1-cuj §3, a1-e2e §3); the live battery was also API-driven. Mitigating live fact: the control plane serves a bundled SPA at its root (`GET /` ? 200 `text/html`, live battery) ? the tester path is ****single-origin****, so the CF-Access CORS-preflight worry applies only to a split-origin deploy. Close it with §1.1 |

| 2 | Alpha service gated behind chosen auth posture + documented disable/rollback path | ****DONE** (as of this review)** | Edge auth ON verified live: 401 sans CF JWT, `api.khelsutra.guru` ? CF login 302 (e2e runlog §6), re-verified at rev 00010 after the battery window (restore posture, header above). Disable/rollback path: §5.6 of this doc (R1?R5) |

| 3 | n=15 corpus portable, no stale absolute paths | ****DONE**** | #452; tracker row A2 + A2 smoke command (`docs/ALPHA\_LAUNCH\_READINESS.md:101,123-130`) |

| 4 | Nightly baseline + heuristic floor refreshed with recorded commands | ****DONE**** | #453/#454; recorded outputs at `docs/ALPHA\_LAUNCH\_READINESS.md:132-168` |

| 5 | R2/tolF1 gate promoted or rejected with recorded ablation | ****DONE**** | #455/#458; A6/A8 records (`docs/ALPHA\_LAUNCH\_READINESS.md:170-177,200-206`) ? R2/tolF1 supported for the baseline-off?served contrast, strict split-count guard still failing |

| 6 | Fusion/person/Gate-A served-default decisions explicit | ****DONE**** | #458; A8 record (`docs/ALPHA\_LAUNCH\_READINESS.md:200-206`) ? person fusion not promoted, audio shadow-only, Gate A served default ****rejected**** |

| 7 | Ship-gate target + allowed alpha claims recorded | ****NOT DONE ? A10 GATED**** | Owner decision open (#172); Gen-0 `ship=False` (sign 9/15, p=0.3036, `docs/ALPHA\_LAUNCH\_READINESS.md:214`) |

| 8 | WASB C3 status recorded and reflected in alpha messaging | ****NOT DONE ? A11 GATED**** | Recorded (Dockerfile warning, tracker row), ****not reflected in messaging****: the SPA wizard still sells a Gemini-key step and no copy carries the no-model-sharing guardrail (§4, khelsutra #154) |

| 9 | P10/P11 reopened or explicitly deferred | ****NOT DONE ? A12 GATED**** | No recorded owner deferral; recommend recording "deferred for alpha-plus" to close this box honestly (§6.3) |

| 10 | Owner records GO/NO-GO for the first alpha wave | ****NOT DONE**** | This review is the input; §0 is the recommendation |

1.1 The 5-minute owner checklist that closes DoD #1

From an allowlisted address, in a normal browser (no gcloud, no SSH):

342. user (2026-07-04T12:33:35.552Z)

warning: in the working copy of 'docs/ALPHA_LAUNCH_REVIEW_2026-07-04.md', LF will be replaced by CRLF the next time Git touches it

```
docs/ALPHA_LAUNCH_READINESS.md      |    5 +-
docs/ALPHA_LAUNCH_REVIEW_2026-07-04.md | 475 +++++
docs/DOC_STATUS.md                  |    3 +-
3 files changed, 480 insertions(+), 3 deletions(-)
```

```
diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md
index 8d73436..4323204 100644
```

```
--- a/docs/ALPHA_LAUNCH_READINESS.md
+++ b/docs/ALPHA_LAUNCH_READINESS.md
```

```
@@ -109,7 +109,7 @@ Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/
```

```
| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5, A8 | Owner decision | GATED | - |
```

```
| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance | GATED | - |
```

```
| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train`/`reeval` job, delta-F1 surface, and alignment with `COMPUTE_DECOUPLED_SERVING` M11 | `khelsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen (`khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
```

```
-| A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure triage, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch decision | TODO | - |
```

```
+| A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure triage, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner
```

launch decision | GATED | review DELIVERED 2026-07-04 ?
`docs/ALPHA\_LAUNCH\_REVIEW\_2026-07-04.md`; GO/NO-GO awaiting owner |

8. Concrete next PR sequence

@@ -224,7 +224,7 @@ Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_
10. Definition of done

- [] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
- [] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path.
- + - [x] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path. (Edge auth verified live ? 401 sans CF JWT, `api.khelsutra.guru` ? CF login 302: e2e runlog \$6, re-verified 2026-07-04; disable/rollback documented in `docs/ALPHA\_LAUNCH\_REVIEW\_2026-07-04.md` §5.6.)
- [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.
- [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
- [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.

@@ -267,3 +267,4 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json

- 2026-07-03 - Created from a synced five-repo review of `NEXT\_STEPS`, `ROADMAP`, golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
- 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default decisions, and A9 Gen-0 n15 shadow-training gate verdict.
- 2026-07-04 - Marked A3/A7/A8/A9 complete after collector #62 and indexer #458 review/CI; clarified that the two Boxhill eval inputs are run-log-only 1080p hashes and that F-drive 4K originals are provenance-only.

+ - 2026-07-04 - A13 launch review DELIVERED (`docs/ALPHA\_LAUNCH\_REVIEW\_2026-07-04.md`): conditional-GO recommendation for the current 4-email allowlist (conditions: khelsutra #154 copy fixes + owner browser-CUJ checklist), NO-GO for wider invites until A10/A11 + SA least-privilege + retention. A13 ? GATED on the owner GO/NO-GO; §10 auth/rollback box checked (browser-CUJ box deliberately left open).

343. assistant (2026-07-04T12:33:43.069Z)

The diffs are exactly right. Committing, pushing, and opening the PR.

344. user (2026-07-04T12:33:52.282Z)

docs/a13-launch-review
122f174 docs(runlog): A1 post-merge e2e ? CUJ green on the final image; flip A1 DONE (#477)
warning: in the working copy of 'docs/ALPHA_LAUNCH_REVIEW_2026-07-04.md', LF will be replaced by CRLF the next time Git touches it
[docs/a13-launch-review ffcbe16] docs(a13): alpha launch review ? conditional-GO recommendation; GO/NO-GO to owner
3 files changed, 480 insertions(+), 3 deletions(-)
create mode 100644 docs/ALPHA_LAUNCH_REVIEW_2026-07-04.md
ffcbe16 docs(a13): alpha launch review ? conditional-GO recommendation; GO/NO-GO to owner
remote:
remote: Create a pull request for 'docs/a13-launch-review' on GitHub by visiting:
remote: <https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/docs/a13-launch-review>
remote:
branch 'docs/a13-launch-review' set up to track 'origin/docs/a13-launch-review'.
To <https://github.com/Khelsutra/badminton-highlight-indexer.git>
* [new branch] docs/a13-launch-review -> docs/a13-launch-review

345. user (2026-07-04T12:34:13.685Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/478>

346. user (2026-07-04T12:34:24.025Z)

Command running in background with ID: bxp3w0bh1. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\tasks\bxp3w0bh1.output. You will be notified when it completes. To check interim output, use Read on that file path.

347. user (2026-07-04T12:37:30.530Z)

<retrieval_status>success</retrieval_status>

<task_id>bxp3w0bh1</task_id>

<task_type>local_bash</task_type>

<status>completed</status>

<exit_code>0</exit_code>

<output>

PR478 CI: ALL GREEN (11 checks)

</output>

348. user (2026-07-04T12:37:45.045Z)

#478 state=MERGED mergedAt=2026-07-04T12:37:41Z

Deleted branch docs/a13-launch-review (was ffcbe16).

create mode 100644 docs/ALPHA_LAUNCH_REVIEW_2026-07-04.md

64fa4a2 docs(a13): alpha launch review ? conditional-GO recommendation; GO/NO-GO to owner (#478)

349. user (2026-07-04T12:37:57.217Z)

Updated task #6 status

350. user (2026-07-04T12:38:05.509Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

351. user (2026-07-04T12:38:22.418Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

352. user (2026-07-04T12:38:44.306Z)

The file c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

353. assistant (2026-07-04T12:39:12.275Z)

****The next stage is done: A13 ? the alpha launch review ? is delivered and merged ([#478](https://github.com/Khelsutra/badminton-highlight-indexer/pull/478)), backed by a thorough live test battery on the deployed stack including the Gemini leg. The launch GO/NO-GO now sits with you, with a concrete evidence-based recommendation.****

How it went

****Live test battery (all measured on the real stack, posture fully restored after):****
- ****11-probe failure-mode battery**** ? the hosted-mode input jail held against a foreign bucket
and a separator-boundary attack (`videos-evil`); a missing gs:// object fails cleanly and client-visibly; bad segmenter/compile/jobs/traversal all return crisp 4xx; uploads reject `.exe`, enforce the (newly confirmed) ****4096 MB cap**** with a 413, and 409 out-of-order parts.

- ****Concurrency smoke**** ? three simultaneous cloud jobs all claimed and completed; no false-empty, no stranded dispatch. The #468 fix holds under a real burst.
- ****Gemini handoff leg**** ? the piece the WASB-only alpha never exercised. Two real finds: the `gemini-api-key` secret ****had no accessor grant**** (the documented mount path had never actually been used ? fixed live), and then the full pipeline ran clean: `skip\_ai\_handoff=false` booted with the key check passing, and the MahadevpuraSingles run produced ****43 AI-confirmed segments vs 48 WASB-only****, with worker logs showing real `gemini-flash-latest` calls for all 48 handoff windows. The Gemini path ***works*** in the hosted stack; the alpha stays WASB-only by design, and I restored that posture (rev `00010`, auth ON, `gemini\_key\_present=false` re-verified, `allUsers` restored for the CF path).
- Security hygiene: during test windows I removed the `allUsers` invoker first, so nothing was ever publicly exposed (note: that briefly takes `api.khelsutra.guru` down for testers ? now documented as the standard practice).

****The review itself**** (`docs/ALPHA\_LAUNCH\_REVIEW\_2026-07-04.md`, 475 lines, built from a 4-agent research fan-out + adversarial critique + the battery evidence, scrubbed of secrets/emails):

- ****Recommendation: conditional GO for the current 4-email allowlist**** ? with two conditions before testers touch it ? and ****NO-GO for any wider invites**** until A10 (ship-gate) and A11 (WASB-C3 provenance) close, plus a least-privilege service account and a retention fix.
- ****The two conditions:**** (1) fix the copy blockers filed as [khelsutra #154](https://github.com/avidullu/khelsutra/issues/154) ? the Studio wizard tells users their video "never leaves this device," which is ****false**** on the hosted alpha, and the policy claims "our own infrastructure" while it runs on Google Cloud Singapore; (2) run the 5-minute ****browser CUJ checklist**** in the review's §1.1 ? the honest gap the critique surfaced is that every CUJ pass so far (mine included) was curl-driven, and the tracker's DoD #1 requires a real browser pass, which only an allowlisted human can do.
- Also inside: a full operator runbook (with the deploy gotchas and cost notes), an observability/triage table built from the actual battery results, the tester one-pager, rollback levers R1?R5, and the SA blast-radius warning (`roles/editor` on a public-invoker service).

****Budget:**** this stage consumed roughly four L4 executions (three short, one full match), one Gemini-enabled run (~48 flash-model calls), and no new image builds ? well inside the \$300; the exact dollar figure needs a billing-console glance, which the review flags rather than guesses.

****What's on you now:**** the GO/NO-GO record, the §1.1 browser run-through, the #154 copy approval, the A10/A11 decisions, and the cert check before ~Sept 30 (healthy as of today). Memory and the handoff are updated so any session can pick up from exactly here; the natural next engineering grab is the khelsutra #154 fix PR.